

EMS Backend Actual Files

Full source and configuration code

Project path: C:\Users\cso\Desktop\Citc-Project\hr-backend\ems

Generated: 2026-07-15 12:55

Included files: 94

Total source/config lines: 5729

Excluded: .git, target, .idea, .agents, output

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- 001. .gitattributes (2 lines)
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- 003. HELP.md (37 lines)
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- 005. mvnw (295 lines)
- 006. mvnw.cmd (190 lines)
- 007. .mvn/wrapper/maven-wrapper.properties (3 lines)
- 008. src/main/resources/application.yml (23 lines)
- 009. src/main/java/com/citec/ems/EmsApplication.java (20 lines)
- 010. src/main/java/com/citec/ems/hr/application/EmployeeService.java (384 lines)
- 011. src/main/java/com/citec/ems/hr/application/HrReferenceService.java (276 lines)
- 012. src/main/java/com/citec/ems/hr/domain/ContractType.java (45 lines)
- 013. src/main/java/com/citec/ems/hr/domain/Employee.java (65 lines)
- 014. src/main/java/com/citec/ems/hr/domain/EmployeeProfile.java (100 lines)
- 015. src/main/java/com/citec/ems/hr/domain/EmployeeStatus.java (32 lines)
- 016. src/main/java/com/citec/ems/hr/domain/EmploymentContract.java (81 lines)
- 017. src/main/java/com/citec/ems/hr/domain/EmploymentRecord.java (74 lines)
- 018. src/main/java/com/citec/ems/hr/domain/JobGrade.java (50 lines)
- 019. src/main/java/com/citec/ems/hr/domain/JobPosition.java (77 lines)
- 020. src/main/java/com/citec/ems/hr/domain/Organization.java (50 lines)
- 021. src/main/java/com/citec/ems/hr/domain/OrganizationProfile.java (94 lines)
- 022. src/main/java/com/citec/ems/hr/domain/OrganizationUnit.java (78 lines)
- 023. src/main/java/com/citec/ems/hr/domain/OrganizationUnitRelation.java (63 lines)
- 024. src/main/java/com/citec/ems/hr/domain/PositionAssignment.java (71 lines)
- 025. src/main/java/com/citec/ems/hr/domain/RelationType.java (42 lines)
- 026. src/main/java/com/citec/ems/hr/domain/UnitType.java (42 lines)
- 027. src/main/java/com/citec/ems/hr/infrastructure/ContractTypeRepository.java (14 lines)

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028. src/main/java/com/citec/ems/hr/infrastructure/EmployeeProfileRepository.java (17 lines)
029. src/main/java/com/citec/ems/hr/infrastructure/EmployeeRepository.java (33 lines)
030. src/main/java/com/citec/ems/hr/infrastructure/EmployeeStatusRepository.java (14 lines)
031. src/main/java/com/citec/ems/hr/infrastructure/EmploymentContractRepository.java (14 lines)
032. src/main/java/com/citec/ems/hr/infrastructure/EmploymentRecordRepository.java (17 lines)
033. src/main/java/com/citec/ems/hr/infrastructure/JobGradeRepository.java (14 lines)
034. src/main/java/com/citec/ems/hr/infrastructure/JobPositionRepository.java (30 lines)
035. src/main/java/com/citec/ems/hr/infrastructure/OrganizationProfileRepository.java (14 lines)
036. src/main/java/com/citec/ems/hr/infrastructure/OrganizationRepository.java (14 lines)
037. src/main/java/com/citec/ems/hr/infrastructure/OrganizationUnitRelationRepository.java (12 lines)
038. src/main/java/com/citec/ems/hr/infrastructure/OrganizationUnitRepository.java (32 lines)
039. src/main/java/com/citec/ems/hr/infrastructure/PositionAssignmentRepository.java (14 lines)
040. src/main/java/com/citec/ems/hr/infrastructure/RelationTypeRepository.java (14 lines)
041. src/main/java/com/citec/ems/hr/infrastructure/UnitTypeRepository.java (14 lines)
042. src/main/java/com/citec/ems/hr/package-info.java (4 lines)
043. src/main/java/com/citec/ems/hr/web/EmployeeController.java (103 lines)
044. src/main/java/com/citec/ems/hr/web/HrDtos.java (299 lines)
045. src/main/java/com/citec/ems/hr/web/HrReferenceController.java (114 lines)
046. src/main/java/com/citec/ems/iam/UserAccountSummary.java (4 lines)
047. src/main/java/com/citec/ems/iam/UserAccounts.java (27 lines)
048. src/main/java/com/citec/ems/iam/application/AuthService.java (295 lines)
049. src/main/java/com/citec/ems/iam/application/IamAdminService.java (228 lines)
050. src/main/java/com/citec/ems/iam/application/IpRuleService.java (108 lines)
051. src/main/java/com/citec/ems/iam/domain/ApplicationModule.java (39 lines)
052. src/main/java/com/citec/ems/iam/domain/ClientApplication.java (33 lines)
053. src/main/java/com/citec/ems/iam/domain/LoginAttempt.java (63 lines)
054. src/main/java/com/citec/ems/iam/domain/Permission.java (52 lines)
055. src/main/java/com/citec/ems/iam/domain/RefreshToken.java (71 lines)
056. src/main/java/com/citec/ems/iam/domain/Role.java (45 lines)
057. src/main/java/com/citec/ems/iam/domain/RolePermission.java (62 lines)
058. src/main/java/com/citec/ems/iam/domain/RolePermissionId.java (32 lines)
059. src/main/java/com/citec/ems/iam/domain/User.java (54 lines)
060. src/main/java/com/citec/ems/iam/domain/UserDevice.java (93 lines)
061. src/main/java/com/citec/ems/iam/domain/UserLoginIpRule.java (71 lines)
062. src/main/java/com/citec/ems/iam/domain/UserRole.java (59 lines)
063. src/main/java/com/citec/ems/iam/domain/UserRoleId.java (32 lines)
064. src/main/java/com/citec/ems/iam/domain/UserSession.java (78 lines)
065. src/main/java/com/citec/ems/iam/domain/UserStatus.java (32 lines)
066. src/main/java/com/citec/ems/iam/infrastructure/ApplicationModuleRepository.java (14 lines)
067. src/main/java/com/citec/ems/iam/infrastructure/ClientApplicationRepository.java (14 lines)
068. src/main/java/com/citec/ems/iam/infrastructure/LoginAttemptRepository.java (12 lines)
069. src/main/java/com/citec/ems/iam/infrastructure/PermissionRepository.java (14 lines)
070. src/main/java/com/citec/ems/iam/infrastructure/RefreshTokenRepository.java (14 lines)
071. src/main/java/com/citec/ems/iam/infrastructure/RolePermissionRepository.java (34 lines)
072. src/main/java/com/citec/ems/iam/infrastructure/RoleRepository.java (14 lines)
073. src/main/java/com/citec/ems/iam/infrastructure/UserDeviceRepository.java (15 lines)
074. src/main/java/com/citec/ems/iam/infrastructure/UserLoginIpRuleRepository.java (31 lines)

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075. src/main/java/com/citec/ems/iam/infrastructure/UserRepository.java (27 lines)
076. src/main/java/com/citec/ems/iam/infrastructure/UserRoleRepository.java (28 lines)
077. src/main/java/com/citec/ems/iam/infrastructure/UserSessionRepository.java (15 lines)
078. src/main/java/com/citec/ems/iam/infrastructure/UserStatusRepository.java (14 lines)
079. src/main/java/com/citec/ems/iam/package-info.java (4 lines)
080. src/main/java/com/citec/ems/iam/security/AccessTokenAuthenticationFilter.java (80 lines)
081. src/main/java/com/citec/ems/iam/security/AccessTokenClaims.java (17 lines)
082. src/main/java/com/citec/ems/iam/security/AccessTokenService.java (175 lines)
083. src/main/java/com/citec/ems/iam/security/AuthenticatedUser.java (15 lines)
084. src/main/java/com/citec/ems/iam/security/EmsSecurityProperties.java (26 lines)
085. src/main/java/com/citec/ems/iam/security/RefreshTokenService.java (33 lines)
086. src/main/java/com/citec/ems/iam/security/SecurityConfig.java (70 lines)
087. src/main/java/com/citec/ems/iam/web/AuthController.java (51 lines)
088. src/main/java/com/citec/ems/iam/web/IamAdminController.java (97 lines)
089. src/main/java/com/citec/ems/iam/web/IamDtos.java (144 lines)
090. src/main/java/com/citec/ems/shared/ApiExceptionHandler.java (50 lines)
091. src/main/java/com/citec/ems/shared/BadRequestException.java (11 lines)
092. src/main/java/com/citec/ems/shared/NotFoundException.java (11 lines)
093. src/main/java/com/citec/ems/shared/package-info.java (4 lines)
094. src/test/java/com/citec/ems/EmsApplicationTests.java (19 lines)

001. .gitattributes

.gitattributes

2 lines

```
00001 | /mvnw text eol=lf
00002 | *.cmd text eol=crlf
```

002. .gitignore

.gitignore

33 lines

```
00001 | HELP.md
00002 | target/
00003 | .mvn/wrapper/maven-wrapper.jar
00004 | !**/src/main/**/target/
00005 | !**/src/test/**/target/
00006 |
00007 | ### STS ###
00008 | .apt_generated
00009 | .classpath
00010 | .factorypath
00011 | .project
00012 | .settings
00013 | .springBeans
00014 | .sts4-cache
00015 |
00016 | ### IntelliJ IDEA ###
00017 | .idea
00018 | *.iws
00019 | *.iml
00020 | *.ipr
00021 |
00022 | ### NetBeans ###
00023 | /nbproject/private/
00024 | /nbbuild/
00025 | /dist/
00026 | /nbdist/
00027 | /.nb-gradle/
00028 | build/
00029 | !**/src/main/**/build/
00030 | !**/src/test/**/build/
00031 |
00032 | ### VS Code ###
00033 | .vscode/
```

```
00001 | # Getting Started
00002 |
00003 | ### Reference Documentation
00004 |
00005 | For further reference, please consider the following sections:
00006 |
00007 | * [Official Apache Maven documentation](https://maven.apache.org/guides/index.html)
00008 | * [Spring Boot Maven Plugin Reference Guide](https://docs.spring.io/spring-boot/4.1.0/maven-plugin)
00009 | * [Create an OCI image](https://docs.spring.io/spring-boot/4.1.0/maven-plugin/build-image.html)
00010 | * [Spring Data JPA](https://docs.spring.io/spring-boot/4.1.0/reference/data/sql.html#data.sql.jpa-and-spring-data)
00011 | * [Spring Boot DevTools](https://docs.spring.io/spring-boot/4.1.0/reference/using/devtools.html)
00012 | * [Spring Modulith](https://docs.spring.io/spring-modulith/reference/)
00013 | * [Spring Security](https://docs.spring.io/spring-boot/4.1.0/reference/web/spring-security.html)
00014 | * [Validation](https://docs.spring.io/spring-boot/4.1.0/reference/io/validation.html)
00015 | * [Spring Web](https://docs.spring.io/spring-boot/4.1.0/reference/web/servlet.html)
00016 |
00017 | ### Guides
00018 |
00019 | The following guides illustrate how to use some features concretely:
00020 |
00021 | * [Accessing Data with JPA](https://spring.io/guides/gs/accessing-data-jpa/)
00022 | * [Securing a Web Application](https://spring.io/guides/gs/securing-web/)
00023 | * [Spring Boot and OAuth2](https://spring.io/guides/tutorials/spring-boot-oauth2/)
00024 | * [Authenticating a User with LDAP](https://spring.io/guides/gs/authenticating-ldap/)
00025 | * [Validation](https://spring.io/guides/gs/validating-form-input/)
00026 | * [Building a RESTful Web Service](https://spring.io/guides/gs/rest-service/)
00027 | * [Serving Web Content with Spring MVC](https://spring.io/guides/gs/serving-web-content/)
00028 | * [Building REST services with Spring](https://spring.io/guides/tutorials/rest/)
00029 |
00030 | ### Maven Parent overrides
00031 |
00032 | Due to Maven's design, elements are inherited from the parent POM to the project POM.
00033 | While most of the inheritance is fine, it also inherits unwanted elements like `` and `` from the
00034 | parent.
00035 | To prevent this, the project POM contains empty overrides for these elements.
00036 | If you manually switch to a different parent and actually want the inheritance, you need to remove those overrides.
00037 |
```

004. pom.xml

pom.xml

184 lines

```
00001 | <?xml version="1.0" encoding="UTF-8"?>
00002 | <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
00003 |     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
00004 |     <modelVersion>4.0.0</modelVersion>
00005 |     <parent>
00006 |         <groupId>org.springframework.boot</groupId>
00007 |         <artifactId>spring-boot-starter-parent</artifactId>
00008 |         <version>4.1.0</version>
00009 |         <relativePath/> <!-- lookup parent from repository -->
00010 |     </parent>
00011 |     <groupId>com.citec</groupId>
00012 |     <artifactId>ems</artifactId>
00013 |     <version>0.0.1-SNAPSHOT</version>
00014 |     <name>ems</name>
00015 |     <description>ems</description>
00016 |     <url/>
00017 |     <licenses>
00018 |         <license/>
00019 |     </licenses>
00020 |     <developers>
00021 |         <developer/>
00022 |     </developers>
00023 |     <scm>
00024 |         <connection/>
00025 |         <developerConnection/>
00026 |         <tag/>
00027 |         <url/>
00028 |     </scm>
00029 |     <properties>
00030 |         <java.version>21</java.version>
00031 |         <spring-modulith.version>2.1.0</spring-modulith.version>
00032 |     </properties>
00033 |     <dependencies>
00034 |         <dependency>
00035 |             <groupId>org.springframework.boot</groupId>
00036 |             <artifactId>spring-boot-starter-data-jpa</artifactId>
00037 |         </dependency>
00038 |         <dependency>
00039 |             <groupId>org.springframework.boot</groupId>
00040 |             <artifactId>spring-boot-starter-security</artifactId>
00041 |         </dependency>
00042 |         <dependency>
00043 |             <groupId>org.springframework.boot</groupId>
00044 |             <artifactId>spring-boot-starter-validation</artifactId>
00045 |         </dependency>
00046 |         <dependency>
00047 |             <groupId>org.springframework.boot</groupId>
00048 |             <artifactId>spring-boot-starter-webmvc</artifactId>
00049 |         </dependency>
00050 |     </dependencies>
```

004. pom.xml

```
00051 |         <groupId>org.springframework.modulith</groupId>
00052 |         <artifactId>spring-modulith-starter-core</artifactId>
00053 |     </dependency>
00054 |     <dependency>
00055 |         <groupId>org.springframework.modulith</groupId>
00056 |         <artifactId>spring-modulith-starter-jdbc</artifactId>
00057 |     </dependency>
00058 |
00059 |     <dependency>
00060 |         <groupId>org.springframework.boot</groupId>
00061 |         <artifactId>spring-boot-devtools</artifactId>
00062 |         <scope>runtime</scope>
00063 |         <optional>true</optional>
00064 |     </dependency>
00065 |     <dependency>
00066 |         <groupId>org.postgresql</groupId>
00067 |         <artifactId>postgresql</artifactId>
00068 |         <scope>runtime</scope>
00069 |     </dependency>
00070 |     <dependency>
00071 |         <groupId>org.projectlombok</groupId>
00072 |         <artifactId>lombok</artifactId>
00073 |         <optional>true</optional>
00074 |     </dependency>
00075 |
00076 |     <dependency>
00077 |         <groupId>io.jsonwebtoken</groupId>
00078 |         <artifactId>jjwt-impl</artifactId>
00079 |         <version>0.12.6</version>
00080 |         <scope>runtime</scope>
00081 |     </dependency>
00082 |     <dependency>
00083 |         <groupId>io.jsonwebtoken</groupId>
00084 |         <artifactId>jjwt-jackson</artifactId>
00085 |         <version>0.12.6</version>
00086 |         <scope>runtime</scope>
00087 |     </dependency>
00088 |     <dependency>
00089 |         <groupId>org.springdoc</groupId>
00090 |         <artifactId>springdoc-openapi-starter-webmvc-ui</artifactId>
00091 |         <version>2.8.13</version>
00092 |     </dependency>
00093 |     <dependency>
00094 |         <groupId>org.springframework.boot</groupId>
00095 |         <artifactId>spring-boot-starter-data-jpa-test</artifactId>
00096 |         <scope>test</scope>
00097 |     </dependency>
00098 |     <dependency>
00099 |         <groupId>org.springframework.boot</groupId>
00100 |         <artifactId>spring-boot-starter-security-test</artifactId>
00101 |         <scope>test</scope>
00102 |     </dependency>
00103 |     <dependency>
00104 |         <groupId>org.springframework.boot</groupId>
00105 |         <artifactId>spring-boot-starter-validation-test</artifactId>
```


004. pom.xml

```
00106 |         <scope>test</scope>
00107 |     </dependency>
00108 |     <dependency>
00109 |         <groupId>org.springframework.boot</groupId>
00110 |         <artifactId>spring-boot-starter-webmvc-test</artifactId>
00111 |         <scope>test</scope>
00112 |     </dependency>
00113 |     <dependency>
00114 |         <groupId>org.springframework.modulith</groupId>
00115 |         <artifactId>spring-modulith-starter-test</artifactId>
00116 |         <scope>test</scope>
00117 |     </dependency>
00118 | </dependencies>
00119 | <dependencyManagement>
00120 |     <dependencies>
00121 |         <dependency>
00122 |             <groupId>org.springframework.modulith</groupId>
00123 |             <artifactId>spring-modulith-bom</artifactId>
00124 |             <version>${spring-modulith.version}</version>
00125 |             <type>pom</type>
00126 |             <scope>import</scope>
00127 |         </dependency>
00128 |     </dependencies>
00129 | </dependencyManagement>
00130 |
00131 | <build>
00132 |     <plugins>
00133 |         <plugin>
00134 |             <groupId>org.springframework.boot</groupId>
00135 |             <artifactId>spring-boot-maven-plugin</artifactId>
00136 |             <configuration>
00137 |                 <excludes>
00138 |                     <exclude>
00139 |                         <groupId>org.project.lombok</groupId>
00140 |                         <artifactId>lombok</artifactId>
00141 |                     </exclude>
00142 |                 </excludes>
00143 |             </configuration>
00144 |         </plugin>
00145 |         <plugin>
00146 |             <groupId>org.apache.maven.plugins</groupId>
00147 |             <artifactId>maven-compiler-plugin</artifactId>
00148 |             <executions>
00149 |                 <execution>
00150 |                     <id>default-compile</id>
00151 |                     <phase>compile</phase>
00152 |                     <goals>
00153 |                         <goal>compile</goal>
00154 |                     </goals>
00155 |                     <configuration>
00156 |                         <annotationProcessorPaths>
00157 |                             <path>
00158 |                                 <groupId>org.project.lombok</groupId>
00159 |                                 <artifactId>lombok</artifactId>
00160 |                             </path>
```

004. pom.xml

```
00161 |                 </annotationProcessorPaths>
00162 |             </configuration>
00163 |         </execution>
00164 |     <execution>
00165 |         <id>default-testCompile</id>
00166 |         <phase>test-compile</phase>
00167 |         <goals>
00168 |             <goal>testCompile</goal>
00169 |         </goals>
00170 |         <configuration>
00171 |             <annotationProcessorPaths>
00172 |                 <path>
00173 |                     <groupId>org.projectlombok</groupId>
00174 |                     <artifactId>lombok</artifactId>
00175 |                 </path>
00176 |             </annotationProcessorPaths>
00177 |         </configuration>
00178 |     </execution>
00179 | </executions>
00180 | </plugin>
00181 | </plugins>
00182 | </build>
00183 |
00184 | </project>
```

```

00001 | #!/bin/sh
00002 | # -----
00003 | # Licensed to the Apache Software Foundation (ASF) under one
00004 | # or more contributor license agreements. See the NOTICE file
00005 | # distributed with this work for additional information
00006 | # regarding copyright ownership. The ASF licenses this file
00007 | # to you under the Apache License, Version 2.0 (the
00008 | # "License"); you may not use this file except in compliance
00009 | # with the License. You may obtain a copy of the License at
00010 | #
00011 | # http://www.apache.org/licenses/LICENSE-2.0
00012 | #
00013 | # Unless required by applicable law or agreed to in writing,
00014 | # software distributed under the License is distributed on an
00015 | # "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY
00016 | # KIND, either express or implied. See the License for the
00017 | # specific language governing permissions and limitations
00018 | # under the License.
00019 | # -----
00020 |
00021 | # -----
00022 | # Apache Maven Wrapper startup batch script, version 3.3.4
00023 | #
00024 | # Optional ENV vars
00025 | # -----
00026 | # JAVA_HOME - location of a JDK home dir, required when download maven via java source
00027 | # MVNW_REPOURL - repo url base for downloading maven distribution
00028 | # MVNW_USERNAME/MVNW_PASSWORD - user and password for downloading maven
00029 | # MVNW_VERBOSE - true: enable verbose log; debug: trace the mvnw script; others: silence the output
00030 | # -----
00031 |
00032 | set -euf
00033 | [ "${MVNW_VERBOSE-}" != debug ] || set -x
00034 |
00035 | # OS specific support.
00036 | native_path() { printf %s\\n "$1"; }
00037 | case "$(uname)" in
00038 | CYGWIN* | MINGW*)
00039 |     [ -z "${JAVA_HOME-}" ] || JAVA_HOME="$(cygpath --unix "$JAVA_HOME")"
00040 |     native_path() { cygpath --path --windows "$1"; }
00041 |     ;;
00042 | esac
00043 |
00044 | # set JAVACMD and JAVACMD
00045 | set_java_home() {
00046 |     # For Cygwin and MinGW, ensure paths are in Unix format before anything is touched
00047 |     if [ -n "${JAVA_HOME-}" ]; then
00048 |         if [ -x "$JAVA_HOME/jre/sh/java" ]; then
00049 |             # IBM's JDK on AIX uses strange locations for the executables
00050 |             JAVACMD="$JAVA_HOME/jre/sh/java"

```

005. mvnw

```
00051 |     JAVACCMD="$JAVA_HOME/jre/sh/javac"
00052 | else
00053 |     JAVACMD="$JAVA_HOME/bin/java"
00054 |     JAVACCMD="$JAVA_HOME/bin/javac"
00055 |
00056 |     if [ ! -x "$JAVACMD" ] || [ ! -x "$JAVACMD" ]; then
00057 |         echo "The JAVA_HOME environment variable is not defined correctly, so mvnw cannot run." >&2
00058 |         echo "JAVA_HOME is set to \"\$JAVA_HOME\", but \"\$JAVA_HOME/bin/java\" or \"\$JAVA_HOME/bin/javac\" does not exist." >&2
00059 |         return 1
00060 |     fi
00061 | fi
00062 | else
00063 |     JAVACMD="$(
00064 |         'set' +e
00065 |         'unset' -f command 2>/dev/null
00066 |         'command' -v java
00067 |     )" || :
00068 |     JAVACCMD="$(
00069 |         'set' +e
00070 |         'unset' -f command 2>/dev/null
00071 |         'command' -v javac
00072 |     )" || :
00073 |
00074 |     if [ ! -x "${JAVACMD-}" ] || [ ! -x "${JAVACMD-}" ]; then
00075 |         echo "The java/javac command does not exist in PATH nor is JAVA_HOME set, so mvnw cannot run." >&2
00076 |         return 1
00077 |     fi
00078 | fi
00079 | }
00080 |
00081 | # hash string like Java String::hashCode
00082 | hash_string() {
00083 |     str="${1:-}" h=0
00084 |     while [ -n "$str" ]; do
00085 |         char="${str%${str#?}}"
00086 |         h=$((h * 31 + $(LC_CTYPE=C printf %d "'$char') % 4294967296))
00087 |         str="${str#?}"
00088 |     done
00089 |     printf %x\\n $h
00090 | }
00091 |
00092 | verbose() { ;; }
00093 | [ "${MVNW_VERBOSE-}" != true ] || verbose() { printf %s\\n "${1-}"; }
00094 |
00095 | die() {
00096 |     printf %s\\n "$1" >&2
00097 |     exit 1
00098 | }
00099 |
00100 | trim() {
00101 |     # MWRAPPER-139:
00102 |     #   Trims trailing and leading whitespace, carriage returns, tabs, and linefeeds.
00103 |     #   Needed for removing poorly interpreted newline sequences when running in more
00104 |     #   exotic environments such as mingw bash on Windows.
00105 |     printf "%s" "${1}" | tr -d '[:space:]'
```

005. mvnw

```
00106 | }
00107 |
00108 | scriptDir="$(dirname "$0")"
00109 | scriptName="$(basename "$0")"
00110 |
00111 | # parse distributionUrl and optional distributionSha256Sum, requires .mvn/wrapper/maven-wrapper.properties
00112 | while IFS="=" read -r key value; do
00113 |     case "${key-}" in
00114 |         distributionUrl) distributionUrl=$(trim "${value-}") ;;
00115 |         distributionSha256Sum) distributionSha256Sum=$(trim "${value-}") ;;
00116 |     esac
00117 | done <"$scriptDir/.mvn/wrapper/maven-wrapper.properties"
00118 | [ -n "${distributionUrl-}" ] || die "cannot read distributionUrl property in $scriptDir/.mvn/wrapper/maven-wrapper.properties"
00119 |
00120 | case "${distributionUrl##*/}" in
00121 | maven-mvnd-*bin.*)
00122 |     MVN_CMD=mvnd.sh _MVNW_REPO_PATTERN=/maven/mvnd/
00123 |     case "${PROCESSOR_ARCHITECTURE-}${PROCESSOR_ARCHITECTURE-}" in $(uname -a) ) in
00124 |         *AMD64:CYGWIN* | *AMD64:MINGW*) distributionPlatform=windows-amd64 ;;
00125 |         :Darwin*x86_64) distributionPlatform=darwin-amd64 ;;
00126 |         :Darwin*arm64) distributionPlatform=darwin-aarch64 ;;
00127 |         :Linux*x86_64) distributionPlatform=linux-amd64 ;;
00128 |     *)
00129 |         echo "Cannot detect native platform for mvnd on $(uname)-$(uname -m), use pure java version" >&2
00130 |         distributionPlatform=linux-amd64
00131 |     ;;
00132 |     esac
00133 |     distributionUrl="${distributionUrl%-bin.*}-${distributionPlatform}.zip"
00134 | ;;
00135 | maven-mvnd-*) MVN_CMD=mvnd.sh _MVNW_REPO_PATTERN=/maven/mvnd/ ;;
00136 | *) MVN_CMD="mvn${scriptName#mvnw}" _MVNW_REPO_PATTERN=/org/apache/maven/ ;;
00137 | esac
00138 |
00139 | # apply MVNW_REPOURL and calculate MAVEN_HOME
00140 | # maven home pattern: ~/.m2/wrapper/dists/{apache-maven-<version>,maven-mvnd-<version>-<platform>}/<hash>
00141 | [ -z "${MVNW_REPOURL-}" ] || distributionUrl="$MVNW_REPOURL${distributionUrl#*"$_MVNW_REPO_PATTERN"}"
00142 | distributionUrlName="${distributionUrl##*/}"
00143 | distributionUrlNameMain="${distributionUrlName%.*}"
00144 | distributionUrlNameMain="${distributionUrlNameMain%-bin}"
00145 | MAVEN_USER_HOME="${MAVEN_USER_HOME:-${HOME}/.m2}"
00146 | MAVEN_HOME="${MAVEN_USER_HOME}/wrapper/dists/${distributionUrlNameMain}-${hash_string "$distributionUrl"}"
00147 |
00148 | exec_maven() {
00149 |     unset MVNW_VERBOSE MVNW_USERNAME MVNW_PASSWORD MVNW_REPOURL || :
00150 |     exec "$MAVEN_HOME/bin/$MVN_CMD" "$@" || die "cannot exec $MAVEN_HOME/bin/$MVN_CMD"
00151 | }
00152 |
00153 | if [ -d "$MAVEN_HOME" ]; then
00154 |     verbose "found existing MAVEN_HOME at $MAVEN_HOME"
00155 |     exec_maven "$@"
00156 | fi
00157 |
00158 | case "${distributionUrl-}" in
00159 | *?-bin.zip | *?maven-mvnd-*?*.zip) ;;
00160 | *) die "distributionUrl is not valid, must match *-bin.zip or maven-mvnd-*.zip, but found '${distributionUrl-}'" ;;
```

005. mvnw

```
00161 | esac
00162 |
00163 | # prepare tmp dir
00164 | if TMP_DOWNLOAD_DIR="$(mktemp -d)" && [ -d "$TMP_DOWNLOAD_DIR" ]; then
00165 |     clean() { rm -rf -- "$TMP_DOWNLOAD_DIR"; }
00166 |     trap clean HUP INT TERM EXIT
00167 | else
00168 |     die "cannot create temp dir"
00169 | fi
00170 |
00171 | mkdir -p -- "${MAVEN_HOME%/*}"
00172 |
00173 | # Download and Install Apache Maven
00174 | verbose "Couldn't find MAVEN_HOME, downloading and installing it ..."
00175 | verbose "Downloading from: $distributionUrl"
00176 | verbose "Downloading to: $TMP_DOWNLOAD_DIR/$distributionUrlName"
00177 |
00178 | # select .zip or .tar.gz
00179 | if ! command -v unzip >/dev/null; then
00180 |     distributionUrl="${distributionUrl%.zip}.tar.gz"
00181 |     distributionUrlName="${distributionUrl##*/}"
00182 | fi
00183 |
00184 | # verbose opt
00185 | __MVNW_QUIET_WGET=--quiet __MVNW_QUIET_CURL=--silent __MVNW_QUIET_UNZIP=-q __MVNW_QUIET_TAR=''
00186 | [ "${MVNW_VERBOSE-}" != true ] || __MVNW_QUIET_WGET='' __MVNW_QUIET_CURL='' __MVNW_QUIET_UNZIP='' __MVNW_QUIET_TAR=v
00187 |
00188 | # normalize http auth
00189 | case "${MVNW_PASSWORD:+has-password}" in
00190 |     '') MVNW_USERNAME='' MVNW_PASSWORD='' ;;
00191 |     has-password) [ -n "${MVNW_USERNAME-}" ] || MVNW_USERNAME='' MVNW_PASSWORD='' ;;
00192 | esac
00193 |
00194 | if [ -z "${MVNW_USERNAME-}" ] && command -v wget >/dev/null; then
00195 |     verbose "Found wget ... using wget"
00196 |     wget ${__MVNW_QUIET_WGET:+$__MVNW_QUIET_WGET} "$distributionUrl" -O "$TMP_DOWNLOAD_DIR/$distributionUrlName" || die "wget: Failed to fetch $distributionUrl"
00197 | elif [ -z "${MVNW_USERNAME-}" ] && command -v curl >/dev/null; then
00198 |     verbose "Found curl ... using curl"
00199 |     curl ${__MVNW_QUIET_CURL:+$__MVNW_QUIET_CURL} -f -L -o "$TMP_DOWNLOAD_DIR/$distributionUrlName" "$distributionUrl" || die "curl: Failed to fetch $distributionUrl"
00200 | elif set_java_home; then
00201 |     verbose "Falling back to use Java to download"
00202 |     javaSource="$TMP_DOWNLOAD_DIR/Downloader.java"
00203 |     targetZip="$TMP_DOWNLOAD_DIR/$distributionUrlName"
00204 |     cat >"$javaSource" <<-END
00205 |     public class Downloader extends java.net.Authenticator
00206 |     {
00207 |         protected java.net.PasswordAuthentication getPasswordAuthentication()
00208 |         {
00209 |             return new java.net.PasswordAuthentication( System.getenv( "MVNW_USERNAME" ), System.getenv( "MVNW_PASSWORD" ).toCharArray() );
00210 |         }
00211 |         public static void main( String[] args ) throws Exception
00212 |         {
00213 |             setDefault( new Downloader() );
00214 |             java.nio.file.Files.copy( java.net.URI.create( args[0] ).toURL().openStream(), java.nio.file.Paths.get( args[1] ).toAbsolutePath().normalize() );
00215 |         }
00216 |     }
```

005. mvnw

```
00216 |     }
00217 |     END
00218 |     # For Cygwin/MinGW, switch paths to Windows format before running javac and java
00219 |     verbose " - Compiling Downloader.java ..."
00220 |     "${native_path "$JAVACMD")" "${(native_path "$javaSource")}" || die "Failed to compile Downloader.java"
00221 |     verbose " - Running Downloader.java ..."
00222 |     "${(native_path "$JAVACMD")" -cp "${(native_path "$TMP_DOWNLOAD_DIR")}" Downloader "$distributionUrl" "${(native_path "$targetZip")}"
00223 | fi
00224 |
00225 | # If specified, validate the SHA-256 sum of the Maven distribution zip file
00226 | if [ -n "${distributionSha256Sum-}" ]; then
00227 |     distributionSha256Result=false
00228 |     if [ "$MVN_CMD" = mvnd.sh ]; then
00229 |         echo "Checksum validation is not supported for maven-mvnd." >&2
00230 |         echo "Please disable validation by removing 'distributionSha256Sum' from your maven-wrapper.properties." >&2
00231 |         exit 1
00232 |     elif command -v sha256sum >/dev/null; then
00233 |         if echo "$distributionSha256Sum $TMP_DOWNLOAD_DIR/$distributionUrlName" | sha256sum -c - >/dev/null 2>&1; then
00234 |             distributionSha256Result=true
00235 |         fi
00236 |     elif command -v shasum >/dev/null; then
00237 |         if echo "$distributionSha256Sum $TMP_DOWNLOAD_DIR/$distributionUrlName" | shasum -a 256 -c >/dev/null 2>&1; then
00238 |             distributionSha256Result=true
00239 |         fi
00240 |     else
00241 |         echo "Checksum validation was requested but neither 'sha256sum' or 'shasum' are available." >&2
00242 |         echo "Please install either command, or disable validation by removing 'distributionSha256Sum' from your maven-wrapper.properties." >&2
00243 |         exit 1
00244 |     fi
00245 |     if [ $distributionSha256Result = false ]; then
00246 |         echo "Error: Failed to validate Maven distribution SHA-256, your Maven distribution might be compromised." >&2
00247 |         echo "If you updated your Maven version, you need to update the specified distributionSha256Sum property." >&2
00248 |         exit 1
00249 |     fi
00250 | fi
00251 |
00252 | # unzip and move
00253 | if command -v unzip >/dev/null; then
00254 |     unzip ${_MVNW_QUIET_UNZIP:+"$_MVNW_QUIET_UNZIP"} "$TMP_DOWNLOAD_DIR/$distributionUrlName" -d "$TMP_DOWNLOAD_DIR" || die "failed to unzip"
00255 | else
00256 |     tar xzf${_MVNW_QUIET_TAR:+"$_MVNW_QUIET_TAR"} "$TMP_DOWNLOAD_DIR/$distributionUrlName" -C "$TMP_DOWNLOAD_DIR" || die "failed to untar"
00257 | fi
00258 |
00259 | # Find the actual extracted directory name (handles snapshots where filename != directory name)
00260 | actualDistributionDir=""
00261 |
00262 | # First try the expected directory name (for regular distributions)
00263 | if [ -d "$TMP_DOWNLOAD_DIR/$distributionUrlNameMain" ]; then
00264 |     if [ -f "$TMP_DOWNLOAD_DIR/$distributionUrlNameMain/bin/$MVN_CMD" ]; then
00265 |         actualDistributionDir="$distributionUrlNameMain"
00266 |     fi
00267 | fi
00268 |
00269 | # If not found, search for any directory with the Maven executable (for snapshots)
00270 | if [ -z "$actualDistributionDir" ]; then
```

005. mvnw

```
00271 | # enable globbing to iterate over items
00272 | set +f
00273 | for dir in "$TMP_DOWNLOAD_DIR"/*; do
00274 |     if [ -d "$dir" ]; then
00275 |         if [ -f "$dir/bin/$MVN_CMD" ]; then
00276 |             actualDistributionDir="$(basename "$dir")"
00277 |             break
00278 |         fi
00279 |     fi
00280 | done
00281 | set -f
00282 | fi
00283 |
00284 | if [ -z "$actualDistributionDir" ]; then
00285 |     verbose "Contents of $TMP_DOWNLOAD_DIR:"
00286 |     verbose "$(ls -la "$TMP_DOWNLOAD_DIR")"
00287 |     die "Could not find Maven distribution directory in extracted archive"
00288 | fi
00289 |
00290 | verbose "Found extracted Maven distribution directory: $actualDistributionDir"
00291 | printf %s\\n "$distributionUrl" >"$TMP_DOWNLOAD_DIR/$actualDistributionDir/mvnw.url"
00292 | mv -- "$TMP_DOWNLOAD_DIR/$actualDistributionDir" "$MAVEN_HOME" || [ -d "$MAVEN_HOME" ] || die "fail to move MAVEN_HOME"
00293 |
00294 | clean || :
00295 | exec_maven "$@"
```


mvnw.cmd

190 lines

```

00001 | <# : batch portion
00002 | @REM -----
00003 | @REM Licensed to the Apache Software Foundation (ASF) under one
00004 | @REM or more contributor license agreements. See the NOTICE file
00005 | @REM distributed with this work for additional information
00006 | @REM regarding copyright ownership. The ASF licenses this file
00007 | @REM to you under the Apache License, Version 2.0 (the
00008 | @REM "License"); you may not use this file except in compliance
00009 | @REM with the License. You may obtain a copy of the License at
00010 | @REM
00011 | @REM      http://www.apache.org/licenses/LICENSE-2.0
00012 | @REM
00013 | @REM Unless required by applicable law or agreed to in writing,
00014 | @REM software distributed under the License is distributed on an
00015 | @REM "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY
00016 | @REM KIND, either express or implied. See the License for the
00017 | @REM specific language governing permissions and limitations
00018 | @REM under the License.
00019 | @REM -----
00020 |
00021 | @REM -----
00022 | @REM Apache Maven Wrapper startup batch script, version 3.3.4
00023 | @REM
00024 | @REM Optional ENV vars
00025 | @REM   MVNW_REPOURL - repo url base for downloading maven distribution
00026 | @REM   MVNW_USERNAME/MVNW_PASSWORD - user and password for downloading maven
00027 | @REM   MVNW_VERBOSE - true: enable verbose log; others: silence the output
00028 | @REM -----
00029 |
00030 | @IF "%__MVNW_ARG0_NAME__"==" (SET __MVNW_ARG0_NAME__=%~nx0)
00031 | @SET __MVNW_CMD__=
00032 | @SET __MVNW_ERROR__=
00033 | @SET __MVNW_PSMODULEP_SAVE=%PSModulePath%
00034 | @SET PSModulePath=
00035 | @FOR /F "usebackq tokens=1* delims==" %%A IN (`powershell -noprofile "& {$scriptDir='%~dp0'; $script='%__MVNW_ARG0_NAME__'; icm -ScriptBlock ([Scriptblock]::Create((Get-
00036 | Content -Raw '%~f0')) -NoNewScope}"`) DO @(
00037 |   IF "%%A"=="MVN_CMD" (set __MVNW_CMD__=%%B) ELSE IF "%%B"==" (echo %%A) ELSE (echo %%A=%%B)
00038 | )
00039 | @SET PSModulePath=%__MVNW_PSMODULEP_SAVE%
00040 | @SET __MVNW_PSMODULEP_SAVE=
00041 | @SET __MVNW_ARG0_NAME__=
00042 | @SET MVNW_USERNAME=
00043 | @SET MVNW_PASSWORD=
00044 | @IF NOT "%__MVNW_CMD__"==" (%__MVNW_CMD__% *)
00045 | @echo Cannot start maven from wrapper >&2 && exit /b 1
00046 | @GOTO :EOF
00047 | : end batch / begin powershell #>
00048 | $ErrorActionPreference = "Stop"
00049 | if ($env:MVNW_VERBOSE -eq "true") {

```

006. mvnw.cmd

```
00050 | $VerbosePreference = "Continue"
00051 | }
00052 |
00053 | # calculate distributionUrl, requires .mvn/wrapper/maven-wrapper.properties
00054 | $distributionUrl = (Get-Content -Raw "$scriptDir/.mvn/wrapper/maven-wrapper.properties" | ConvertFrom-StringData).distributionUrl
00055 | if (!$distributionUrl) {
00056 |     Write-Error "cannot read distributionUrl property in $scriptDir/.mvn/wrapper/maven-wrapper.properties"
00057 | }
00058 |
00059 | switch -wildcard -casesensitive ( $($distributionUrl -replace '^.*/','') ) {
00060 |     "maven-mvnd-*" {
00061 |         $USE_MVND = $true
00062 |         $distributionUrl = $distributionUrl -replace '-bin\[^\.]*$', "-windows-amd64.zip"
00063 |         $MVN_CMD = "mvnd.cmd"
00064 |         break
00065 |     }
00066 |     default {
00067 |         $USE_MVND = $false
00068 |         $MVN_CMD = $script -replace '^mvnw','mvn'
00069 |         break
00070 |     }
00071 | }
00072 |
00073 | # apply MVNW_REPOURL and calculate MAVEN_HOME
00074 | # maven home pattern: ~/.m2/wrapper/dists/{apache-maven-<version>,maven-mvnd-<version>-<platform>}/<hash>
00075 | if ($env:MVNW_REPOURL) {
00076 |     $MVNW_REPO_PATTERN = if ($USE_MVND -eq $False) { "/org/apache/maven/" } else { "/maven/mvnd/" }
00077 |     $distributionUrl = "$env:MVNW_REPOURL$MVNW_REPO_PATTERN($distributionUrl -replace "^.*$MVNW_REPO_PATTERN",'')"
00078 | }
00079 | $distributionUrlName = $distributionUrl -replace '^.*/',''
00080 | $distributionUrlNameMain = $distributionUrlName -replace '\[^\.]*$', '' -replace '-bin$', ''
00081 |
00082 | $MAVEN_M2_PATH = "$HOME/.m2"
00083 | if ($env:MAVEN_USER_HOME) {
00084 |     $MAVEN_M2_PATH = "$env:MAVEN_USER_HOME"
00085 | }
00086 |
00087 | if (-not (Test-Path -Path $MAVEN_M2_PATH)) {
00088 |     New-Item -Path $MAVEN_M2_PATH -ItemType Directory | Out-Null
00089 | }
00090 |
00091 | $MAVEN_WRAPPER_DIST = $null
00092 | $MAVEN_M2_ITEM = Get-Item $MAVEN_M2_PATH
00093 | if (!$MAVEN_M2_ITEM.Target -or $MAVEN_M2_ITEM.Target.Count -eq 0 -or $MAVEN_M2_ITEM.Target[0] -eq $null) {
00094 |     $MAVEN_WRAPPER_DIST = "$MAVEN_M2_PATH/wrapper/dists"
00095 | } else {
00096 |     $MAVEN_WRAPPER_DIST = $MAVEN_M2_ITEM.Target[0] + "/wrapper/dists"
00097 | }
00098 |
00099 | $MAVEN_HOME_PARENT = "$MAVEN_WRAPPER_DIST/$distributionUrlNameMain"
00100 | $MAVEN_HOME_NAME = ([System.Security.Cryptography.SHA256]::Create().ComputeHash([byte[]][char[]]$distributionUrl) | ForEach-Object {$_.ToString("x2")}) -join ''
00101 | $MAVEN_HOME = "$MAVEN_HOME_PARENT/$MAVEN_HOME_NAME"
00102 |
00103 | if (Test-Path -Path "$MAVEN_HOME" -PathType Container) {
00104 |     Write-Verbose "found existing MAVEN_HOME at $MAVEN_HOME"
```

006. mvnw.cmd

```
00105 | Write-Output "MVN_CMD=$MAVEN_HOME/bin/$MVN_CMD"
00106 | exit $?
00107 | }
00108 |
00109 | if (! $distributionUrlNameMain -or ($distributionUrlName -eq $distributionUrlNameMain)) {
00110 |     Write-Error "distributionUrl is not valid, must end with *-bin.zip, but found $distributionUrl"
00111 | }
00112 |
00113 | # prepare tmp dir
00114 | $TMP_DOWNLOAD_DIR_HOLDER = New-TemporaryFile
00115 | $TMP_DOWNLOAD_DIR = New-Item -ItemType Directory -Path "$TMP_DOWNLOAD_DIR_HOLDER.dir"
00116 | $TMP_DOWNLOAD_DIR_HOLDER.Delete() | Out-Null
00117 | trap {
00118 |     if ($TMP_DOWNLOAD_DIR.Exists) {
00119 |         try { Remove-Item $TMP_DOWNLOAD_DIR -Recurse -Force | Out-Null }
00120 |         catch { Write-Warning "Cannot remove $TMP_DOWNLOAD_DIR" }
00121 |     }
00122 | }
00123 |
00124 | New-Item -ItemType Directory -Path "$MAVEN_HOME_PARENT" -Force | Out-Null
00125 |
00126 | # Download and Install Apache Maven
00127 | Write-Verbose "Couldn't find MAVEN_HOME, downloading and installing it ..."
00128 | Write-Verbose "Downloading from: $distributionUrl"
00129 | Write-Verbose "Downloading to: $TMP_DOWNLOAD_DIR/$distributionUrlName"
00130 |
00131 | $webclient = New-Object System.Net.WebClient
00132 | if ($env:MVNW_USERNAME -and $env:MVNW_PASSWORD) {
00133 |     $webclient.Credentials = New-Object System.Net.NetworkCredential($env:MVNW_USERNAME, $env:MVNW_PASSWORD)
00134 | }
00135 | [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
00136 | $webclient.DownloadFile($distributionUrl, "$TMP_DOWNLOAD_DIR/$distributionUrlName") | Out-Null
00137 |
00138 | # If specified, validate the SHA-256 sum of the Maven distribution zip file
00139 | $distributionSha256Sum = (Get-Content -Raw "$scriptDir/.mvn/wrapper/maven-wrapper.properties" | ConvertFrom-StringData).distributionSha256Sum
00140 | if ($distributionSha256Sum) {
00141 |     if ($USE_MVND) {
00142 |         Write-Error "Checksum validation is not supported for maven-mvnd. `nPlease disable validation by removing 'distributionSha256Sum' from your maven-wrapper.properties."
00143 |     }
00144 |     Import-Module $PSHOME\Modules\Microsoft.PowerShell.Utility -Function Get-FileHash
00145 |     if ((Get-FileHash "$TMP_DOWNLOAD_DIR/$distributionUrlName" -Algorithm SHA256).Hash.ToLower() -ne $distributionSha256Sum) {
00146 |         Write-Error "Error: Failed to validate Maven distribution SHA-256, your Maven distribution might be compromised. If you updated your Maven version, you need to update the
specified distributionSha256Sum property."
00147 |     }
00148 | }
00149 |
00150 | # unzip and move
00151 | Expand-Archive "$TMP_DOWNLOAD_DIR/$distributionUrlName" -DestinationPath "$TMP_DOWNLOAD_DIR" | Out-Null
00152 |
00153 | # Find the actual extracted directory name (handles snapshots where filename != directory name)
00154 | $actualDistributionDir = ""
00155 |
00156 | # First try the expected directory name (for regular distributions)
00157 | $expectedPath = Join-Path "$TMP_DOWNLOAD_DIR" "$distributionUrlNameMain"
00158 | $expectedMvnPath = Join-Path "$expectedPath" "bin/$MVN_CMD"
```

006. mvnw.cmd

```
00159 | if ((Test-Path -Path $expectedPath -PathType Container) -and (Test-Path -Path $expectedMvnPath -PathType Leaf)) {
00160 |     $actualDistributionDir = $distributionUrlNameMain
00161 | }
00162 |
00163 | # If not found, search for any directory with the Maven executable (for snapshots)
00164 | if (!$actualDistributionDir) {
00165 |     Get-ChildItem -Path "$TMP_DOWNLOAD_DIR" -Directory | ForEach-Object {
00166 |         $testPath = Join-Path $_.FullName "bin/$MVN_CMD"
00167 |         if (Test-Path -Path $testPath -PathType Leaf) {
00168 |             $actualDistributionDir = $_.Name
00169 |         }
00170 |     }
00171 | }
00172 |
00173 | if (!$actualDistributionDir) {
00174 |     Write-Error "Could not find Maven distribution directory in extracted archive"
00175 | }
00176 |
00177 | Write-Verbose "Found extracted Maven distribution directory: $actualDistributionDir"
00178 | Rename-Item -Path "$TMP_DOWNLOAD_DIR/$actualDistributionDir" -NewName $MAVEN_HOME_NAME | Out-Null
00179 | try {
00180 |     Move-Item -Path "$TMP_DOWNLOAD_DIR/$MAVEN_HOME_NAME" -Destination $MAVEN_HOME_PARENT | Out-Null
00181 | } catch {
00182 |     if (!(Test-Path -Path "$MAVEN_HOME" -PathType Container)) {
00183 |         Write-Error "fail to move MAVEN_HOME"
00184 |     }
00185 | } finally {
00186 |     try { Remove-Item $TMP_DOWNLOAD_DIR -Recurse -Force | Out-Null }
00187 |     catch { Write-Warning "Cannot remove $TMP_DOWNLOAD_DIR" }
00188 | }
00189 |
00190 | Write-Output "MVN_CMD=$MAVEN_HOME/bin/$MVN_CMD"
```

007. .mvn/wrapper/maven-wrapper.properties

.mvn/wrapper/maven-wrapper.properties

3 lines

```
00001 | wrapperVersion=3.3.4
00002 | distributionType=only-script
00003 | distributionUrl=https://repo.maven.apache.org/maven2/org/apache/maven/apache-maven/3.9.16/apache-maven-3.9.16-bin.zip
```

008. src/main/resources/application.yml

src/main/resources/application.yml

23 lines

```
00001 | spring:
00002 |   datasource:
00003 |     url: jdbc:postgresql://localhost:5432/emsdb
00004 |     username: postgres
00005 |     password: "0000"
00006 |   jpa:
00007 |     open-in-view: false
00008 |     hibernate:
00009 |       ddl-auto: validate
00010 |     show-sql: true
00011 |   modulith:
00012 |     events:
00013 |       jdbc:
00014 |         schema-initialization:
00015 |           enabled: true
00016 |   server:
00017 |     port: 8080
00018 |
00019 |   ems:
00020 |     security:
00021 |       token-secret: ${EMS_SECURITY_TOKEN_SECRET:change-this-local-development-secret-before-production}
00022 |       access-token-ttl: 15m
00023 |       refresh-token-ttl: 7d
```

009. src/main/java/com/citec/ems/EmsApplication.java

src/main/java/com/citec/ems/EmsApplication.java

20 lines

```
00001 | package com.citec.ems;
00002 |
00003 | import org.springframework.boot.SpringApplication;
00004 | import org.springframework.boot.autoconfigure.SpringBootApplication;
00005 | import org.springframework.boot.context.properties.ConfigurationPropertiesScan;
00006 | import org.springframework.modulith.Modulithic;
00007 |
00008 | @SpringBootApplication
00009 | @ConfigurationPropertiesScan
00010 | @Modulithic(systemName = "EMS", sharedModules = "shared")
00011 | public class EmsApplication {
00012 |
00013 |     public static void main(String[] args) {
00014 |         SpringApplication.run(EmsApplication.class, args);
00015 |     }
00016 |
00017 | }
00018 |
00019 |
00020 |
```

010. src/main/java/com/citec/ems/hr/application/EmployeeService.java

src/main/java/com/citec/ems/hr/application/EmployeeService.java

384 lines

```
00001 | package com.citec.ems.hr.application;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import com.citec.ems.hr.infrastructure.*;
00006 | import com.citec.ems.shared.BadRequestException;
00007 | import com.citec.ems.shared.NotFoundException;
00008 | import com.citec.ems.hr.web.HrDtos.EmployeeCreateRequest;
00009 | import com.citec.ems.hr.web.HrDtos.EmployeeDetail;
00010 | import com.citec.ems.hr.web.HrDtos.EmployeeProfileRequest;
00011 | import com.citec.ems.hr.web.HrDtos.EmployeeProfileResponse;
00012 | import com.citec.ems.hr.web.HrDtos.EmployeeSummary;
00013 | import com.citec.ems.hr.web.HrDtos.EmployeeUpdateRequest;
00014 | import com.citec.ems.hr.web.HrDtos.EmploymentContractRequest;
00015 | import com.citec.ems.hr.web.HrDtos.EmploymentContractResponse;
00016 | import com.citec.ems.hr.web.HrDtos.EmploymentRecordRequest;
00017 | import com.citec.ems.hr.web.HrDtos.EmploymentRecordResponse;
00018 | import com.citec.ems.hr.web.HrDtos.PositionAssignmentRequest;
00019 | import com.citec.ems.hr.web.HrDtos.PositionAssignmentResponse;
00020 | import com.citec.ems.iam.UserAccountSummary;
00021 | import com.citec.ems.iam.UserAccounts;
00022 | import java.time.LocalDate;
00023 | import java.time.LocalDateTime;
00024 | import org.springframework.data.domain.Page;
00025 | import org.springframework.data.domain.Pageable;
00026 | import org.springframework.stereotype.Service;
00027 | import org.springframework.transaction.annotation.Transactional;
00028 |
00029 | @Service
00030 | public class EmployeeService {
00031 |
00032 |     private final EmployeeRepository employeeRepository;
00033 |     private final EmployeeProfileRepository employeeProfileRepository;
00034 |     private final EmployeeStatusRepository employeeStatusRepository;
00035 |     private final OrganizationUnitRepository organizationUnitRepository;
00036 |     private final JobPositionRepository jobPositionRepository;
00037 |     private final ContractTypeRepository contractTypeRepository;
00038 |     private final EmploymentContractRepository employmentContractRepository;
00039 |     private final PositionAssignmentRepository positionAssignmentRepository;
00040 |     private final EmploymentRecordRepository employmentRecordRepository;
00041 |     private final UserAccounts userAccounts;
00042 |
00043 |     public EmployeeService(
00044 |         EmployeeRepository employeeRepository,
00045 |         EmployeeProfileRepository employeeProfileRepository,
00046 |         EmployeeStatusRepository employeeStatusRepository,
00047 |         OrganizationUnitRepository organizationUnitRepository,
00048 |         JobPositionRepository jobPositionRepository,
00049 |         ContractTypeRepository contractTypeRepository,
00050 |         EmploymentContractRepository employmentContractRepository,
```


010. src/main/java/com/citec/ems/hr/application/EmployeeService.java

```
00051 |         PositionAssignmentRepository positionAssignmentRepository,
00052 |         EmploymentRecordRepository employmentRecordRepository,
00053 |         UserAccounts userAccounts) {
00054 |     this.employeeRepository = employeeRepository;
00055 |     this.employeeProfileRepository = employeeProfileRepository;
00056 |     this.employeeStatusRepository = employeeStatusRepository;
00057 |     this.organizationUnitRepository = organizationUnitRepository;
00058 |     this.jobPositionRepository = jobPositionRepository;
00059 |     this.contractTypeRepository = contractTypeRepository;
00060 |     this.employmentContractRepository = employmentContractRepository;
00061 |     this.positionAssignmentRepository = positionAssignmentRepository;
00062 |     this.employmentRecordRepository = employmentRecordRepository;
00063 |     this.userAccounts = userAccounts;
00064 | }
00065 |
00066 | @Transactional(readOnly = true)
00067 | public Page<EmployeeSummary> listEmployees(String q, Long orgUnitId, String statusCode, Pageable pageable) {
00068 |     return employeeRepository.search(blankToNull(q), orgUnitId, blankToNull(statusCode), pageable)
00069 |         .map(this::employeeSummary);
00070 | }
00071 |
00072 | @Transactional(readOnly = true)
00073 | public EmployeeDetail getEmployee(Long employeeId) {
00074 |     Employee employee = getEmployeeEntity(employeeId);
00075 |     return new EmployeeDetail(
00076 |         employeeSummary(employee),
00077 |         employeeProfileRepository.findByEmployeeEmployeeIdOrderByPrimaryProfileDescCreatedAtDesc(employeeId)
00078 |             .stream().map(this::profileResponse).toList(),
00079 |         employmentContractRepository.findByEmployeeEmployeeIdOrderByStartDateDesc(employeeId)
00080 |             .stream().map(this::contractResponse).toList(),
00081 |         positionAssignmentRepository.findByEmployeeEmployeeIdOrderByStartDateDesc(employeeId)
00082 |             .stream().map(this::assignmentResponse).toList(),
00083 |         employmentRecordRepository.findByEmployeeEmployeeIdOrderByStartDateDesc(employeeId)
00084 |             .stream().map(this::recordResponse).toList(),
00085 |         employee.getCreatedAt(),
00086 |         employee.getUpdatedAt());
00087 | }
00088 |
00089 | @Transactional
00090 | public EmployeeSummary createEmployee(EmployeeCreateRequest request) {
00091 |     employeeRepository.findByEmployeeNumberIgnoreCase(request.employeeNumber()).ifPresent(employee -> {
00092 |         throw new BadRequestException("Employee number already exists.");
00093 |     });
00094 |     Employee employee = new Employee();
00095 |     employee.setEmployeeNumber(request.employeeNumber().trim());
00096 |     employee.setUserId(request.userId() == null ? null : requireUserId(request.userId()));
00097 |     employee.setCurrentOrgUnit(request.currentOrgUnitId() == null ? null : getOrgUnit(request.currentOrgUnitId()));
00098 |     employee.setHireDate(request.hireDate());
00099 |     employee.setStartDate(request.startDate());
00100 |     employee.setStatus(resolveStatus(request.employeeStatusId()));
00101 |     Employee saved = employeeRepository.save(employee);
00102 |
00103 |     if (request.profile() != null) {
00104 |         EmployeeProfile profile = buildProfile(saved, request.profile());
00105 |         profile.setPrimaryProfile(request.profile().primary() == null || request.profile().primary());
```

010. src/main/java/com/citec/ems/hr/application/EmployeeService.java

```
00106 |         employeeProfileRepository.save(profile);
00107 |     }
00108 |     return employeeSummary(saved);
00109 | }
00110 |
00111 | @Transactional
00112 | public EmployeeSummary updateEmployee(Long employeeId, EmployeeUpdateRequest request) {
00113 |     Employee employee = getEmployeeEntity(employeeId);
00114 |     if (request.employeeNumber() != null && !request.employeeNumber().equalsIgnoreCase(employee.getEmployeeNumber())) {
00115 |         employeeRepository.findByEmployeeNumberIgnoreCase(request.employeeNumber()).ifPresent(existing -> {
00116 |             throw new BadRequestException("Employee number already exists.");
00117 |         });
00118 |         employee.setEmployeeNumber(request.employeeNumber().trim());
00119 |     }
00120 |     if (request.userId() != null) {
00121 |         employee.setUserId(requireUserId(request.userId()));
00122 |     }
00123 |     if (request.currentOrgUnitId() != null) {
00124 |         employee.setCurrentOrgUnit(getOrgUnit(request.currentOrgUnitId()));
00125 |     }
00126 |     if (request.hireDate() != null) {
00127 |         employee.setHireDate(request.hireDate());
00128 |     }
00129 |     if (request.startDate() != null) {
00130 |         employee.setStartDate(request.startDate());
00131 |     }
00132 |     if (request.terminationDate() != null) {
00133 |         employee.setTerminationDate(request.terminationDate());
00134 |     }
00135 |     if (request.employeeStatusId() != null) {
00136 |         employee.setStatus(resolveStatus(request.employeeStatusId()));
00137 |     }
00138 |     return employeeSummary(employee);
00139 | }
00140 |
00141 | @Transactional
00142 | public EmployeeProfileResponse addProfile(Long employeeId, EmployeeProfileRequest request) {
00143 |     Employee employee = getEmployeeEntity(employeeId);
00144 |     boolean primary = request.primary() == null || request.primary();
00145 |     if (primary) {
00146 |         employeeProfileRepository.findFirstByEmployeeEmployeeIdAndPrimaryProfileTrue(employeeId)
00147 |             .ifPresent(existing -> existing.setPrimaryProfile(false));
00148 |     }
00149 |     EmployeeProfile profile = buildProfile(employee, request);
00150 |     profile.setPrimaryProfile(primary);
00151 |     return profileResponse(employeeProfileRepository.save(profile));
00152 | }
00153 |
00154 | @Transactional
00155 | public EmploymentContractResponse addContract(Long employeeId, EmploymentContractRequest request) {
00156 |     Employee employee = getEmployeeEntity(employeeId);
00157 |     EmploymentContract contract = new EmploymentContract();
00158 |     contract.setEmployee(employee);
00159 |     contract.setContractType(request.contractTypeId() == null ? null : getContractType(request.contractTypeId()));
00160 |     contract.setContractNumber(request.contractNumber());
```

010. src/main/java/com/citec/ems/hr/application/EmployeeService.java

```
00161 |         contract.setStartDate(request.startDate());
00162 |         contract.setEndDate(request.endDate());
00163 |         contract.setSalary(request.salary());
00164 |         contract.setSalaryCurrency(request.salaryCurrency() == null || request.salaryCurrency().isBlank()
00165 |             ? "EGP"
00166 |             : request.salaryCurrency().trim().toUpperCase());
00167 |         contract.setWorkingHoursPerWeek(request.workingHoursPerWeek());
00168 |         contract.setWorkingHoursPerMonth(request.workingHoursPerMonth());
00169 |         contract.setProbationPeriodDays(request.probationPeriodDays());
00170 |         contract.setFulltime(request.fulltime());
00171 |         contract.setActive(request.active() == null || request.active());
00172 |         return contractResponse(employmentContractRepository.save(contract));
00173 |     }
00174 |
00175 |     @Transactional
00176 |     public PositionAssignmentResponse addPositionAssignment(Long employeeId, PositionAssignmentRequest request) {
00177 |         Employee employee = getEmployeeEntity(employeeId);
00178 |         JobPosition position = getPosition(request.positionId());
00179 |         PositionAssignment assignment = new PositionAssignment();
00180 |         assignment.setEmployee(employee);
00181 |         assignment.setPosition(position);
00182 |         assignment.setAssignmentType(request.assignmentType() == null ? (short) 1 : request.assignmentType());
00183 |         assignment.setPrimaryAssignment(request.primary() == null || request.primary());
00184 |         assignment.setStartDate(request.startDate() == null ? LocalDate.now() : request.startDate());
00185 |         assignment.setEndDate(request.endDate());
00186 |         assignment.setActive(request.active() == null || request.active());
00187 |         assignment.setAssignedByUserId(request.assignedByUserId() == null ? null : requireUserId(request.assignedByUserId()));
00188 |         assignment.setAssignedAt(LocalDateTime.now());
00189 |         if (Boolean.TRUE.equals(assignment.getActive()) && Boolean.TRUE.equals(assignment.getPrimaryAssignment())) {
00190 |             employee.setCurrentOrgUnit(position.getOrgUnit());
00191 |         }
00192 |         return assignmentResponse(positionAssignmentRepository.save(assignment));
00193 |     }
00194 |
00195 |     @Transactional
00196 |     public EmploymentRecordResponse addEmploymentRecord(Long employeeId, EmploymentRecordRequest request) {
00197 |         Employee employee = getEmployeeEntity(employeeId);
00198 |         boolean current = request.current() == null || request.current();
00199 |         if (current) {
00200 |             employmentRecordRepository.findFirstByEmployeeEmployeeIdAndCurrentTrue(employeeId)
00201 |                 .ifPresent(existing -> {
00202 |                     existing.setCurrent(false);
00203 |                     if (existing.getEndDate() == null) {
00204 |                         existing.setEndDate(request.startDate().minusDays(1));
00205 |                     }
00206 |                 });
00207 |         }
00208 |         EmploymentRecord record = new EmploymentRecord();
00209 |         record.setEmployee(employee);
00210 |         record.setOrgUnit(request.orgUnitId() == null ? null : getOrgUnit(request.orgUnitId()));
00211 |         record.setPosition(request.positionId() == null ? null : getPosition(request.positionId()));
00212 |         record.setRecordType(request.recordType());
00213 |         record.setStartDate(request.startDate());
00214 |         record.setEndDate(request.endDate());
00215 |         record.setContract(request.contractId() == null ? null : getContract(request.contractId()));
```

010. src/main/java/com/citec/ems/hr/application/EmployeeService.java

```
00216 |         record.setReportingTo(request.reportingToEmployeeId() == null ? null : getEmployeeEntity(request.reportingToEmployeeId()));
00217 |         record.setCurrent(current);
00218 |         if (current && record.getOrgUnit() != null) {
00219 |             employee.setCurrentOrgUnit(record.getOrgUnit());
00220 |         }
00221 |         return recordResponse(employmentRecordRepository.save(record));
00222 |     }
00223 |
00224 |     private EmployeeProfile buildProfile(Employee employee, EmployeeProfileRequest request) {
00225 |         EmployeeProfile profile = new EmployeeProfile();
00226 |         profile.setEmployee(employee);
00227 |         profile.setFirstName(request.firstName().trim());
00228 |         profile.setOtherName(request.otherName().trim());
00229 |         profile.setDisplayName(request.displayName());
00230 |         profile.setGender(request.gender());
00231 |         profile.setBirthDate(request.birthDate());
00232 |         profile.setNationalId(request.nationalId());
00233 |         profile.setPersonalEmail(request.personalEmail());
00234 |         profile.setBusinessEmail(request.businessEmail());
00235 |         profile.setPhoneNumber(request.phoneNumber());
00236 |         profile.setMobileNumber(request.mobileNumber());
00237 |         profile.setCountryId(request.countryId());
00238 |         profile.setStateId(request.stateId());
00239 |         profile.setCityId(request.cityId());
00240 |         profile.setAddressLine1(request.addressLine1());
00241 |         profile.setAddressLine2(request.addressLine2());
00242 |         profile.setPostalCode(request.postalCode());
00243 |         return profile;
00244 |     }
00245 |
00246 |     private EmployeeStatus resolveStatus(Short statusId) {
00247 |         if (statusId != null) {
00248 |             return employeeStatusRepository.findById(statusId)
00249 |                 .orElseThrow(() -> new NotFoundException("Employee status was not found."));
00250 |         }
00251 |         return employeeStatusRepository.findByStatusCodeIgnoreCase(EmployeeStatus.ACTIVE)
00252 |             .orElseThrow(() -> new NotFoundException("ACTIVE employee status was not found."));
00253 |     }
00254 |
00255 |     private Employee getEmployeeEntity(Long employeeId) {
00256 |         return employeeRepository.findById(employeeId)
00257 |             .orElseThrow(() -> new NotFoundException("Employee was not found."));
00258 |     }
00259 |
00260 |     private Long requireUserId(Long userId) {
00261 |         if (!userAccounts.existsById(userId)) {
00262 |             throw new NotFoundException("User was not found.");
00263 |         }
00264 |         return userId;
00265 |     }
00266 |
00267 |     private OrganizationUnit getOrgUnit(Long orgUnitId) {
00268 |         return organizationUnitRepository.findById(orgUnitId)
00269 |             .orElseThrow(() -> new NotFoundException("Organization unit was not found."));
00270 |     }
```

010. src/main/java/com/citec/ems/hr/application/EmployeeService.java

```
00271 |
00272 | private JobPosition getPosition(Long positionId) {
00273 |     return jobPositionRepository.findById(positionId)
00274 |         .orElseThrow(() -> new NotFoundException("Position was not found."));
00275 | }
00276 |
00277 | private ContractType getContractType(Integer contractTypeId) {
00278 |     return contractTypeRepository.findById(contractTypeId)
00279 |         .orElseThrow(() -> new NotFoundException("Contract type was not found."));
00280 | }
00281 |
00282 | private EmploymentContract getContract(Long contractId) {
00283 |     return employmentContractRepository.findById(contractId)
00284 |         .orElseThrow(() -> new NotFoundException("Employment contract was not found."));
00285 | }
00286 |
00287 | private EmployeeSummary employeeSummary(Employee employee) {
00288 |     EmployeeProfile profile = employeeProfileRepository.findFirstByEmployeeEmployeeIdAndPrimaryProfileTrue(employee.getEmployeeId())
00289 |         .orElse(null);
00290 |     UserAccountSummary user = employee.getUserId() == null
00291 |         ? null
00292 |         : userAccounts.findSummary(employee.getUserId()).orElse(null);
00293 |     return new EmployeeSummary(
00294 |         employee.getEmployeeId(),
00295 |         employee.getEmployeeNumber(),
00296 |         user == null ? null : user.userId(),
00297 |         user == null ? null : user.username(),
00298 |         employee.getCurrentOrgUnit() == null ? null : employee.getCurrentOrgUnit().getOrgUnitId(),
00299 |         employee.getCurrentOrgUnit() == null ? null : employee.getCurrentOrgUnit().getUnitName(),
00300 |         employee.getStatus().getStatusCode(),
00301 |         profile == null ? null : profile.getDisplayName(),
00302 |         profile == null ? null : profile.getBusinessEmail(),
00303 |         employee.getHireDate(),
00304 |         employee.getStartDate(),
00305 |         employee.getTerminationDate());
00306 | }
00307 |
00308 | private EmployeeProfileResponse profileResponse(EmployeeProfile profile) {
00309 |     return new EmployeeProfileResponse(
00310 |         profile.getProfileId(),
00311 |         profile.getPrimaryProfile(),
00312 |         profile.getFirstName(),
00313 |         profile.getOtherName(),
00314 |         profile.getDisplayName(),
00315 |         profile.getGender(),
00316 |         profile.getBirthDate(),
00317 |         profile.getNationalId(),
00318 |         profile.getPersonalEmail(),
00319 |         profile.getBusinessEmail(),
00320 |         profile.getPhoneNumber(),
00321 |         profile.getMobileNumber(),
00322 |         profile.getCountryId(),
00323 |         profile.getStateId(),
00324 |         profile.getCityId(),
00325 |         profile.getAddressLine1(),
```

010. src/main/java/com/citec/ems/hr/application/EmployeeService.java

```
00326 |         profile.getAddressLine2(),
00327 |         profile.getPostalCode());
00328 |     }
00329 |
00330 |     private EmploymentContractResponse contractResponse(EmploymentContract contract) {
00331 |         return new EmploymentContractResponse(
00332 |             contract.getContractId(),
00333 |             contract.getContractType() == null ? null : contract.getContractType().getContractTypeId(),
00334 |             contract.getContractType() == null ? null : contract.getContractType().getContractTypeCode(),
00335 |             contract.getContractNumber(),
00336 |             contract.getStartDate(),
00337 |             contract.getEndDate(),
00338 |             contract.getSalary(),
00339 |             contract.getSalaryCurrency(),
00340 |             contract.getWorkingHoursPerWeek(),
00341 |             contract.getWorkingHoursPerMonth(),
00342 |             contract.getProbationPeriodDays(),
00343 |             contract.getFulltime(),
00344 |             contract.getActive());
00345 |     }
00346 |
00347 |     private PositionAssignmentResponse assignmentResponse(PositionAssignment assignment) {
00348 |         return new PositionAssignmentResponse(
00349 |             assignment.getPositionAssignmentId(),
00350 |             assignment.getPosition().getPositionId(),
00351 |             assignment.getPosition().getPositionCode(),
00352 |             assignment.getPosition().getPositionTitle(),
00353 |             assignment.getAssignmentType(),
00354 |             assignment.getPrimaryAssignment(),
00355 |             assignment.getStartDate(),
00356 |             assignment.getEndDate(),
00357 |             assignment.getActive(),
00358 |             assignment.getAssignedByUserId(),
00359 |             assignment.getAssignedAt());
00360 |     }
00361 |
00362 |     private EmploymentRecordResponse recordResponse(EmploymentRecord record) {
00363 |         return new EmploymentRecordResponse(
00364 |             record.getEmploymentId(),
00365 |             record.getOrgUnit() == null ? null : record.getOrgUnit().getOrgUnitId(),
00366 |             record.getOrgUnit() == null ? null : record.getOrgUnit().getUnitName(),
00367 |             record.getPosition() == null ? null : record.getPosition().getPositionId(),
00368 |             record.getPosition() == null ? null : record.getPosition().getPositionTitle(),
00369 |             record.getRecordType(),
00370 |             record.getStartDate(),
00371 |             record.getEndDate(),
00372 |             record.getContract() == null ? null : record.getContract().getContractId(),
00373 |             record.getReportingTo() == null ? null : record.getReportingTo().getEmployeeId(),
00374 |             record.getCurrent());
00375 |     }
00376 |
00377 |     private String blankToNull(String value) {
00378 |         return value == null || value.isBlank() ? null : value.trim();
00379 |     }
00380 | }
```

010. src/main/java/com/citec/ems/hr/application/EmployeeService.java

```
00381 |  
00382 |  
00383 |  
00384 |
```

011. src/main/java/com/citec/ems/hr/application/HrReferenceService.java

src/main/java/com/citec/ems/hr/application/HrReferenceService.java

276 lines

```
00001 | package com.citec.ems.hr.application;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import com.citec.ems.hr.infrastructure.*;
00006 | import com.citec.ems.shared.BadRequestException;
00007 | import com.citec.ems.shared.NotFoundException;
00008 | import com.citec.ems.hr.web.HrDtos.ContractTypeCreateRequest;
00009 | import com.citec.ems.hr.web.HrDtos.ContractTypeResponse;
00010 | import com.citec.ems.hr.web.HrDtos.EmployeeStatusResponse;
00011 | import com.citec.ems.hr.web.HrDtos.JobGradeCreateRequest;
00012 | import com.citec.ems.hr.web.HrDtos.JobGradeResponse;
00013 | import com.citec.ems.hr.web.HrDtos.JobPositionCreateRequest;
00014 | import com.citec.ems.hr.web.HrDtos.JobPositionResponse;
00015 | import com.citec.ems.hr.web.HrDtos.OrganizationCreateRequest;
00016 | import com.citec.ems.hr.web.HrDtos.OrganizationResponse;
00017 | import com.citec.ems.hr.web.HrDtos.OrganizationUnitCreateRequest;
00018 | import com.citec.ems.hr.web.HrDtos.OrganizationUnitResponse;
00019 | import com.citec.ems.hr.web.HrDtos.UnitTypeResponse;
00020 | import java.time.LocalDate;
00021 | import org.springframework.data.domain.Page;
00022 | import org.springframework.data.domain.Pageable;
00023 | import org.springframework.stereotype.Service;
00024 | import org.springframework.transaction.annotation.Transactional;
00025 |
00026 | @Service
00027 | public class HrReferenceService {
00028 |
00029 |     private final OrganizationRepository organizationRepository;
00030 |     private final OrganizationUnitRepository organizationUnitRepository;
00031 |     private final UnitTypeRepository unitTypeRepository;
00032 |     private final JobGradeRepository jobGradeRepository;
00033 |     private final JobPositionRepository jobPositionRepository;
00034 |     private final ContractTypeRepository contractTypeRepository;
00035 |     private final EmployeeStatusRepository employeeStatusRepository;
00036 |
00037 |     public HrReferenceService(
00038 |         OrganizationRepository organizationRepository,
00039 |         OrganizationUnitRepository organizationUnitRepository,
00040 |         UnitTypeRepository unitTypeRepository,
00041 |         JobGradeRepository jobGradeRepository,
00042 |         JobPositionRepository jobPositionRepository,
00043 |         ContractTypeRepository contractTypeRepository,
00044 |         EmployeeStatusRepository employeeStatusRepository) {
00045 |         this.organizationRepository = organizationRepository;
00046 |         this.organizationUnitRepository = organizationUnitRepository;
00047 |         this.unitTypeRepository = unitTypeRepository;
00048 |         this.jobGradeRepository = jobGradeRepository;
00049 |         this.jobPositionRepository = jobPositionRepository;
00050 |         this.contractTypeRepository = contractTypeRepository;
```


011. src/main/java/com/citec/ems/hr/application/HrReferenceService.java

```
00051 |         this.employeeStatusRepository = employeeStatusRepository;
00052 |     }
00053 |
00054 |     @Transactional(readOnly = true)
00055 |     public Page<OrganizationResponse> listOrganizations(Pageable pageable) {
00056 |         return organizationRepository.findAll(pageable).map(this::organizationResponse);
00057 |     }
00058 |
00059 |     @Transactional
00060 |     public OrganizationResponse createOrganization(OrganizationCreateRequest request) {
00061 |         organizationRepository.findByOrganizationCodeIgnoreCase(request.organizationCode()).ifPresent(organization -> {
00062 |             throw new BadRequestException("Organization code already exists.");
00063 |         });
00064 |         Organization organization = new Organization();
00065 |         organization.setOrganizationCode(request.organizationCode().trim().toUpperCase());
00066 |         organization.setOrganizationNameEn(request.organizationNameEn().trim());
00067 |         organization.setOrganizationNameAr(request.organizationNameAr().trim());
00068 |         organization.setActive(request.active() == null || request.active());
00069 |         return organizationResponse(organizationRepository.save(organization));
00070 |     }
00071 |
00072 |     @Transactional(readOnly = true)
00073 |     public Page<OrganizationUnitResponse> listOrganizationUnits(
00074 |         Integer organizationId,
00075 |         Long parentOrgUnitId,
00076 |         String q,
00077 |         Pageable pageable) {
00078 |         return organizationUnitRepository.search(organizationId, parentOrgUnitId, blankToNull(q), pageable)
00079 |             .map(this::organizationUnitResponse);
00080 |     }
00081 |
00082 |     @Transactional
00083 |     public OrganizationUnitResponse createOrganizationUnit(OrganizationUnitCreateRequest request) {
00084 |         Organization organization = organizationRepository.findById(request.organizationId())
00085 |             .orElseThrow(() -> new NotFoundException("Organization was not found."));
00086 |         organizationUnitRepository.findByOrganizationIdAndUnitCodeIgnoreCase(
00087 |             request.organizationId(),
00088 |             request.unitCode()).ifPresent(unit -> {
00089 |             throw new BadRequestException("Organization unit code already exists for this organization.");
00090 |         });
00091 |         OrganizationUnit unit = new OrganizationUnit();
00092 |         unit.setOrganization(organization);
00093 |         unit.setUnitType(unitTypeRepository.findById(request.unitTypeId())
00094 |             .orElseThrow(() -> new NotFoundException("Unit type was not found.")));
00095 |         unit.setParentOrgUnit(request.parentOrgUnitId() == null ? null : getOrganizationUnit(request.parentOrgUnitId()));
00096 |         unit.setUnitCode(request.unitCode().trim().toUpperCase());
00097 |         unit.setUnitName(request.unitName().trim());
00098 |         unit.setUnitNameAr(request.unitNameAr().trim());
00099 |         unit.setDescription(request.description());
00100 |         unit.setDescriptionAr(request.descriptionAr());
00101 |         unit.setStartDate(request.startDate() == null ? LocalDate.now() : request.startDate());
00102 |         unit.setEndDate(request.endDate());
00103 |         unit.setActive(request.active() == null || request.active());
00104 |         return organizationUnitResponse(organizationUnitRepository.save(unit));
00105 |     }
```

011. src/main/java/com/citec/ems/hr/application/HrReferenceService.java

```
00106 |
00107 |     @Transactional(readOnly = true)
00108 |     public Page<UnitTypeResponse> listUnitTypes(Pageable pageable) {
00109 |         return unitTypeRepository.findAll(pageable).map(this::unitTypeResponse);
00110 |     }
00111 |
00112 |     @Transactional(readOnly = true)
00113 |     public Page<EmployeeStatusResponse> listEmployeeStatuses(Pageable pageable) {
00114 |         return employeeStatusRepository.findAll(pageable).map(this::employeeStatusResponse);
00115 |     }
00116 |
00117 |     @Transactional(readOnly = true)
00118 |     public Page<JobGradeResponse> listJobGrades(Pageable pageable) {
00119 |         return jobGradeRepository.findAll(pageable).map(this::jobGradeResponse);
00120 |     }
00121 |
00122 |     @Transactional
00123 |     public JobGradeResponse createJobGrade(JobGradeCreateRequest request) {
00124 |         jobGradeRepository.findByGradeCodeIgnoreCase(request.gradeCode()).ifPresent(grade -> {
00125 |             throw new BadRequestException("Job grade code already exists.");
00126 |         });
00127 |         JobGrade grade = new JobGrade();
00128 |         grade.setGradeCode(request.gradeCode().trim().toUpperCase());
00129 |         grade.setGradeName(request.gradeName());
00130 |         grade.setGradeRank(request.gradeRank());
00131 |         grade.setActive(request.active() == null || request.active());
00132 |         return jobGradeResponse(jobGradeRepository.save(grade));
00133 |     }
00134 |
00135 |     @Transactional(readOnly = true)
00136 |     public Page<JobPositionResponse> listJobPositions(Long orgUnitId, String q, Pageable pageable) {
00137 |         return jobPositionRepository.search(orgUnitId, blankToNull(q), pageable).map(this::jobPositionResponse);
00138 |     }
00139 |
00140 |     @Transactional
00141 |     public JobPositionResponse createJobPosition(JobPositionCreateRequest request) {
00142 |         jobPositionRepository.findByPositionCodeIgnoreCase(request.positionCode()).ifPresent(position -> {
00143 |             throw new BadRequestException("Position code already exists.");
00144 |         });
00145 |         JobPosition position = new JobPosition();
00146 |         position.setPositionCode(request.positionCode().trim().toUpperCase());
00147 |         position.setPositionTitle(request.positionTitle().trim());
00148 |         position.setPositionTitleAr(request.positionTitleAr().trim());
00149 |         position.setOrgUnit(getOrganizationUnit(request.orgUnitId()));
00150 |         position.setGrade(request.gradeId() == null ? null : getJobGrade(request.gradeId()));
00151 |         position.setPositionLevel(request.positionLevel() == null ? 1 : request.positionLevel());
00152 |         position.setReportsToPosition(request.reportsToPositionId() == null ? null : getJobPosition(request.reportsToPositionId()));
00153 |         position.setPositionDescription(request.positionDescription());
00154 |         position.setPositionDescriptionAr(request.positionDescriptionAr());
00155 |         position.setOpen(request.open() == null || request.open());
00156 |         position.setActive(request.active() == null || request.active());
00157 |         return jobPositionResponse(jobPositionRepository.save(position));
00158 |     }
00159 |
00160 |     @Transactional(readOnly = true)
```

011. src/main/java/com/citec/ems/hr/application/HrReferenceService.java

```
00161 | public Page<ContractTypeResponse> listContractTypes(Pageable pageable) {
00162 |     return contractTypeRepository.findAll(pageable).map(this::contractTypeResponse);
00163 | }
00164 |
00165 | @Transactional
00166 | public ContractTypeResponse createContractType(ContractTypeCreateRequest request) {
00167 |     contractTypeRepository.findByContractTypeCodeIgnoreCase(request.contractTypeCode()).ifPresent(contractType -> {
00168 |         throw new BadRequestException("Contract type code already exists.");
00169 |     });
00170 |     ContractType contractType = new ContractType();
00171 |     contractType.setContractTypeCode(request.contractTypeCode().trim().toUpperCase());
00172 |     contractType.setContractTypeName(request.contractTypeName().trim());
00173 |     contractType.setDescription(request.description());
00174 |     contractType.setActive(request.active() == null || request.active());
00175 |     return contractTypeResponse(contractTypeRepository.save(contractType));
00176 | }
00177 |
00178 | private OrganizationUnit getOrganizationUnit(Long orgUnitId) {
00179 |     return organizationUnitRepository.findById(orgUnitId)
00180 |         .orElseThrow(() -> new NotFoundException("Organization unit was not found."));
00181 | }
00182 |
00183 | private JobGrade getJobGrade(Long gradeId) {
00184 |     return jobGradeRepository.findById(gradeId)
00185 |         .orElseThrow(() -> new NotFoundException("Job grade was not found."));
00186 | }
00187 |
00188 | private JobPosition getJobPosition(Long positionId) {
00189 |     return jobPositionRepository.findById(positionId)
00190 |         .orElseThrow(() -> new NotFoundException("Job position was not found."));
00191 | }
00192 |
00193 | private OrganizationResponse organizationResponse(Organization organization) {
00194 |     return new OrganizationResponse(
00195 |         organization.getOrganizationId(),
00196 |         organization.getOrganizationCode(),
00197 |         organization.getOrganizationNameEn(),
00198 |         organization.getOrganizationNameAr(),
00199 |         organization.getActive());
00200 | }
00201 |
00202 | private OrganizationUnitResponse organizationUnitResponse(OrganizationUnit unit) {
00203 |     return new OrganizationUnitResponse(
00204 |         unit.getOrgUnitId(),
00205 |         unit.getOrganization().getOrganizationId(),
00206 |         unit.getOrganization().getOrganizationCode(),
00207 |         unit.getUnitType().getUnitTypeId(),
00208 |         unit.getUnitType().getUnitCode(),
00209 |         unit.getParentOrgUnit() == null ? null : unit.getParentOrgUnit().getOrgUnitId(),
00210 |         unit.getUnitCode(),
00211 |         unit.getUnitName(),
00212 |         unit.getUnitNameAr(),
00213 |         unit.getStartDate(),
00214 |         unit.getEndDate(),
00215 |         unit.getActive());
```

011. src/main/java/com/citec/ems/hr/application/HrReferenceService.java

```
00216 |     }
00217 |
00218 |     private UnitTypeResponse unitTypeResponse(UnitType unitType) {
00219 |         return new UnitTypeResponse(
00220 |             unitType.getUnitTypeId(),
00221 |             unitType.getUnitCode(),
00222 |             unitType.getUnitNameEn(),
00223 |             unitType.getUnitNameAr(),
00224 |             unitType.getDescription(),
00225 |             unitType.getDescriptionAr());
00226 |     }
00227 |
00228 |     private EmployeeStatusResponse employeeStatusResponse(EmployeeStatus status) {
00229 |         return new EmployeeStatusResponse(
00230 |             status.getEmployeeStatusId(),
00231 |             status.getStatusCode(),
00232 |             status.getDescription());
00233 |     }
00234 |
00235 |     private JobGradeResponse jobGradeResponse(JobGrade grade) {
00236 |         return new JobGradeResponse(
00237 |             grade.getGradeId(),
00238 |             grade.getGradeCode(),
00239 |             grade.getGradeName(),
00240 |             grade.getGradeRank(),
00241 |             grade.getActive());
00242 |     }
00243 |
00244 |     private JobPositionResponse jobPositionResponse(JobPosition position) {
00245 |         return new JobPositionResponse(
00246 |             position.getPositionId(),
00247 |             position.getPositionCode(),
00248 |             position.getPositionTitle(),
00249 |             position.getPositionTitleAr(),
00250 |             position.getOrgUnit().getOrgUnitId(),
00251 |             position.getOrgUnit().getUnitName(),
00252 |             position.getGrade() == null ? null : position.getGrade().getGradeId(),
00253 |             position.getGrade() == null ? null : position.getGrade().getGradeCode(),
00254 |             position.getPositionLevel(),
00255 |             position.getReportsToPosition() == null ? null : position.getReportsToPosition().getPositionId(),
00256 |             position.getOpen(),
00257 |             position.getActive());
00258 |     }
00259 |
00260 |     private ContractTypeResponse contractTypeResponse(ContractType contractType) {
00261 |         return new ContractTypeResponse(
00262 |             contractType.getContractTypeId(),
00263 |             contractType.getContractTypeCode(),
00264 |             contractType.getContractTypeName(),
00265 |             contractType.getDescription(),
00266 |             contractType.getActive());
00267 |     }
00268 |
00269 |     private String blankToNull(String value) {
00270 |         return value == null || value.isBlank() ? null : value.trim();
```

011. src/main/java/com/citec/ems/hr/application/HrReferenceService.java

```
00271 |     }  
00272 | }  
00273 |  
00274 |  
00275 |  
00276 |
```

012. src/main/java/com/citec/ems/hr/domain/ContractType.java

src/main/java/com/citec/ems/hr/domain/ContractType.java

45 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.GeneratedValue;
00006 | import jakarta.persistence.GenerationType;
00007 | import jakarta.persistence.Id;
00008 | import jakarta.persistence.Table;
00009 | import java.time.LocalDateTime;
00010 | import lombok.Getter;
00011 | import lombok.NoArgsConstructor;
00012 | import lombok.Setter;
00013 | import org.hibernate.annotations.CreationTimestamp;
00014 |
00015 | @Entity
00016 | @Table(name = "contract_type", schema = "hr")
00017 | @Getter
00018 | @Setter
00019 | @NoArgsConstructor
00020 | public class ContractType {
00021 |
00022 |     @Id
00023 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00024 |     @Column(name = "contract_type_id")
00025 |     private Integer contractTypeId;
00026 |
00027 |     @Column(name = "contract_type_code", nullable = false, length = 100)
00028 |     private String contractTypeCode;
00029 |
00030 |     @Column(name = "contract_type_name", nullable = false)
00031 |     private String contractTypeName;
00032 |
00033 |     @Column(name = "description", columnDefinition = "text")
00034 |     private String description;
00035 |
00036 |     @CreationTimestamp
00037 |     @Column(name = "created_at", nullable = false, updatable = false)
00038 |     private LocalDateTime createdAt;
00039 |
00040 |     @Column(name = "is_active", nullable = false)
00041 |     private Boolean active = true;
00042 | }
00043 |
00044 |
00045 |
```

013. src/main/java/com/citec/ems/hr/domain/Employee.java

src/main/java/com/citec/ems/hr/domain/Employee.java

65 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDate;
00013 | import java.time.LocalDateTime;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 | import org.hibernate.annotations.UpdateTimestamp;
00019 |
00020 | @Entity
00021 | @Table(name = "employee", schema = "hr")
00022 | @Getter
00023 | @Setter
00024 | @NoArgsConstructor
00025 | public class Employee {
00026 |
00027 |     @Id
00028 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00029 |     @Column(name = "employee_id")
00030 |     private Long employeeId;
00031 |
00032 |     @Column(name = "user_id")
00033 |     private Long userId;
00034 |
00035 |     @Column(name = "employee_number", nullable = false, length = 50)
00036 |     private String employeeNumber;
00037 |
00038 |     @ManyToOne(fetch = FetchType.LAZY)
00039 |     @JoinColumn(name = "current_org_unit_id")
00040 |     private OrganizationUnit currentOrgUnit;
00041 |
00042 |     @Column(name = "hire_date")
00043 |     private LocalDate hireDate;
00044 |
00045 |     @Column(name = "start_date")
00046 |     private LocalDate startDate;
00047 |
00048 |     @Column(name = "termination_date")
00049 |     private LocalDate terminationDate;
00050 | }
```

013. src/main/java/com/citec/ems/hr/domain/Employee.java

```
00051 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00052 |     @JoinColumn(name = "employee_status_id", nullable = false)
00053 |     private EmployeeStatus status;
00054 |
00055 |     @CreationTimestamp
00056 |     @Column(name = "created_at", nullable = false, updatable = false)
00057 |     private LocalDateTime createdAt;
00058 |
00059 |     @UpdateTimestamp
00060 |     @Column(name = "updated_at")
00061 |     private LocalDateTime updatedAt;
00062 | }
00063 |
00064 |
00065 |
```


014. src/main/java/com/citec/ems/hr/domain/EmployeeProfile.java

src/main/java/com/citec/ems/hr/domain/EmployeeProfile.java

100 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDate;
00013 | import java.time.LocalDateTime;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 | import org.hibernate.annotations.UpdateTimestamp;
00019 |
00020 | @Entity
00021 | @Table(name = "employee_profile", schema = "hr")
00022 | @Getter
00023 | @Setter
00024 | @NoArgsConstructor
00025 | public class EmployeeProfile {
00026 |
00027 |     @Id
00028 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00029 |     @Column(name = "profile_id")
00030 |     private Long profileId;
00031 |
00032 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00033 |     @JoinColumn(name = "employee_id", nullable = false)
00034 |     private Employee employee;
00035 |
00036 |     @Column(name = "is_primary", nullable = false)
00037 |     private Boolean primaryProfile = true;
00038 |
00039 |     @Column(name = "first_name", nullable = false, length = 100)
00040 |     private String firstName;
00041 |
00042 |     @Column(name = "other_name", nullable = false, length = 200)
00043 |     private String otherName;
00044 |
00045 |     @Column(name = "display_name", length = 100)
00046 |     private String displayName;
00047 |
00048 |     @Column(name = "gender", nullable = false)
00049 |     private Short gender;
00050 | }
```

014. src/main/java/com/citec/ems/hr/domain/EmployeeProfile.java

```
00051 |     @Column(name = "birth_date")
00052 |     private LocalDate birthDate;
00053 |
00054 |     @Column(name = "national_id", length = 14)
00055 |     private String nationalId;
00056 |
00057 |     @Column(name = "personal_email", length = 100)
00058 |     private String personalEmail;
00059 |
00060 |     @Column(name = "business_email", length = 100)
00061 |     private String businessEmail;
00062 |
00063 |     @Column(name = "phone_number", length = 100)
00064 |     private String phoneNumber;
00065 |
00066 |     @Column(name = "mobile_number", length = 100)
00067 |     private String mobileNumber;
00068 |
00069 |     @Column(name = "country_id")
00070 |     private Short countryId;
00071 |
00072 |     @Column(name = "state_id")
00073 |     private Integer stateId;
00074 |
00075 |     @Column(name = "city_id")
00076 |     private Integer cityId;
00077 |
00078 |     @Column(name = "address_line1")
00079 |     private String addressLine1;
00080 |
00081 |     @Column(name = "address_line2")
00082 |     private String addressLine2;
00083 |
00084 |     @Column(name = "postal_code", length = 20)
00085 |     private String postalCode;
00086 |
00087 |     @Column(name = "profile_photo", columnDefinition = "bytea")
00088 |     private byte[] profilePhoto;
00089 |
00090 |     @CreationTimestamp
00091 |     @Column(name = "created_at", nullable = false, updatable = false)
00092 |     private LocalDateTime createdAt;
00093 |
00094 |     @UpdateTimestamp
00095 |     @Column(name = "updated_at")
00096 |     private LocalDateTime updatedAt;
00097 | }
00098 |
00099 |
00100 |
```

015. src/main/java/com/citec/ems/hr/domain/EmployeeStatus.java

src/main/java/com/citec/ems/hr/domain/EmployeeStatus.java

32 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.Id;
00006 | import jakarta.persistence.Table;
00007 | import lombok.Getter;
00008 | import lombok.NoArgsConstructor;
00009 | import lombok.Setter;
00010 |
00011 | @Entity
00012 | @Table(name = "employee_status", schema = "hr")
00013 | @Getter
00014 | @Setter
00015 | @NoArgsConstructor
00016 | public class EmployeeStatus {
00017 |
00018 |     public static final String ACTIVE = "ACTIVE";
00019 |
00020 |     @Id
00021 |     @Column(name = "employee_status_id")
00022 |     private Short employeeStatusId;
00023 |
00024 |     @Column(name = "status_code", nullable = false, length = 50)
00025 |     private String statusCode;
00026 |
00027 |     @Column(name = "description")
00028 |     private String description;
00029 | }
00030 |
00031 |
00032 |
```

016. src/main/java/com/citec/ems/hr/domain/EmploymentContract.java

src/main/java/com/citec/ems/hr/domain/EmploymentContract.java

81 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.math.BigDecimal;
00013 | import java.time.LocalDate;
00014 | import java.time.LocalDateTime;
00015 | import lombok.Getter;
00016 | import lombok.NoArgsConstructor;
00017 | import lombok.Setter;
00018 | import org.hibernate.annotations.CreationTimestamp;
00019 | import org.hibernate.annotations.UpdateTimestamp;
00020 |
00021 | @Entity
00022 | @Table(name = "employment_contract", schema = "hr")
00023 | @Getter
00024 | @Setter
00025 | @NoArgsConstructor
00026 | public class EmploymentContract {
00027 |
00028 |     @Id
00029 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00030 |     @Column(name = "contract_id")
00031 |     private Long contractId;
00032 |
00033 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00034 |     @JoinColumn(name = "employee_id", nullable = false)
00035 |     private Employee employee;
00036 |
00037 |     @ManyToOne(fetch = FetchType.LAZY)
00038 |     @JoinColumn(name = "contract_type_id")
00039 |     private ContractType contractType;
00040 |
00041 |     @Column(name = "contract_number", length = 50)
00042 |     private String contractNumber;
00043 |
00044 |     @Column(name = "start_date", nullable = false)
00045 |     private LocalDate startDate;
00046 |
00047 |     @Column(name = "end_date")
00048 |     private LocalDate endDate;
00049 |
00050 |     @Column(name = "salary", precision = 14, scale = 2)
```

016. src/main/java/com/citec/ems/hr/domain/EmploymentContract.java

```
00051 |     private BigDecimal salary;
00052 |
00053 |     @Column(name = "salary_currency", nullable = false, length = 10)
00054 |     private String salaryCurrency = "EGP";
00055 |
00056 |     @Column(name = "working_hours_per_week", precision = 5, scale = 2)
00057 |     private BigDecimal workingHoursPerWeek;
00058 |
00059 |     @Column(name = "working_hours_per_month", precision = 5, scale = 2)
00060 |     private BigDecimal workingHoursPerMonth;
00061 |
00062 |     @Column(name = "probation_period_days")
00063 |     private Integer probationPeriodDays;
00064 |
00065 |     @Column(name = "is_fulltime")
00066 |     private Boolean fulltime;
00067 |
00068 |     @Column(name = "is_active", nullable = false)
00069 |     private Boolean active = true;
00070 |
00071 |     @CreationTimestamp
00072 |     @Column(name = "created_at", nullable = false, updatable = false)
00073 |     private LocalDateTime createdAt;
00074 |
00075 |     @UpdateTimestamp
00076 |     @Column(name = "updated_at")
00077 |     private LocalDateTime updatedAt;
00078 | }
00079 |
00080 |
00081 |
```

017. src/main/java/com/citec/ems/hr/domain/EmploymentRecord.java

src/main/java/com/citec/ems/hr/domain/EmploymentRecord.java

74 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDate;
00013 | import java.time.LocalDateTime;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 | import org.hibernate.annotations.UpdateTimestamp;
00019 |
00020 | @Entity
00021 | @Table(name = "employment_record", schema = "hr")
00022 | @Getter
00023 | @Setter
00024 | @NoArgsConstructor
00025 | public class EmploymentRecord {
00026 |
00027 |     @Id
00028 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00029 |     @Column(name = "employment_id")
00030 |     private Long employmentId;
00031 |
00032 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00033 |     @JoinColumn(name = "employee_id", nullable = false)
00034 |     private Employee employee;
00035 |
00036 |     @ManyToOne(fetch = FetchType.LAZY)
00037 |     @JoinColumn(name = "org_unit_id")
00038 |     private OrganizationUnit orgUnit;
00039 |
00040 |     @ManyToOne(fetch = FetchType.LAZY)
00041 |     @JoinColumn(name = "position_id")
00042 |     private JobPosition position;
00043 |
00044 |     @Column(name = "record_type")
00045 |     private Short recordType;
00046 |
00047 |     @Column(name = "start_date", nullable = false)
00048 |     private LocalDate startDate;
00049 |
00050 |     @Column(name = "end_date")
```

017. src/main/java/com/citec/ems/hr/domain/EmploymentRecord.java

```
00051 |     private LocalDate endDate;
00052 |
00053 |     @ManyToOne(fetch = FetchType.LAZY)
00054 |     @JoinColumn(name = "contract_id")
00055 |     private EmploymentContract contract;
00056 |
00057 |     @ManyToOne(fetch = FetchType.LAZY)
00058 |     @JoinColumn(name = "reporting_to")
00059 |     private Employee reportingTo;
00060 |
00061 |     @Column(name = "is_current", nullable = false)
00062 |     private Boolean current = true;
00063 |
00064 |     @CreationTimestamp
00065 |     @Column(name = "created_at", nullable = false, updatable = false)
00066 |     private LocalDateTime createdAt;
00067 |
00068 |     @UpdateTimestamp
00069 |     @Column(name = "updated_at")
00070 |     private LocalDateTime updatedAt;
00071 | }
00072 |
00073 |
00074 |
```

src/main/java/com/citec/ems/hr/domain/JobGrade.java

50 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.GeneratedValue;
00006 | import jakarta.persistence.GenerationType;
00007 | import jakarta.persistence.Id;
00008 | import jakarta.persistence.Table;
00009 | import java.time.LocalDateTime;
00010 | import lombok.Getter;
00011 | import lombok.NoArgsConstructor;
00012 | import lombok.Setter;
00013 | import org.hibernate.annotations.CreationTimestamp;
00014 | import org.hibernate.annotations.UpdateTimestamp;
00015 |
00016 | @Entity
00017 | @Table(name = "job_grade", schema = "hr")
00018 | @Getter
00019 | @Setter
00020 | @NoArgsConstructor
00021 | public class JobGrade {
00022 |
00023 |     @Id
00024 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00025 |     @Column(name = "grade_id")
00026 |     private Long gradeId;
00027 |
00028 |     @Column(name = "grade_code", nullable = false, length = 100)
00029 |     private String gradeCode;
00030 |
00031 |     @Column(name = "grade_name")
00032 |     private String gradeName;
00033 |
00034 |     @Column(name = "grade_rank", nullable = false)
00035 |     private Short gradeRank;
00036 |
00037 |     @CreationTimestamp
00038 |     @Column(name = "created_at", nullable = false, updatable = false)
00039 |     private LocalDateTime createdAt;
00040 |
00041 |     @Column(name = "is_active", nullable = false)
00042 |     private Boolean active = true;
00043 |
00044 |     @UpdateTimestamp
00045 |     @Column(name = "updated_at")
00046 |     private LocalDateTime updatedAt;
00047 | }
00048 |
00049 |
00050 |
```


019. src/main/java/com/citec/ems/hr/domain/JobPosition.java

src/main/java/com/citec/ems/hr/domain/JobPosition.java

77 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDateTime;
00013 | import lombok.Getter;
00014 | import lombok.NoArgsConstructor;
00015 | import lombok.Setter;
00016 | import org.hibernate.annotations.CreationTimestamp;
00017 | import org.hibernate.annotations.UpdateTimestamp;
00018 |
00019 | @Entity
00020 | @Table(name = "job_position", schema = "hr")
00021 | @Getter
00022 | @Setter
00023 | @NoArgsConstructor
00024 | public class JobPosition {
00025 |
00026 |     @Id
00027 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00028 |     @Column(name = "position_id")
00029 |     private Long positionId;
00030 |
00031 |     @Column(name = "position_code", nullable = false, length = 100)
00032 |     private String positionCode;
00033 |
00034 |     @Column(name = "position_title", nullable = false)
00035 |     private String positionTitle;
00036 |
00037 |     @Column(name = "position_title_ar", nullable = false)
00038 |     private String positionTitleAr;
00039 |
00040 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00041 |     @JoinColumn(name = "org_unit_id", nullable = false)
00042 |     private OrganizationUnit orgUnit;
00043 |
00044 |     @ManyToOne(fetch = FetchType.LAZY)
00045 |     @JoinColumn(name = "grade_id")
00046 |     private JobGrade grade;
00047 |
00048 |     @Column(name = "position_level")
00049 |     private Integer positionLevel = 1;
00050 | }
```

019. src/main/java/com/citec/ems/hr/domain/JobPosition.java

```
00051 |     @ManyToOne(fetch = FetchType.LAZY)
00052 |     @JoinColumn(name = "reports_to_position_id")
00053 |     private JobPosition reportsToPosition;
00054 |
00055 |     @Column(name = "position_description", columnDefinition = "text")
00056 |     private String positionDescription;
00057 |
00058 |     @Column(name = "position_description_ar", columnDefinition = "text")
00059 |     private String positionDescriptionAr;
00060 |
00061 |     @Column(name = "is_open", nullable = false)
00062 |     private Boolean open = true;
00063 |
00064 |     @Column(name = "is_active", nullable = false)
00065 |     private Boolean active = true;
00066 |
00067 |     @CreationTimestamp
00068 |     @Column(name = "created_at", nullable = false, updatable = false)
00069 |     private LocalDateTime createdAt;
00070 |
00071 |     @UpdateTimestamp
00072 |     @Column(name = "updated_at")
00073 |     private LocalDateTime updatedAt;
00074 | }
00075 |
00076 |
00077 |
```

020. src/main/java/com/citec/ems/hr/domain/Organization.java

src/main/java/com/citec/ems/hr/domain/Organization.java

50 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.GeneratedValue;
00006 | import jakarta.persistence.GenerationType;
00007 | import jakarta.persistence.Id;
00008 | import jakarta.persistence.Table;
00009 | import java.time.LocalDateTime;
00010 | import lombok.Getter;
00011 | import lombok.NoArgsConstructor;
00012 | import lombok.Setter;
00013 | import org.hibernate.annotations.CreationTimestamp;
00014 | import org.hibernate.annotations.UpdateTimestamp;
00015 |
00016 | @Entity
00017 | @Table(name = "organization", schema = "hr")
00018 | @Getter
00019 | @Setter
00020 | @NoArgsConstructor
00021 | public class Organization {
00022 |
00023 |     @Id
00024 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00025 |     @Column(name = "organization_id")
00026 |     private Integer organizationId;
00027 |
00028 |     @Column(name = "organization_code", nullable = false, length = 100)
00029 |     private String organizationCode;
00030 |
00031 |     @Column(name = "organization_name_en", nullable = false)
00032 |     private String organizationNameEn;
00033 |
00034 |     @Column(name = "organization_name_ar", nullable = false)
00035 |     private String organizationNameAr;
00036 |
00037 |     @Column(name = "is_active", nullable = false)
00038 |     private Boolean active = true;
00039 |
00040 |     @CreationTimestamp
00041 |     @Column(name = "created_at", nullable = false, updatable = false)
00042 |     private LocalDateTime createdAt;
00043 |
00044 |     @UpdateTimestamp
00045 |     @Column(name = "updated_at")
00046 |     private LocalDateTime updatedAt;
00047 | }
00048 |
00049 |
00050 |
```

021. src/main/java/com/citec/ems/hr/domain/OrganizationProfile.java

src/main/java/com/citec/ems/hr/domain/OrganizationProfile.java

94 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDate;
00013 | import java.time.LocalDateTime;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 | import org.hibernate.annotations.UpdateTimestamp;
00019 |
00020 | @Entity
00021 | @Table(name = "organization_profile", schema = "hr")
00022 | @Getter
00023 | @Setter
00024 | @NoArgsConstructor
00025 | public class OrganizationProfile {
00026 |
00027 |     @Id
00028 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00029 |     @Column(name = "organization_profile_id")
00030 |     private Integer organizationProfileId;
00031 |
00032 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00033 |     @JoinColumn(name = "organization_id", nullable = false)
00034 |     private Organization organization;
00035 |
00036 |     @Column(name = "organization_name_en", nullable = false)
00037 |     private String organizationNameEn;
00038 |
00039 |     @Column(name = "organization_name_ar", nullable = false)
00040 |     private String organizationNameAr;
00041 |
00042 |     @Column(name = "established_date")
00043 |     private LocalDate establishedDate;
00044 |
00045 |     @Column(name = "tax_number")
00046 |     private String taxNumber;
00047 |
00048 |     @Column(name = "registration_number")
00049 |     private String registrationNumber;
00050 | }
```

021. src/main/java/com/citec/ems/hr/domain/OrganizationProfile.java

```
00051 |     @Column(name = "phone")
00052 |     private String phone;
00053 |
00054 |     @Column(name = "email")
00055 |     private String email;
00056 |
00057 |     @Column(name = "website")
00058 |     private String website;
00059 |
00060 |     @Column(name = "country_id")
00061 |     private Short countryId;
00062 |
00063 |     @Column(name = "state_id")
00064 |     private Integer stateId;
00065 |
00066 |     @Column(name = "city_id")
00067 |     private Integer cityId;
00068 |
00069 |     @Column(name = "address_line1")
00070 |     private String addressLine1;
00071 |
00072 |     @Column(name = "address_line2")
00073 |     private String addressLine2;
00074 |
00075 |     @Column(name = "postal_code", length = 20)
00076 |     private String postalCode;
00077 |
00078 |     @Column(name = "is_active", nullable = false)
00079 |     private Boolean active = true;
00080 |
00081 |     @CreationTimestamp
00082 |     @Column(name = "created_at", nullable = false, updatable = false)
00083 |     private LocalDateTime createdAt;
00084 |
00085 |     @UpdateTimestamp
00086 |     @Column(name = "updated_at")
00087 |     private LocalDateTime updatedAt;
00088 |
00089 |     @Column(name = "organization_logo", columnDefinition = "bytea")
00090 |     private byte[] organizationLogo;
00091 | }
00092 |
00093 |
00094 |
```

022. src/main/java/com/citec/ems/hr/domain/OrganizationUnit.java

src/main/java/com/citec/ems/hr/domain/OrganizationUnit.java

78 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDate;
00013 | import java.time.LocalDateTime;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 | import org.hibernate.annotations.UpdateTimestamp;
00019 |
00020 | @Entity
00021 | @Table(name = "organization_unit", schema = "hr")
00022 | @Getter
00023 | @Setter
00024 | @NoArgsConstructor
00025 | public class OrganizationUnit {
00026 |
00027 |     @Id
00028 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00029 |     @Column(name = "org_unit_id")
00030 |     private Long orgUnitId;
00031 |
00032 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00033 |     @JoinColumn(name = "organization_id", nullable = false)
00034 |     private Organization organization;
00035 |
00036 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00037 |     @JoinColumn(name = "unit_type_id", nullable = false)
00038 |     private UnitType unitType;
00039 |
00040 |     @ManyToOne(fetch = FetchType.LAZY)
00041 |     @JoinColumn(name = "parent_org_unit_id")
00042 |     private OrganizationUnit parentOrgUnit;
00043 |
00044 |     @Column(name = "unit_code", nullable = false, length = 50)
00045 |     private String unitCode;
00046 |
00047 |     @Column(name = "unit_name", nullable = false)
00048 |     private String unitName;
00049 |
00050 |     @Column(name = "unit_name_ar", nullable = false)
```

022. src/main/java/com/citec/ems/hr/domain/OrganizationUnit.java

```
00051 |     private String unitNameAr;
00052 |
00053 |     @Column(name = "description", columnDefinition = "text")
00054 |     private String description;
00055 |
00056 |     @Column(name = "description_ar", columnDefinition = "text")
00057 |     private String descriptionAr;
00058 |
00059 |     @Column(name = "start_date", nullable = false)
00060 |     private LocalDate startDate = LocalDate.now();
00061 |
00062 |     @Column(name = "end_date")
00063 |     private LocalDate endDate;
00064 |
00065 |     @Column(name = "is_active", nullable = false)
00066 |     private Boolean active = true;
00067 |
00068 |     @CreationTimestamp
00069 |     @Column(name = "created_at", nullable = false, updatable = false)
00070 |     private LocalDateTime createdAt;
00071 |
00072 |     @UpdateTimestamp
00073 |     @Column(name = "updated_at")
00074 |     private LocalDateTime updatedAt;
00075 | }
00076 |
00077 |
00078 |
```

023. src/main/java/com/citec/ems/hr/domain/OrganizationUnitRelation.java

src/main/java/com/citec/ems/hr/domain/OrganizationUnitRelation.java

63 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDate;
00013 | import java.time.LocalDateTime;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 | import org.hibernate.annotations.UpdateTimestamp;
00019 |
00020 | @Entity
00021 | @Table(name = "organization_unit_relation", schema = "hr")
00022 | @Getter
00023 | @Setter
00024 | @NoArgsConstructor
00025 | public class OrganizationUnitRelation {
00026 |
00027 |     @Id
00028 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00029 |     @Column(name = "relation_id")
00030 |     private Long relationId;
00031 |
00032 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00033 |     @JoinColumn(name = "parent_org_unit_id", nullable = false)
00034 |     private OrganizationUnit parentOrgUnit;
00035 |
00036 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00037 |     @JoinColumn(name = "child_org_unit_id", nullable = false)
00038 |     private OrganizationUnit childOrgUnit;
00039 |
00040 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00041 |     @JoinColumn(name = "relation_type_id", nullable = false)
00042 |     private RelationType relationType;
00043 |
00044 |     @Column(name = "start_date", nullable = false)
00045 |     private LocalDate startDate = LocalDate.now();
00046 |
00047 |     @Column(name = "end_date")
00048 |     private LocalDate endDate;
00049 |
00050 |     @Column(name = "is_active", nullable = false)
```


023. src/main/java/com/citec/ems/hr/domain/OrganizationUnitRelation.java

```
00051 |     private Boolean active = true;
00052 |
00053 |     @CreationTimestamp
00054 |     @Column(name = "created_at", nullable = false, updatable = false)
00055 |     private LocalDateTime createdAt;
00056 |
00057 |     @UpdateTimestamp
00058 |     @Column(name = "updated_at")
00059 |     private LocalDateTime updatedAt;
00060 | }
00061 |
00062 |
00063 |
```

024. src/main/java/com/citec/ems/hr/domain/PositionAssignment.java

src/main/java/com/citec/ems/hr/domain/PositionAssignment.java

71 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDate;
00013 | import java.time.LocalDateTime;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 | import org.hibernate.annotations.UpdateTimestamp;
00019 |
00020 | @Entity
00021 | @Table(name = "position_assignment", schema = "hr")
00022 | @Getter
00023 | @Setter
00024 | @NoArgsConstructor
00025 | public class PositionAssignment {
00026 |
00027 |     @Id
00028 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00029 |     @Column(name = "position_assignment_id")
00030 |     private Long positionAssignmentId;
00031 |
00032 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00033 |     @JoinColumn(name = "employee_id", nullable = false)
00034 |     private Employee employee;
00035 |
00036 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00037 |     @JoinColumn(name = "position_id", nullable = false)
00038 |     private JobPosition position;
00039 |
00040 |     @Column(name = "assignment_type", nullable = false)
00041 |     private Short assignmentType = 1;
00042 |
00043 |     @Column(name = "is_primary", nullable = false)
00044 |     private Boolean primaryAssignment = true;
00045 |
00046 |     @Column(name = "start_date", nullable = false)
00047 |     private LocalDate startDate = LocalDate.now();
00048 |
00049 |     @Column(name = "end_date")
00050 |     private LocalDate endDate;
```

024. src/main/java/com/citec/ems/hr/domain/PositionAssignment.java

```
00051 |  
00052 |     @Column(name = "is_active", nullable = false)  
00053 |     private Boolean active = true;  
00054 |  
00055 |     @CreationTimestamp  
00056 |     @Column(name = "created_at", nullable = false, updatable = false)  
00057 |     private LocalDateTime createdAt;  
00058 |  
00059 |     @Column(name = "assigned_by")  
00060 |     private Long assignedByUserId;  
00061 |  
00062 |     @Column(name = "assigned_at")  
00063 |     private LocalDateTime assignedAt;  
00064 |  
00065 |     @UpdateTimestamp  
00066 |     @Column(name = "updated_at")  
00067 |     private LocalDateTime updatedAt;  
00068 | }  
00069 |  
00070 |  
00071 |
```

025. src/main/java/com/citec/ems/hr/domain/RelationType.java

src/main/java/com/citec/ems/hr/domain/RelationType.java

42 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.GeneratedValue;
00006 | import jakarta.persistence.GenerationType;
00007 | import jakarta.persistence.Id;
00008 | import jakarta.persistence.Table;
00009 | import lombok.Getter;
00010 | import lombok.NoArgsConstructor;
00011 | import lombok.Setter;
00012 |
00013 | @Entity
00014 | @Table(name = "relation_type", schema = "hr")
00015 | @Getter
00016 | @Setter
00017 | @NoArgsConstructor
00018 | public class RelationType {
00019 |
00020 |     @Id
00021 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00022 |     @Column(name = "relation_type_id")
00023 |     private Integer relationTypeId;
00024 |
00025 |     @Column(name = "relation_code", nullable = false, length = 50)
00026 |     private String relationCode;
00027 |
00028 |     @Column(name = "relation_name", nullable = false)
00029 |     private String relationName;
00030 |
00031 |     @Column(name = "relation_name_ar", nullable = false)
00032 |     private String relationNameAr;
00033 |
00034 |     @Column(name = "description", columnDefinition = "text")
00035 |     private String description;
00036 |
00037 |     @Column(name = "description_ar", columnDefinition = "text")
00038 |     private String descriptionAr;
00039 | }
00040 |
00041 |
00042 |
```

026. src/main/java/com/citec/ems/hr/domain/UnitType.java

src/main/java/com/citec/ems/hr/domain/UnitType.java

42 lines

```
00001 | package com.citec.ems.hr.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.GeneratedValue;
00006 | import jakarta.persistence.GenerationType;
00007 | import jakarta.persistence.Id;
00008 | import jakarta.persistence.Table;
00009 | import lombok.Getter;
00010 | import lombok.NoArgsConstructor;
00011 | import lombok.Setter;
00012 |
00013 | @Entity
00014 | @Table(name = "unit_type", schema = "hr")
00015 | @Getter
00016 | @Setter
00017 | @NoArgsConstructor
00018 | public class UnitType {
00019 |
00020 |     @Id
00021 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00022 |     @Column(name = "unit_type_id")
00023 |     private Integer unitTypeId;
00024 |
00025 |     @Column(name = "unit_code", nullable = false, length = 50)
00026 |     private String unitCode;
00027 |
00028 |     @Column(name = "unit_name_en", nullable = false, length = 200)
00029 |     private String unitNameEn;
00030 |
00031 |     @Column(name = "unit_name_ar", nullable = false, length = 200)
00032 |     private String unitNameAr;
00033 |
00034 |     @Column(name = "description", columnDefinition = "text")
00035 |     private String description;
00036 |
00037 |     @Column(name = "description_ar", columnDefinition = "text")
00038 |     private String descriptionAr;
00039 | }
00040 |
00041 |
00042 |
```

027. src/main/java/com/citec/ems/hr/infrastructure/ContractTypeRepository.java

src/main/java/com/citec/ems/hr/infrastructure/ContractTypeRepository.java

14 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface ContractTypeRepository extends JpaRepository<ContractType, Integer> {
00009 |     Optional<ContractType> findByContractTypeCodeIgnoreCase(String contractTypeCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

028. src/main/java/com/citec/ems/hr/infrastructure/EmployeeProfileRepository.java

src/main/java/com/citec/ems/hr/infrastructure/EmployeeProfileRepository.java

17 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.List;
00006 | import java.util.Optional;
00007 | import org.springframework.data.jpa.repository.JpaRepository;
00008 |
00009 | public interface EmployeeProfileRepository extends JpaRepository<EmployeeProfile, Long> {
00010 |     Optional<EmployeeProfile> findFirstByEmployeeEmployeeIdAndPrimaryProfileTrue(Long employeeId);
00011 |
00012 |     List<EmployeeProfile> findByEmployeeEmployeeIdOrderByPrimaryProfileDescCreatedAtDesc(Long employeeId);
00013 | }
00014 |
00015 |
00016 |
00017 |
```

029. src/main/java/com/citec/ems/hr/infrastructure/EmployeeRepository.java

src/main/java/com/citec/ems/hr/infrastructure/EmployeeRepository.java

33 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.EntityGraph;
00007 | import org.springframework.data.jpa.repository.JpaRepository;
00008 | import org.springframework.data.jpa.repository.Query;
00009 | import org.springframework.data.repository.query.Param;
00010 |
00011 | public interface EmployeeRepository extends JpaRepository<Employee, Long> {
00012 |
00013 |     Optional<Employee> findByEmployeeNumberIgnoreCase(String employeeNumber);
00014 |
00015 |     @EntityGraph(attributePaths = {"user", "status", "currentOrgUnit"})
00016 |     @Query("""
00017 |         select employee
00018 |         from Employee employee
00019 |         where (:orgUnitId is null or employee.currentOrgUnit.orgUnitId = :orgUnitId)
00020 |             and (:statusCode is null or lower(employee.status.statusCode) = lower(:statusCode))
00021 |             and (:q is null
00022 |                 or lower(employee.employeeNumber) like lower(concat('%', :q, '%')))
00023 |         """)
00024 |     org.springframework.data.domain.Page<Employee> search(
00025 |         @Param("q") String q,
00026 |         @Param("orgUnitId") Long orgUnitId,
00027 |         @Param("statusCode") String statusCode,
00028 |         org.springframework.data.domain.Pageable pageable);
00029 | }
00030 |
00031 |
00032 |
00033 |
```


030. src/main/java/com/citec/ems/hr/infrastructure/EmployeeStatusRepository.java

src/main/java/com/citec/ems/hr/infrastructure/EmployeeStatusRepository.java

14 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface EmployeeStatusRepository extends JpaRepository<EmployeeStatus, Short> {
00009 |     Optional<EmployeeStatus> findByStatusCodeIgnoreCase(String statusCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

031. src/main/java/com/citec/ems/hr/infrastructure/EmploymentContractRepository.java

src/main/java/com/citec/ems/hr/infrastructure/EmploymentContractRepository.java

14 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.List;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface EmploymentContractRepository extends JpaRepository<EmploymentContract, Long> {
00009 |     List<EmploymentContract> findByEmployeeEmployeeIdOrderByStartDateDesc(Long employeeId);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

032. src/main/java/com/citec/ems/hr/infrastructure/EmploymentRecordRepository.java

src/main/java/com/citec/ems/hr/infrastructure/EmploymentRecordRepository.java

17 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.List;
00006 | import java.util.Optional;
00007 | import org.springframework.data.jpa.repository.JpaRepository;
00008 |
00009 | public interface EmploymentRecordRepository extends JpaRepository<EmploymentRecord, Long> {
00010 |     Optional<EmploymentRecord> findFirstByEmployeeEmployeeIdAndCurrentTrue(Long employeeId);
00011 |
00012 |     List<EmploymentRecord> findByEmployeeEmployeeIdOrderByStartDateDesc(Long employeeId);
00013 | }
00014 |
00015 |
00016 |
00017 |
```

033. src/main/java/com/citec/ems/hr/infrastructure/JobGradeRepository.java

src/main/java/com/citec/ems/hr/infrastructure/JobGradeRepository.java

14 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface JobGradeRepository extends JpaRepository<JobGrade, Long> {
00009 |     Optional<JobGrade> findByGradeCodeIgnoreCase(String gradeCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

034. src/main/java/com/citec/ems/hr/infrastructure/JobPositionRepository.java

src/main/java/com/citec/ems/hr/infrastructure/JobPositionRepository.java

30 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 | import org.springframework.data.jpa.repository.Query;
00008 | import org.springframework.data.repository.query.Param;
00009 |
00010 | public interface JobPositionRepository extends JpaRepository<JobPosition, Long> {
00011 |
00012 |     Optional<JobPosition> findByPositionCodeIgnoreCase(String positionCode);
00013 |
00014 |     @Query("""
00015 |         select position
00016 |         from JobPosition position
00017 |         where (:orgUnitId is null or position.orgUnit.orgUnitId = :orgUnitId)
00018 |             and (:q is null
00019 |                 or lower(position.positionCode) like lower(concat('%', :q, '%'))
00020 |                 or lower(position.positionTitle) like lower(concat('%', :q, '%')))
00021 |         """)
00022 |     org.springframework.data.domain.Page<JobPosition> search(
00023 |         @Param("orgUnitId") Long orgUnitId,
00024 |         @Param("q") String q,
00025 |         org.springframework.data.domain.Pageable pageable);
00026 | }
00027 |
00028 |
00029 |
00030 |
```

035. src/main/java/com/citec/ems/hr/infrastructure/OrganizationProfileRepository.java

src/main/java/com/citec/ems/hr/infrastructure/OrganizationProfileRepository.java

14 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface OrganizationProfileRepository extends JpaRepository<OrganizationProfile, Integer> {
00009 |     Optional<OrganizationProfile> findFirstByOrganizationOrganizationIdAndActiveTrue(Integer organizationId);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

036. src/main/java/com/citec/ems/hr/infrastructure/OrganizationRepository.java

src/main/java/com/citec/ems/hr/infrastructure/OrganizationRepository.java

14 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface OrganizationRepository extends JpaRepository<Organization, Integer> {
00009 |     Optional<Organization> findByOrganizationCodeIgnoreCase(String organizationCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

037. src/main/java/com/citec/ems/hr/infrastructure/OrganizationUnitRelationRepository.java

src/main/java/com/citec/ems/hr/infrastructure/OrganizationUnitRelationRepository.java

12 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import org.springframework.data.jpa.repository.JpaRepository;
00006 |
00007 | public interface OrganizationUnitRelationRepository extends JpaRepository<OrganizationUnitRelation, Long> {
00008 | }
00009 |
00010 |
00011 |
00012 |
```


038. src/main/java/com/citec/ems/hr/infrastructure/OrganizationUnitRepository.java

src/main/java/com/citec/ems/hr/infrastructure/OrganizationUnitRepository.java

32 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 | import org.springframework.data.jpa.repository.Query;
00008 | import org.springframework.data.repository.query.Param;
00009 |
00010 | public interface OrganizationUnitRepository extends JpaRepository<OrganizationUnit, Long> {
00011 |
00012 |     Optional<OrganizationUnit> findByOrganizationOrganizationIdAndUnitCodeIgnoreCase(Integer organizationId, String unitCode);
00013 |
00014 |     @Query("""
00015 |         select unit
00016 |         from OrganizationUnit unit
00017 |         where (:organizationId is null or unit.organization.organizationId = :organizationId)
00018 |             and (:parentOrgUnitId is null or unit.parentOrgUnit.orgUnitId = :parentOrgUnitId)
00019 |             and (:q is null
00020 |                 or lower(unit.unitCode) like lower(concat('%', :q, '%'))
00021 |                 or lower(unit.unitName) like lower(concat('%', :q, '%')))
00022 |         """)
00023 |     org.springframework.data.domain.Page<OrganizationUnit> search(
00024 |         @Param("organizationId") Integer organizationId,
00025 |         @Param("parentOrgUnitId") Long parentOrgUnitId,
00026 |         @Param("q") String q,
00027 |         org.springframework.data.domain.Pageable pageable);
00028 | }
00029 |
00030 |
00031 |
00032 |
```

039. src/main/java/com/citec/ems/hr/infrastructure/PositionAssignmentRepository.java

src/main/java/com/citec/ems/hr/infrastructure/PositionAssignmentRepository.java

14 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.List;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface PositionAssignmentRepository extends JpaRepository<PositionAssignment, Long> {
00009 |     List<PositionAssignment> findByEmployeeEmployeeIdOrderByStartDateDesc(Long employeeId);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

040. src/main/java/com/citec/ems/hr/infrastructure/RelationTypeRepository.java

src/main/java/com/citec/ems/hr/infrastructure/RelationTypeRepository.java

14 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface RelationTypeRepository extends JpaRepository<RelationType, Integer> {
00009 |     Optional<RelationType> findByRelationCodeIgnoreCase(String relationCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

041. src/main/java/com/citec/ems/hr/infrastructure/UnitTypeRepository.java

src/main/java/com/citec/ems/hr/infrastructure/UnitTypeRepository.java

14 lines

```
00001 | package com.citec.ems.hr.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.hr.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface UnitTypeRepository extends JpaRepository<UnitType, Integer> {
00009 |     Optional<UnitType> findByUnitCodeIgnoreCase(String unitCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

042. src/main/java/com/citec/ems/hr/package-info.java

src/main/java/com/citec/ems/hr/package-info.java

4 lines

```
00001 | @org.springframework.modulith.ApplicationModule(  
00002 |     displayName = "Human Resources",  
00003 |     allowedDependencies = "iam")  
00004 | package com.citec.ems.hr;
```

043. src/main/java/com/citec/ems/hr/web/EmployeeController.java

src/main/java/com/citec/ems/hr/web/EmployeeController.java

103 lines

```
00001 | package com.citec.ems.hr.web;
00002 |
00003 |
00004 | import com.citec.ems.hr.application.*;
00005 | import com.citec.ems.hr.web.HrDtos.EmployeeCreateRequest;
00006 | import com.citec.ems.hr.web.HrDtos.EmployeeDetail;
00007 | import com.citec.ems.hr.web.HrDtos.EmployeeProfileRequest;
00008 | import com.citec.ems.hr.web.HrDtos.EmployeeProfileResponse;
00009 | import com.citec.ems.hr.web.HrDtos.EmployeeSummary;
00010 | import com.citec.ems.hr.web.HrDtos.EmployeeUpdateRequest;
00011 | import com.citec.ems.hr.web.HrDtos.EmploymentContractRequest;
00012 | import com.citec.ems.hr.web.HrDtos.EmploymentContractResponse;
00013 | import com.citec.ems.hr.web.HrDtos.EmploymentRecordRequest;
00014 | import com.citec.ems.hr.web.HrDtos.EmploymentRecordResponse;
00015 | import com.citec.ems.hr.web.HrDtos.PositionAssignmentRequest;
00016 | import com.citec.ems.hr.web.HrDtos.PositionAssignmentResponse;
00017 | import jakarta.validation.Valid;
00018 | import org.springframework.data.domain.Page;
00019 | import org.springframework.data.domain.Pageable;
00020 | import org.springframework.http.HttpStatus;
00021 | import org.springframework.web.bind.annotation.GetMapping;
00022 | import org.springframework.web.bind.annotation.PathVariable;
00023 | import org.springframework.web.bind.annotation.PostMapping;
00024 | import org.springframework.web.bind.annotation.PutMapping;
00025 | import org.springframework.web.bind.annotation.RequestBody;
00026 | import org.springframework.web.bind.annotation.RequestMapping;
00027 | import org.springframework.web.bind.annotation.RequestParam;
00028 | import org.springframework.web.bind.annotation.ResponseStatus;
00029 | import org.springframework.web.bind.annotation.RestController;
00030 |
00031 | @RestController
00032 | @RequestMapping("/api/hr/employees")
00033 | public class EmployeeController {
00034 |
00035 |     private final EmployeeService employeeService;
00036 |
00037 |     public EmployeeController(EmployeeService employeeService) {
00038 |         this.employeeService = employeeService;
00039 |     }
00040 |
00041 |     @GetMapping
00042 |     public Page<EmployeeSummary> listEmployees(
00043 |         @RequestParam(required = false) String q,
00044 |         @RequestParam(required = false) Long orgUnitId,
00045 |         @RequestParam(required = false) String statusCode,
00046 |         Pageable pageable) {
00047 |         return employeeService.listEmployees(q, orgUnitId, statusCode, pageable);
00048 |     }
00049 |
00050 |     @PostMapping
```

043. src/main/java/com/citec/ems/hr/web/EmployeeController.java

```
00051 | @ResponseStatus(HttpStatus.CREATED)
00052 | public EmployeeSummary createEmployee(@Valid @RequestBody EmployeeCreateRequest request) {
00053 |     return employeeService.createEmployee(request);
00054 | }
00055 |
00056 | @GetMapping("/{employeeId}")
00057 | public EmployeeDetail getEmployee(@PathVariable Long employeeId) {
00058 |     return employeeService.getEmployee(employeeId);
00059 | }
00060 |
00061 | @PutMapping("/{employeeId}")
00062 | public EmployeeSummary updateEmployee(
00063 |     @PathVariable Long employeeId,
00064 |     @Valid @RequestBody EmployeeUpdateRequest request) {
00065 |     return employeeService.updateEmployee(employeeId, request);
00066 | }
00067 |
00068 | @PostMapping("/{employeeId}/profiles")
00069 | @ResponseStatus(HttpStatus.CREATED)
00070 | public EmployeeProfileResponse addProfile(
00071 |     @PathVariable Long employeeId,
00072 |     @Valid @RequestBody EmployeeProfileRequest request) {
00073 |     return employeeService.addProfile(employeeId, request);
00074 | }
00075 |
00076 | @PostMapping("/{employeeId}/contracts")
00077 | @ResponseStatus(HttpStatus.CREATED)
00078 | public EmploymentContractResponse addContract(
00079 |     @PathVariable Long employeeId,
00080 |     @Valid @RequestBody EmploymentContractRequest request) {
00081 |     return employeeService.addContract(employeeId, request);
00082 | }
00083 |
00084 | @PostMapping("/{employeeId}/position-assignments")
00085 | @ResponseStatus(HttpStatus.CREATED)
00086 | public PositionAssignmentResponse addPositionAssignment(
00087 |     @PathVariable Long employeeId,
00088 |     @Valid @RequestBody PositionAssignmentRequest request) {
00089 |     return employeeService.addPositionAssignment(employeeId, request);
00090 | }
00091 |
00092 | @PostMapping("/{employeeId}/employment-records")
00093 | @ResponseStatus(HttpStatus.CREATED)
00094 | public EmploymentRecordResponse addEmploymentRecord(
00095 |     @PathVariable Long employeeId,
00096 |     @Valid @RequestBody EmploymentRecordRequest request) {
00097 |     return employeeService.addEmploymentRecord(employeeId, request);
00098 | }
00099 | }
00100 |
00101 |
00102 |
00103 |
```

044. src/main/java/com/citec/ems/hr/web/HrDtos.java

src/main/java/com/citec/ems/hr/web/HrDtos.java

299 lines

```
00001 | package com.citec.ems.hr.web;
00002 |
00003 |
00004 | import com.citec.ems.hr.application.*;
00005 | import jakarta.validation.Valid;
00006 | import jakarta.validation.constraints.DecimalMin;
00007 | import jakarta.validation.constraints.Email;
00008 | import jakarta.validation.constraints.NotBlank;
00009 | import jakarta.validation.constraints.NotNull;
00010 | import jakarta.validation.constraints.Size;
00011 | import java.math.BigDecimal;
00012 | import java.time.LocalDate;
00013 | import java.time.LocalDateTime;
00014 | import java.util.List;
00015 |
00016 | public final class HrDtos {
00017 |
00018 |     private HrDtos() {
00019 |     }
00020 |
00021 |     public record EmployeeCreateRequest(
00022 |         @NotBlank @Size(max = 50) String employeeNumber,
00023 |         Long userId,
00024 |         Long currentOrgUnitId,
00025 |         LocalDate hireDate,
00026 |         LocalDate startDate,
00027 |         Short employeeStatusId,
00028 |         @Valid EmployeeProfileRequest profile) {
00029 |     }
00030 |
00031 |     public record EmployeeUpdateRequest(
00032 |         @Size(max = 50) String employeeNumber,
00033 |         Long userId,
00034 |         Long currentOrgUnitId,
00035 |         LocalDate hireDate,
00036 |         LocalDate startDate,
00037 |         LocalDate terminationDate,
00038 |         Short employeeStatusId) {
00039 |     }
00040 |
00041 |     public record EmployeeProfileRequest(
00042 |         Boolean primary,
00043 |         @NotBlank @Size(max = 100) String firstName,
00044 |         @NotBlank @Size(max = 200) String otherName,
00045 |         @Size(max = 100) String displayName,
00046 |         @NotNull Short gender,
00047 |         LocalDate birthDate,
00048 |         @Size(max = 14) String nationalId,
00049 |         @Email @Size(max = 100) String personalEmail,
00050 |         @Email @Size(max = 100) String businessEmail,
```


044. src/main/java/com/citec/ems/hr/web/HrDtos.java

```
00051 |         @Size(max = 100) String phoneNumber,
00052 |         @Size(max = 100) String mobileNumber,
00053 |         Short countryId,
00054 |         Integer stateId,
00055 |         Integer cityId,
00056 |         @Size(max = 255) String addressLine1,
00057 |         @Size(max = 255) String addressLine2,
00058 |         @Size(max = 20) String postalCode) {
00059 |     }
00060 |
00061 |     public record EmployeeSummary(
00062 |         Long employeeId,
00063 |         String employeeNumber,
00064 |         Long userId,
00065 |         String username,
00066 |         Long currentOrgUnitId,
00067 |         String currentOrgUnitName,
00068 |         String statusCode,
00069 |         String displayName,
00070 |         String businessEmail,
00071 |         LocalDate hireDate,
00072 |         LocalDate startDate,
00073 |         LocalDate terminationDate) {
00074 |     }
00075 |
00076 |     public record EmployeeDetail(
00077 |         EmployeeSummary employee,
00078 |         List<EmployeeProfileResponse> profiles,
00079 |         List<EmploymentContractResponse> contracts,
00080 |         List<PositionAssignmentResponse> positionAssignments,
00081 |         List<EmploymentRecordResponse> employmentRecords,
00082 |         LocalDateTime createdAt,
00083 |         LocalDateTime updatedAt) {
00084 |     }
00085 |
00086 |     public record EmployeeProfileResponse(
00087 |         Long profileId,
00088 |         Boolean primary,
00089 |         String firstName,
00090 |         String otherName,
00091 |         String displayName,
00092 |         Short gender,
00093 |         LocalDate birthDate,
00094 |         String nationalId,
00095 |         String personalEmail,
00096 |         String businessEmail,
00097 |         String phoneNumber,
00098 |         String mobileNumber,
00099 |         Short countryId,
00100 |         Integer stateId,
00101 |         Integer cityId,
00102 |         String addressLine1,
00103 |         String addressLine2,
00104 |         String postalCode) {
00105 |     }
```

044. src/main/java/com/citec/ems/hr/web/HrDtos.java

```
00106 |
00107 |     public record EmploymentContractRequest(
00108 |         Integer contractTypeId,
00109 |         @Size(max = 50) String contractNumber,
00110 |         @NotNull LocalDate startDate,
00111 |         LocalDate endDate,
00112 |         @DecimalMin("0.00") BigDecimal salary,
00113 |         @Size(max = 10) String salaryCurrency,
00114 |         BigDecimal workingHoursPerWeek,
00115 |         BigDecimal workingHoursPerMonth,
00116 |         Integer probationPeriodDays,
00117 |         Boolean fulltime,
00118 |         Boolean active) {
00119 |     }
00120 |
00121 |     public record EmploymentContractResponse(
00122 |         Long contractId,
00123 |         Integer contractTypeId,
00124 |         String contractTypeCode,
00125 |         String contractNumber,
00126 |         LocalDate startDate,
00127 |         LocalDate endDate,
00128 |         BigDecimal salary,
00129 |         String salaryCurrency,
00130 |         BigDecimal workingHoursPerWeek,
00131 |         BigDecimal workingHoursPerMonth,
00132 |         Integer probationPeriodDays,
00133 |         Boolean fulltime,
00134 |         Boolean active) {
00135 |     }
00136 |
00137 |     public record PositionAssignmentRequest(
00138 |         @NotNull Long positionId,
00139 |         Short assignmentType,
00140 |         Boolean primary,
00141 |         LocalDate startDate,
00142 |         LocalDate endDate,
00143 |         Boolean active,
00144 |         Long assignedByUserId) {
00145 |     }
00146 |
00147 |     public record PositionAssignmentResponse(
00148 |         Long positionAssignmentId,
00149 |         Long positionId,
00150 |         String positionCode,
00151 |         String positionTitle,
00152 |         Short assignmentType,
00153 |         Boolean primary,
00154 |         LocalDate startDate,
00155 |         LocalDate endDate,
00156 |         Boolean active,
00157 |         Long assignedByUserId,
00158 |         LocalDateTime assignedAt) {
00159 |     }
00160 |
```

044. src/main/java/com/citec/ems/hr/web/HrDtos.java

```
00161 |     public record EmploymentRecordRequest(  
00162 |         Long orgUnitId,  
00163 |         Long positionId,  
00164 |         Short recordType,  
00165 |         @NotNull LocalDate startDate,  
00166 |         LocalDate endDate,  
00167 |         Long contractId,  
00168 |         Long reportingToEmployeeId,  
00169 |         Boolean current) {  
00170 |     }  
00171 |  
00172 |     public record EmploymentRecordResponse(  
00173 |         Long employmentId,  
00174 |         Long orgUnitId,  
00175 |         String orgUnitName,  
00176 |         Long positionId,  
00177 |         String positionTitle,  
00178 |         Short recordType,  
00179 |         LocalDate startDate,  
00180 |         LocalDate endDate,  
00181 |         Long contractId,  
00182 |         Long reportingToEmployeeId,  
00183 |         Boolean current) {  
00184 |     }  
00185 |  
00186 |     public record EmployeeStatusResponse(Short employeeStatusId, String statusCode, String description) {  
00187 |     }  
00188 |  
00189 |     public record OrganizationCreateRequest(  
00190 |         @NotBlank @Size(max = 100) String organizationCode,  
00191 |         @NotBlank @Size(max = 255) String organizationNameEn,  
00192 |         @NotBlank @Size(max = 255) String organizationNameAr,  
00193 |         Boolean active) {  
00194 |     }  
00195 |  
00196 |     public record OrganizationResponse(  
00197 |         Integer organizationId,  
00198 |         String organizationCode,  
00199 |         String organizationNameEn,  
00200 |         String organizationNameAr,  
00201 |         Boolean active) {  
00202 |     }  
00203 |  
00204 |     public record UnitTypeResponse(  
00205 |         Integer unitTypeId,  
00206 |         String unitCode,  
00207 |         String unitNameEn,  
00208 |         String unitNameAr,  
00209 |         String description,  
00210 |         String descriptionAr) {  
00211 |     }  
00212 |  
00213 |     public record OrganizationUnitCreateRequest(  
00214 |         @NotNull Integer organizationId,  
00215 |         @NotNull Integer unitTypeId,
```

044. src/main/java/com/citec/ems/hr/web/HrDtos.java

```
00216 |         Long parentOrgUnitId,
00217 |         @NotBlank @Size(max = 50) String unitCode,
00218 |         @NotBlank @Size(max = 255) String unitName,
00219 |         @NotBlank @Size(max = 255) String unitNameAr,
00220 |         String description,
00221 |         String descriptionAr,
00222 |         LocalDate startDate,
00223 |         LocalDate endDate,
00224 |         Boolean active) {
00225 |     }
00226 |
00227 |     public record OrganizationUnitResponse(
00228 |         Long orgUnitId,
00229 |         Integer organizationId,
00230 |         String organizationCode,
00231 |         Integer unitTypeId,
00232 |         String unitTypeCode,
00233 |         Long parentOrgUnitId,
00234 |         String unitCode,
00235 |         String unitName,
00236 |         String unitNameAr,
00237 |         LocalDate startDate,
00238 |         LocalDate endDate,
00239 |         Boolean active) {
00240 |     }
00241 |
00242 |     public record JobGradeCreateRequest(
00243 |         @NotBlank @Size(max = 100) String gradeCode,
00244 |         @Size(max = 255) String gradeName,
00245 |         @NotNull Short gradeRank,
00246 |         Boolean active) {
00247 |     }
00248 |
00249 |     public record JobGradeResponse(Long gradeId, String gradeCode, String gradeName, Short gradeRank, Boolean active) {
00250 |     }
00251 |
00252 |     public record JobPositionCreateRequest(
00253 |         @NotBlank @Size(max = 100) String positionCode,
00254 |         @NotBlank @Size(max = 255) String positionTitle,
00255 |         @NotBlank @Size(max = 255) String positionTitleAr,
00256 |         @NotNull Long orgUnitId,
00257 |         Long gradeId,
00258 |         Integer positionLevel,
00259 |         Long reportsToPositionId,
00260 |         String positionDescription,
00261 |         String positionDescriptionAr,
00262 |         Boolean open,
00263 |         Boolean active) {
00264 |     }
00265 |
00266 |     public record JobPositionResponse(
00267 |         Long positionId,
00268 |         String positionCode,
00269 |         String positionTitle,
00270 |         String positionTitleAr,
```

044. src/main/java/com/citec/ems/hr/web/HrDtos.java

```
00271 |         Long orgUnitId,
00272 |         String orgUnitName,
00273 |         Long gradeId,
00274 |         String gradeCode,
00275 |         Integer positionLevel,
00276 |         Long reportsToPositionId,
00277 |         Boolean open,
00278 |         Boolean active) {
00279 |     }
00280 |
00281 |     public record ContractTypeCreateRequest(
00282 |         @NotBlank @Size(max = 100) String contractTypeCode,
00283 |         @NotBlank @Size(max = 255) String contractTypeName,
00284 |         String description,
00285 |         Boolean active) {
00286 |     }
00287 |
00288 |     public record ContractTypeResponse(
00289 |         Integer contractTypeId,
00290 |         String contractTypeCode,
00291 |         String contractTypeName,
00292 |         String description,
00293 |         Boolean active) {
00294 |     }
00295 | }
00296 |
00297 |
00298 |
00299 |
```

045. src/main/java/com/citec/ems/hr/web/HrReferenceController.java

src/main/java/com/citec/ems/hr/web/HrReferenceController.java

114 lines

```
00001 | package com.citec.ems.hr.web;
00002 |
00003 |
00004 | import com.citec.ems.hr.application.*;
00005 | import com.citec.ems.hr.web.HrDtos.ContractTypeCreateRequest;
00006 | import com.citec.ems.hr.web.HrDtos.ContractTypeResponse;
00007 | import com.citec.ems.hr.web.HrDtos.EmployeeStatusResponse;
00008 | import com.citec.ems.hr.web.HrDtos.JobGradeCreateRequest;
00009 | import com.citec.ems.hr.web.HrDtos.JobGradeResponse;
00010 | import com.citec.ems.hr.web.HrDtos.JobPositionCreateRequest;
00011 | import com.citec.ems.hr.web.HrDtos.JobPositionResponse;
00012 | import com.citec.ems.hr.web.HrDtos.OrganizationCreateRequest;
00013 | import com.citec.ems.hr.web.HrDtos.OrganizationResponse;
00014 | import com.citec.ems.hr.web.HrDtos.OrganizationUnitCreateRequest;
00015 | import com.citec.ems.hr.web.HrDtos.OrganizationUnitResponse;
00016 | import com.citec.ems.hr.web.HrDtos.UnitTypeResponse;
00017 | import jakarta.validation.Valid;
00018 | import org.springframework.data.domain.Page;
00019 | import org.springframework.data.domain.Pageable;
00020 | import org.springframework.http.HttpStatus;
00021 | import org.springframework.web.bind.annotation.GetMapping;
00022 | import org.springframework.web.bind.annotation.PostMapping;
00023 | import org.springframework.web.bind.annotation.RequestBody;
00024 | import org.springframework.web.bind.annotation.RequestMapping;
00025 | import org.springframework.web.bind.annotation.RequestParam;
00026 | import org.springframework.web.bind.annotation.ResponseStatus;
00027 | import org.springframework.web.bind.annotation.RestController;
00028 |
00029 | @RestController
00030 | @RequestMapping("/api/hr")
00031 | public class HrReferenceController {
00032 |
00033 |     private final HrReferenceService hrReferenceService;
00034 |
00035 |     public HrReferenceController(HrReferenceService hrReferenceService) {
00036 |         this.hrReferenceService = hrReferenceService;
00037 |     }
00038 |
00039 |     @GetMapping("/organizations")
00040 |     public Page<OrganizationResponse> listOrganizations(Pageable pageable) {
00041 |         return hrReferenceService.listOrganizations(pageable);
00042 |     }
00043 |
00044 |     @PostMapping("/organizations")
00045 |     @ResponseStatus(HttpStatus.CREATED)
00046 |     public OrganizationResponse createOrganization(@Valid @RequestBody OrganizationCreateRequest request) {
00047 |         return hrReferenceService.createOrganization(request);
00048 |     }
00049 |
00050 |     @GetMapping("/org-units")
```

045. src/main/java/com/citec/ems/hr/web/HrReferenceController.java

```
00051 |     public Page<OrganizationUnitResponse> listOrganizationUnits(
00052 |         @RequestParam(required = false) Integer organizationId,
00053 |         @RequestParam(required = false) Long parentOrgUnitId,
00054 |         @RequestParam(required = false) String q,
00055 |         Pageable pageable) {
00056 |         return hrReferenceService.listOrganizationUnits(organizationId, parentOrgUnitId, q, pageable);
00057 |     }
00058 |
00059 |     @PostMapping("/org-units")
00060 |     @ResponseStatus(HttpStatus.CREATED)
00061 |     public OrganizationUnitResponse createOrganizationUnit(@Valid @RequestBody OrganizationUnitCreateRequest request) {
00062 |         return hrReferenceService.createOrganizationUnit(request);
00063 |     }
00064 |
00065 |     @GetMapping("/unit-types")
00066 |     public Page<UnitTypeResponse> listUnitTypes(Pageable pageable) {
00067 |         return hrReferenceService.listUnitTypes(pageable);
00068 |     }
00069 |
00070 |     @GetMapping("/employee-statuses")
00071 |     public Page<EmployeeStatusResponse> listEmployeeStatuses(Pageable pageable) {
00072 |         return hrReferenceService.listEmployeeStatuses(pageable);
00073 |     }
00074 |
00075 |     @GetMapping("/job-grades")
00076 |     public Page<JobGradeResponse> listJobGrades(Pageable pageable) {
00077 |         return hrReferenceService.listJobGrades(pageable);
00078 |     }
00079 |
00080 |     @PostMapping("/job-grades")
00081 |     @ResponseStatus(HttpStatus.CREATED)
00082 |     public JobGradeResponse createJobGrade(@Valid @RequestBody JobGradeCreateRequest request) {
00083 |         return hrReferenceService.createJobGrade(request);
00084 |     }
00085 |
00086 |     @GetMapping("/positions")
00087 |     public Page<JobPositionResponse> listJobPositions(
00088 |         @RequestParam(required = false) Long orgUnitId,
00089 |         @RequestParam(required = false) String q,
00090 |         Pageable pageable) {
00091 |         return hrReferenceService.listJobPositions(orgUnitId, q, pageable);
00092 |     }
00093 |
00094 |     @PostMapping("/positions")
00095 |     @ResponseStatus(HttpStatus.CREATED)
00096 |     public JobPositionResponse createJobPosition(@Valid @RequestBody JobPositionCreateRequest request) {
00097 |         return hrReferenceService.createJobPosition(request);
00098 |     }
00099 |
00100 |     @GetMapping("/contract-types")
00101 |     public Page<ContractTypeResponse> listContractTypes(Pageable pageable) {
00102 |         return hrReferenceService.listContractTypes(pageable);
00103 |     }
00104 |
00105 |     @PostMapping("/contract-types")
```

045. src/main/java/com/citec/ems/hr/web/HrReferenceController.java

```
00106 |     @ResponseStatus(HttpStatus.CREATED)
00107 |     public ContractTypeResponse createContractType(@Valid @RequestBody ContractTypeCreateRequest request) {
00108 |         return hrReferenceService.createContractType(request);
00109 |     }
00110 | }
00111 |
00112 |
00113 |
00114 |
```


046. src/main/java/com/citec/ems/iam/UserAccountSummary.java

src/main/java/com/citec/ems/iam/UserAccountSummary.java

4 lines

```
00001 | package com.citec.ems.iam;
00002 |
00003 | public record UserAccountSummary(Long userId, String username, String email) {
00004 | }
```

047. src/main/java/com/citec/ems/iam/UserAccounts.java

src/main/java/com/citec/ems/iam/UserAccounts.java

27 lines

```
00001 | package com.citec.ems.iam;
00002 |
00003 | import com.citec.ems.iam.infrastructure.UserRepository;
00004 | import java.util.Optional;
00005 | import org.springframework.stereotype.Service;
00006 | import org.springframework.transaction.annotation.Transactional;
00007 |
00008 | @Service
00009 | public class UserAccounts {
00010 |
00011 |     private final UserRepository userRepository;
00012 |
00013 |     public UserAccounts(UserRepository userRepository) {
00014 |         this.userRepository = userRepository;
00015 |     }
00016 |
00017 |     @Transactional(readOnly = true)
00018 |     public boolean existsById(Long userId) {
00019 |         return userRepository.existsById(userId);
00020 |     }
00021 |
00022 |     @Transactional(readOnly = true)
00023 |     public Optional<UserAccountSummary> findSummary(Long userId) {
00024 |         return userRepository.findById(userId)
00025 |             .map(user -> new UserAccountSummary(user.getUserId(), user.getUsername(), user.getEmail()));
00026 |     }
00027 | }
```

048. src/main/java/com/citec/ems/iam/application/AuthService.java

src/main/java/com/citec/ems/iam/application/AuthService.java

295 lines

```
00001 | package com.citec.ems.iam.application;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import com.citec.ems.iam.infrastructure.*;
00006 | import com.citec.ems.shared.BadRequestException;
00007 | import com.citec.ems.iam.web.IamDtos.AuthResponse;
00008 | import com.citec.ems.iam.web.IamDtos.AuthUserResponse;
00009 | import com.citec.ems.iam.web.IamDtos.LoginRequest;
00010 | import com.citec.ems.iam.security.AccessTokenService;
00011 | import com.citec.ems.iam.security.AccessTokenService.GeneratedAccessToken;
00012 | import com.citec.ems.iam.security.AuthenticatedUser;
00013 | import com.citec.ems.iam.security.EmsSecurityProperties;
00014 | import com.citec.ems.iam.security.RefreshTokenService;
00015 | import jakarta.servlet.http.HttpServletRequest;
00016 | import java.time.LocalDate;
00017 | import java.time.LocalDateTime;
00018 | import java.util.List;
00019 | import java.util.UUID;
00020 | import org.springframework.security.authentication.BadCredentialsException;
00021 | import org.springframework.security.crypto.password.PasswordEncoder;
00022 | import org.springframework.stereotype.Service;
00023 | import org.springframework.transaction.annotation.Transactional;
00024 |
00025 | @Service
00026 | public class AuthService {
00027 |
00028 |     private static final String WEB_CLIENT = "WEB_APPLICATION";
00029 |     private static final String MOBILE_CLIENT = "MOBILE_APPLICATION";
00030 |     private static final String DEFAULT_APPLICATION = "ADMIN";
00031 |
00032 |     private final UserRepository userRepository;
00033 |     private final ClientApplicationRepository clientApplicationRepository;
00034 |     private final UserDeviceRepository userDeviceRepository;
00035 |     private final UserSessionRepository userSessionRepository;
00036 |     private final RefreshTokenRepository refreshTokenRepository;
00037 |     private final LoginAttemptRepository loginAttemptRepository;
00038 |     private final UserRoleRepository userRoleRepository;
00039 |     private final RolePermissionRepository rolePermissionRepository;
00040 |     private final PasswordEncoder passwordEncoder;
00041 |     private final AccessTokenService accessTokenService;
00042 |     private final RefreshTokenService refreshTokenService;
00043 |     private final EmsSecurityProperties securityProperties;
00044 |     private final IpRuleService ipRuleService;
00045 |
00046 |     public AuthService(
00047 |         UserRepository userRepository,
00048 |         ClientApplicationRepository clientApplicationRepository,
00049 |         UserDeviceRepository userDeviceRepository,
00050 |         UserSessionRepository userSessionRepository,
```

048. src/main/java/com/citec/ems/iam/application/AuthService.java

```
00051 |         RefreshTokenRepository refreshTokenRepository,
00052 |         LoginAttemptRepository loginAttemptRepository,
00053 |         UserRoleRepository userRoleRepository,
00054 |         RolePermissionRepository rolePermissionRepository,
00055 |         PasswordEncoder passwordEncoder,
00056 |         AccessTokenService accessTokenService,
00057 |         RefreshTokenService refreshTokenService,
00058 |         EmsSecurityProperties securityProperties,
00059 |         IpRuleService ipRuleService) {
00060 |     this.userRepository = userRepository;
00061 |     this.clientApplicationRepository = clientApplicationRepository;
00062 |     this.userDeviceRepository = userDeviceRepository;
00063 |     this.userSessionRepository = userSessionRepository;
00064 |     this.refreshTokenRepository = refreshTokenRepository;
00065 |     this.loginAttemptRepository = loginAttemptRepository;
00066 |     this.userRoleRepository = userRoleRepository;
00067 |     this.rolePermissionRepository = rolePermissionRepository;
00068 |     this.passwordEncoder = passwordEncoder;
00069 |     this.accessTokenService = accessTokenService;
00070 |     this.refreshTokenService = refreshTokenService;
00071 |     this.securityProperties = securityProperties;
00072 |     this.ipRuleService = ipRuleService;
00073 | }
00074 |
00075 | @Transactional
00076 | public AuthResponse login(LoginRequest request, HttpServletRequest servletRequest) {
00077 |     String identifier = request.usernameOrEmail().trim();
00078 |     String clientCode = request.clientCode() == null || request.clientCode().isBlank()
00079 |         ? WEB_CLIENT
00080 |         : request.clientCode().trim();
00081 |     ClientApplication clientApplication = activeClientApplication(clientCode);
00082 |     User user = userRepository.findByUsernameOrEmail(identifier).orElse(null);
00083 |     String ipAddress = clientIp(servletRequest);
00084 |     String userAgent = servletRequest.getHeader("User-Agent");
00085 |
00086 |     if (user == null) {
00087 |         logAttempt(identifier, null, request.deviceUuid(), false, "USER_NOT_FOUND", ipAddress, userAgent, clientApplication);
00088 |         throw invalidCredentials();
00089 |     }
00090 |     if (!passwordEncoder.matches(request.password(), user.getPasswordHash())) {
00091 |         logAttempt(identifier, user, request.deviceUuid(), false, "WRONG_PASSWORD", ipAddress, userAgent, clientApplication);
00092 |         throw invalidCredentials();
00093 |     }
00094 |     if (!UserStatus.ACTIVE.equalsIgnoreCase(user.getStatus().getStatusCode())) {
00095 |         logAttempt(identifier, user, request.deviceUuid(), false, "USER_INACTIVE", ipAddress, userAgent, clientApplication);
00096 |         throw new BadRequestException("User is not active.");
00097 |     }
00098 |
00099 |     UserDevice userDevice = null;
00100 |     if (MOBILE_CLIENT.equalsIgnoreCase(clientApplication.getClientCode())) {
00101 |         userDevice = authorizedDevice(user, request.deviceUuid(), clientApplication);
00102 |     } else {
00103 |         String applicationCode = request.applicationCode() == null || request.applicationCode().isBlank()
00104 |             ? DEFAULT_APPLICATION
00105 |             : request.applicationCode().trim();
```

048. src/main/java/com/citec/ems/iam/application/AuthService.java

```
00106 |         ipRuleService.assertLoginAllowed(user.getUserId(), applicationCode, ipAddress);
00107 |     }
00108 |
00109 |     LocalDate today = LocalDate.now();
00110 |     List<String> roles = userRoleRepository.findActiveRoleCodesByUserId(user.getUserId(), today);
00111 |     List<String> permissions = rolePermissionRepository.findAllowedPermissionCodesByUserId(user.getUserId(), today);
00112 |     LocalDateTime now = LocalDateTime.now();
00113 |
00114 |     UserSession session = new UserSession();
00115 |     session.setUser(user);
00116 |     session.setClientApplication(clientApplication);
00117 |     session.setUserDevice(userDevice);
00118 |     session.setSessionUuid(UUID.randomUUID());
00119 |     session.setIpAddress(ipAddress);
00120 |     session.setUserAgent(userAgent);
00121 |     session.setLastSeenAt(now);
00122 |     session.setExpiresAt(now.plus(securityProperties.refreshTokenTtl()));
00123 |     userSessionRepository.save(session);
00124 |
00125 |     String rawRefreshToken = refreshTokenService.generateRawToken();
00126 |     RefreshToken refreshToken = createRefreshToken(user, session, rawRefreshToken, UUID.randomUUID(), now);
00127 |     refreshTokenRepository.save(refreshToken);
00128 |
00129 |     GeneratedAccessToken accessToken = accessTokenService.generate(user, session.getSessionUuid(), roles, permissions, now);
00130 |     logAttempt(identifier, user, request.deviceUuid(), true, null, ipAddress, userAgent, clientApplication);
00131 |
00132 |     return response(user, roles, permissions, accessToken, rawRefreshToken, refreshToken.getExpiresAt());
00133 | }
00134 |
00135 | @Transactional
00136 | public AuthResponse refresh(String rawRefreshToken) {
00137 |     String tokenHash = refreshTokenService.hash(rawRefreshToken);
00138 |     RefreshToken currentToken = refreshTokenRepository.findByTokenHash(tokenHash)
00139 |         .orElseThrow(() -> new BadCredentialsException("Invalid refresh token."));
00140 |     LocalDateTime now = LocalDateTime.now();
00141 |     if (currentToken.getRevokedAt() != null || currentToken.getExpiresAt().isBefore(now)) {
00142 |         throw new BadCredentialsException("Refresh token is no longer valid.");
00143 |     }
00144 |     if (currentToken.getUsedAt() != null) {
00145 |         revokeSession(currentToken.getUserSession(), "REFRESH_TOKEN_REUSED");
00146 |         throw new BadCredentialsException("Refresh token has already been used.");
00147 |     }
00148 |
00149 |     User user = currentToken.getUser();
00150 |     UserSession session = currentToken.getUserSession();
00151 |     if (session.getRevokedAt() != null || session.getExpiresAt().isBefore(now)) {
00152 |         throw new BadCredentialsException("Session is no longer valid.");
00153 |     }
00154 |
00155 |     LocalDate today = LocalDate.now();
00156 |     List<String> roles = userRoleRepository.findActiveRoleCodesByUserId(user.getUserId(), today);
00157 |     List<String> permissions = rolePermissionRepository.findAllowedPermissionCodesByUserId(user.getUserId(), today);
00158 |
00159 |     currentToken.setUsedAt(now);
00160 |     String nextRawRefreshToken = refreshTokenService.generateRawToken();
```

048. src/main/java/com/citec/ems/iam/application/AuthService.java

```
00161 |         RefreshToken nextToken = createRefreshToken(
00162 |             user,
00163 |             session,
00164 |             nextRawRefreshToken,
00165 |             currentToken.getTokenFamilyUuid(),
00166 |             now);
00167 |         refreshTokenRepository.save(nextToken);
00168 |         currentToken.setReplacedByToken(nextToken);
00169 |         session.setLastSeenAt(now);
00170 |
00171 |         GeneratedAccessToken accessToken = accessTokenService.generate(user, session.getSessionUuid(), roles, permissions, now);
00172 |         return response(user, roles, permissions, accessToken, nextRawRefreshToken, nextToken.getExpiresAt());
00173 |     }
00174 |
00175 |     @Transactional
00176 |     public void logout(AuthenticatedUser authenticatedUser, String rawRefreshToken) {
00177 |         LocalDateTime now = LocalDateTime.now();
00178 |         if (authenticatedUser != null) {
00179 |             userSessionRepository.findBySessionUuid(authenticatedUser.sessionUuid())
00180 |                 .ifPresent(session -> {
00181 |                     session.setRevokedAt(now);
00182 |                     session.setRevokeReason("LOGOUT");
00183 |                 });
00184 |         }
00185 |         if (rawRefreshToken != null && !rawRefreshToken.isBlank()) {
00186 |             refreshTokenRepository.findByTokenHash(refreshTokenService.hash(rawRefreshToken))
00187 |                 .ifPresent(token -> {
00188 |                     token.setRevokedAt(now);
00189 |                     token.setRevokeReason("LOGOUT");
00190 |                 });
00191 |         }
00192 |     }
00193 |
00194 |     private RefreshToken createRefreshToken(
00195 |         User user,
00196 |         UserSession session,
00197 |         String rawToken,
00198 |         UUID tokenFamilyUuid,
00199 |         LocalDateTime now) {
00200 |         RefreshToken token = new RefreshToken();
00201 |         token.setUser(user);
00202 |         token.setUserSession(session);
00203 |         token.setTokenHash(refreshTokenService.hash(rawToken));
00204 |         token.setTokenFamilyUuid(tokenFamilyUuid);
00205 |         token.setExpiresAt(now.plus(securityProperties.refreshTokenTtl()));
00206 |         return token;
00207 |     }
00208 |
00209 |     private ClientApplication activeClientApplication(String clientCode) {
00210 |         ClientApplication clientApplication = clientApplicationRepository.findByClientCodeIgnoreCase(clientCode)
00211 |             .orElseThrow(() -> new BadRequestException("Unknown client application."));
00212 |         if (!Boolean.TRUE.equals(clientApplication.getActive())) {
00213 |             throw new BadRequestException("Client application is not active.");
00214 |         }
00215 |         return clientApplication;
    
```

048. src/main/java/com/citec/ems/iam/application/AuthService.java

```
00216 |     }
00217 |
00218 |     private UserDevice authorizedDevice(User user, UUID deviceUuid, ClientApplication clientApplication) {
00219 |         if (deviceUuid == null) {
00220 |             throw new BadRequestException("Mobile login requires a device UUID.");
00221 |         }
00222 |         UserDevice userDevice = userDeviceRepository.findByDeviceUuid(deviceUuid)
00223 |             .orElseThrow(() -> new BadRequestException("Device is not authorized."));
00224 |         if (!userDevice.getUser().getUserId().equals(user.getUserId())
00225 |             || !userDevice.getClientApplication().getClientApplicationId().equals(clientApplication.getClientApplicationId())
00226 |             || !Short.valueOf(UserDevice.AUTHORIZED).equals(userDevice.getAuthorizationStatus())) {
00227 |             throw new BadRequestException("Device is not authorized.");
00228 |         }
00229 |         return userDevice;
00230 |     }
00231 |
00232 |     private void logAttempt(
00233 |         String identifier,
00234 |         User user,
00235 |         UUID deviceUuid,
00236 |         boolean success,
00237 |         String failureReason,
00238 |         String ipAddress,
00239 |         String userAgent,
00240 |         ClientApplication clientApplication) {
00241 |         LoginAttempt attempt = new LoginAttempt();
00242 |         attempt.setUsernameOrEmail(identifier);
00243 |         attempt.setUser(user);
00244 |         attempt.setDeviceUuid(deviceUuid);
00245 |         attempt.setSuccess(success);
00246 |         attempt.setFailureReason(failureReason);
00247 |         attempt.setIpAddress(ipAddress);
00248 |         attempt.setUserAgent(userAgent);
00249 |         attempt.setClientApplication(clientApplication);
00250 |         loginAttemptRepository.save(attempt);
00251 |     }
00252 |
00253 |     private BadCredentialsException invalidCredentials() {
00254 |         return new BadCredentialsException("Invalid username/email or password.");
00255 |     }
00256 |
00257 |     private String clientIp(HttpServletRequest request) {
00258 |         String forwardedFor = request.getHeader("X-Forwarded-For");
00259 |         if (forwardedFor != null && !forwardedFor.isBlank()) {
00260 |             return forwardedFor;
00261 |         }
00262 |         return request.getRemoteAddr();
00263 |     }
00264 |
00265 |     private void revokeSession(UserSession session, String reason) {
00266 |         session.setRevokedAt(LocalDateTime.now());
00267 |         session.setRevokeReason(reason);
00268 |     }
00269 |
00270 |     private AuthResponse response(
```

048. src/main/java/com/citec/ems/iam/application/AuthService.java

```
00271 |         User user,
00272 |         List<String> roles,
00273 |         List<String> permissions,
00274 |         GeneratedAccessToken accessToken,
00275 |         String refreshToken,
00276 |         LocalDateTime refreshTokenExpiresAt) {
00277 |     return new AuthResponse(
00278 |         "Bearer",
00279 |         accessToken.token(),
00280 |         refreshToken,
00281 |         accessToken.expiresAt(),
00282 |         refreshTokenExpiresAt,
00283 |         new AuthUserResponse(
00284 |             user.getUserId(),
00285 |             user.getUsername(),
00286 |             user.getEmail(),
00287 |             user.getStatus().getStatusCode()),
00288 |         roles,
00289 |         permissions);
00290 |     }
00291 | }
00292 |
00293 |
00294 |
00295 |
```


049. src/main/java/com/citec/ems/iam/application/IamAdminService.java

src/main/java/com/citec/ems/iam/application/IamAdminService.java

228 lines

```
00001 | package com.citec.ems.iam.application;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import com.citec.ems.iam.infrastructure.*;
00006 | import com.citec.ems.shared.BadRequestException;
00007 | import com.citec.ems.shared.NotFoundException;
00008 | import com.citec.ems.iam.web.IamDtos.PermissionCreateRequest;
00009 | import com.citec.ems.iam.web.IamDtos.PermissionResponse;
00010 | import com.citec.ems.iam.web.IamDtos.RoleCreateRequest;
00011 | import com.citec.ems.iam.web.IamDtos.RolePermissionGrantRequest;
00012 | import com.citec.ems.iam.web.IamDtos.RoleResponse;
00013 | import com.citec.ems.iam.web.IamDtos.UserCreateRequest;
00014 | import com.citec.ems.iam.web.IamDtos.UserDetail;
00015 | import com.citec.ems.iam.web.IamDtos.UserRoleAssignRequest;
00016 | import com.citec.ems.iam.web.IamDtos.UserSummary;
00017 | import java.time.LocalDate;
00018 | import java.util.List;
00019 | import org.springframework.data.domain.Page;
00020 | import org.springframework.data.domain.Pageable;
00021 | import org.springframework.security.crypto.password.PasswordEncoder;
00022 | import org.springframework.stereotype.Service;
00023 | import org.springframework.transaction.annotation.Transactional;
00024 |
00025 | @Service
00026 | public class IamAdminService {
00027 |
00028 |     private final UserRepository userRepository;
00029 |     private final UserStatusRepository userStatusRepository;
00030 |     private final RoleRepository roleRepository;
00031 |     private final PermissionRepository permissionRepository;
00032 |     private final ApplicationModuleRepository applicationModuleRepository;
00033 |     private final UserRoleRepository userRoleRepository;
00034 |     private final RolePermissionRepository rolePermissionRepository;
00035 |     private final PasswordEncoder passwordEncoder;
00036 |
00037 |     public IamAdminService(
00038 |         UserRepository userRepository,
00039 |         UserStatusRepository userStatusRepository,
00040 |         RoleRepository roleRepository,
00041 |         PermissionRepository permissionRepository,
00042 |         ApplicationModuleRepository applicationModuleRepository,
00043 |         UserRoleRepository userRoleRepository,
00044 |         RolePermissionRepository rolePermissionRepository,
00045 |         PasswordEncoder passwordEncoder) {
00046 |         this.userRepository = userRepository;
00047 |         this.userStatusRepository = userStatusRepository;
00048 |         this.roleRepository = roleRepository;
00049 |         this.permissionRepository = permissionRepository;
00050 |         this.applicationModuleRepository = applicationModuleRepository;
```

049. src/main/java/com/citec/ems/lam/application/lamAdminService.java

```
00051 |         this.userRoleRepository = userRoleRepository;
00052 |         this.rolePermissionRepository = rolePermissionRepository;
00053 |         this.passwordEncoder = passwordEncoder;
00054 |     }
00055 |
00056 |     @Transactional(readOnly = true)
00057 |     public Page<UserSummary> listUsers(Pageable pageable) {
00058 |         return userRepository.findAll(pageable).map(this::userSummary);
00059 |     }
00060 |
00061 |     @Transactional(readOnly = true)
00062 |     public UserDetail getUser(Long userId) {
00063 |         User user = getUserEntity(userId);
00064 |         LocalDate today = LocalDate.now();
00065 |         return new UserDetail(
00066 |             user.getId(),
00067 |             user.getUsername(),
00068 |             user.getEmail(),
00069 |             user.getStatus().getStatusCode(),
00070 |             userRoleRepository.findActiveRoleCodesByUserId(userId, today),
00071 |             rolePermissionRepository.findAllowedPermissionCodesByUserId(userId, today),
00072 |             user.getCreatedAt(),
00073 |             user.getUpdatedAt());
00074 |     }
00075 |
00076 |     @Transactional
00077 |     public UserSummary createUser(UserCreateRequest request) {
00078 |         userRepository.findByUsernameIgnoreCase(request.username()).ifPresent(user -> {
00079 |             throw new BadRequestException("Username already exists.");
00080 |         });
00081 |         userRepository.findByEmailIgnoreCase(request.email()).ifPresent(user -> {
00082 |             throw new BadRequestException("Email already exists.");
00083 |         });
00084 |
00085 |         UserStatus status = request.userStatusId() == null
00086 |             ? userStatusRepository.findByStatusCodeIgnoreCase(UserStatus.ACTIVE)
00087 |                 .orElseThrow(() -> new NotFoundException("ACTIVE user status was not found. "))
00088 |             : userStatusRepository.findById(request.userStatusId())
00089 |                 .orElseThrow(() -> new NotFoundException("User status was not found. "));
00090 |
00091 |         User user = new User();
00092 |         user.setUsername(request.username().trim());
00093 |         user.setEmail(request.email().trim());
00094 |         user.setPasswordHash(passwordEncoder.encode(request.password()));
00095 |         user.setStatus(status);
00096 |         return userSummary(userRepository.save(user));
00097 |     }
00098 |
00099 |     @Transactional
00100 |     public void assignRole(Long userId, UserRoleAssignRequest request) {
00101 |         User user = getUserEntity(userId);
00102 |         Role role = roleRepository.findById(request.roleId())
00103 |             .orElseThrow(() -> new NotFoundException("Role was not found. "));
00104 |         LocalDate startDate = request.startDate() == null ? LocalDate.now() : request.startDate();
00105 |         UserRoleId id = new UserRoleId(user.getId(), role.getId(), startDate);
```

049. src/main/java/com/citec/ems/lam/application/lamAdminService.java

```
00106 |         if (userRepository.existsById(id)) {
00107 |             throw new BadRequestException("Role assignment already exists for this start date.");
00108 |         }
00109 |         UserRole assignment = new UserRole();
00110 |         assignment.setId(id);
00111 |         assignment.setUser(user);
00112 |         assignment.setRole(role);
00113 |         assignment.setEndDate(request.endDate());
00114 |         assignment.setAssignedBy(request.assignedByUserId() == null ? null : getUserEntity(request.assignedByUserId()));
00115 |         userRepository.save(assignment);
00116 |     }
00117 |
00118 |     @Transactional(readOnly = true)
00119 |     public Page<RoleResponse> listRoles(Pageable pageable) {
00120 |         return roleRepository.findAll(pageable).map(this::roleResponse);
00121 |     }
00122 |
00123 |     @Transactional
00124 |     public RoleResponse createRole(RoleCreateRequest request) {
00125 |         roleRepository.findByRoleCodeIgnoreCase(request.roleCode()).ifPresent(role -> {
00126 |             throw new BadRequestException("Role code already exists.");
00127 |         });
00128 |         Role role = new Role();
00129 |         role.setRoleCode(request.roleCode().trim().toUpperCase());
00130 |         role.setRoleName(request.roleName().trim());
00131 |         role.setDescription(request.description());
00132 |         role.setActive(request.active() == null || request.active());
00133 |         return roleResponse(roleRepository.save(role));
00134 |     }
00135 |
00136 |     @Transactional(readOnly = true)
00137 |     public Page<PermissionResponse> listPermissions(Pageable pageable) {
00138 |         return permissionRepository.findAll(pageable).map(this::permissionResponse);
00139 |     }
00140 |
00141 |     @Transactional
00142 |     public PermissionResponse createPermission(PermissionCreateRequest request) {
00143 |         permissionRepository.findByPermissionCodeIgnoreCase(request.permissionCode()).ifPresent(permission -> {
00144 |             throw new BadRequestException("Permission code already exists.");
00145 |         });
00146 |         Permission permission = new Permission();
00147 |         permission.setPermissionCode(request.permissionCode().trim().toUpperCase());
00148 |         permission.setPermissionName(request.permissionName().trim());
00149 |         permission.setDescription(request.description());
00150 |         permission.setActive(request.active() == null || request.active());
00151 |         permission.setApplicationModule(request.applicationModuleId() == null
00152 |             ? null
00153 |             : applicationModuleRepository.findById(request.applicationModuleId())
00154 |                 .orElseThrow(() -> new NotFoundException("Application module was not found.")));
00155 |         return permissionResponse(permissionRepository.save(permission));
00156 |     }
00157 |
00158 |     @Transactional
00159 |     public void grantPermission(Long roleId, RolePermissionGrantRequest request) {
00160 |         Role role = roleRepository.findById(roleId)
```

049. src/main/java/com/citec/ems/iam/application/IamAdminService.java

```
00161 |         .orElseThrow(() -> new NotFoundException("Role was not found."));
00162 |     Permission permission = permissionRepository.findById(request.permissionId());
00163 |     .orElseThrow(() -> new NotFoundException("Permission was not found."));
00164 |     LocalDate startDate = request.startDate() == null ? LocalDate.now() : request.startDate();
00165 |     RolePermissionId id = new RolePermissionId(role.getRoleId(), permission.getPermissionId(), startDate);
00166 |     if (rolePermissionRepository.existsById(id)) {
00167 |         throw new BadRequestException("Role permission grant already exists for this start date.");
00168 |     }
00169 |     RolePermission rolePermission = new RolePermission();
00170 |     rolePermission.setId(id);
00171 |     rolePermission.setRole(role);
00172 |     rolePermission.setPermission(permission);
00173 |     rolePermission.setEndDate(request.endDate());
00174 |     rolePermission.setAllowed(request.allowed() == null || request.allowed());
00175 |     rolePermission.setActive(request.active() == null || request.active());
00176 |     rolePermission.setGrantedBy(request.grantedByUserId() == null ? null : getUserEntity(request.grantedByUserId()));
00177 |     rolePermissionRepository.save(rolePermission);
00178 | }
00179 |
00180 | @Transactional(readOnly = true)
00181 | public List<PermissionResponse> applicationPermissions(Integer applicationModuleId) {
00182 |     ApplicationModule module = applicationModuleRepository.findById(applicationModuleId)
00183 |         .orElseThrow(() -> new NotFoundException("Application module was not found."));
00184 |     return permissionRepository.findAll().stream()
00185 |         .filter(permission -> permission.getApplicationModule() != null
00186 |             && permission.getApplicationModule().getApplicationModuleId().equals(module.getApplicationModuleId()))
00187 |         .map(this::permissionResponse)
00188 |         .toList();
00189 | }
00190 |
00191 | private User getUserEntity(Long userId) {
00192 |     return userRepository.findById(userId)
00193 |         .orElseThrow(() -> new NotFoundException("User was not found."));
00194 | }
00195 |
00196 | private UserSummary userSummary(User user) {
00197 |     return new UserSummary(
00198 |         user.getUserId(),
00199 |         user.getUsername(),
00200 |         user.getEmail(),
00201 |         user.getStatus().getStatusCode(),
00202 |         user.getCreatedAt(),
00203 |         user.getUpdatedAt());
00204 | }
00205 |
00206 | private RoleResponse roleResponse(Role role) {
00207 |     return new RoleResponse(
00208 |         role.getRoleId(),
00209 |         role.getRoleCode(),
00210 |         role.getRoleName(),
00211 |         role.getDescription(),
00212 |         role.getActive());
00213 | }
00214 |
00215 | private PermissionResponse permissionResponse(Permission permission) {
```

049. src/main/java/com/citec/ems/iam/application/IamAdminService.java

```
00216 |         return new PermissionResponse(  
00217 |             permission.getPermissionId(),  
00218 |             permission.getApplicationModule() == null ? null : permission.getApplicationModule().getApplicationModuleId(),  
00219 |             permission.getPermissionCode(),  
00220 |             permission.getPermissionName(),  
00221 |             permission.getDescription(),  
00222 |             permission.getActive());  
00223 |     }  
00224 | }  
00225 |  
00226 |  
00227 |  
00228 |
```

050. src/main/java/com/citec/ems/iam/application/IpRuleService.java

src/main/java/com/citec/ems/iam/application/IpRuleService.java

108 lines

```
00001 | package com.citec.ems.iam.application;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import com.citec.ems.iam.infrastructure.*;
00006 | import com.citec.ems.shared.BadRequestException;
00007 | import java.net.InetAddress;
00008 | import java.time.LocalDate;
00009 | import java.util.List;
00010 | import org.springframework.stereotype.Service;
00011 |
00012 | @Service
00013 | public class IpRuleService {
00014 |
00015 |     private final ApplicationModuleRepository applicationModuleRepository;
00016 |     private final UserLoginIpRuleRepository userLoginIpRuleRepository;
00017 |
00018 |     public IpRuleService(
00019 |         ApplicationModuleRepository applicationModuleRepository,
00020 |         UserLoginIpRuleRepository userLoginIpRuleRepository) {
00021 |         this.applicationModuleRepository = applicationModuleRepository;
00022 |         this.userLoginIpRuleRepository = userLoginIpRuleRepository;
00023 |     }
00024 |
00025 |     public void assertLoginAllowed(Long userId, String applicationCode, String ipAddress) {
00026 |         Integer moduleId = applicationModuleRepository.findByApplicationCodeIgnoreCase(applicationCode)
00027 |             .map(ApplicationModule::getApplicationModuleId)
00028 |             .orElse(null);
00029 |         List<UserLoginIpRule> rules = userLoginIpRuleRepository.findActiveRules(userId, moduleId, LocalDate.now());
00030 |         if (rules.isEmpty()) {
00031 |             return;
00032 |         }
00033 |
00034 |         boolean hasAllowRules = false;
00035 |         boolean matchesAllow = false;
00036 |         for (UserLoginIpRule rule : rules) {
00037 |             boolean matches = matches(rule, ipAddress);
00038 |             if (Short.valueOf(UserLoginIpRule.BLOCK).equals(rule.getRuleType()) && matches) {
00039 |                 throw new BadRequestException("Login is blocked from this IP address.");
00040 |             }
00041 |             if (Short.valueOf(UserLoginIpRule.ALLOW).equals(rule.getRuleType())) {
00042 |                 hasAllowRules = true;
00043 |                 matchesAllow = matchesAllow || matches;
00044 |             }
00045 |         }
00046 |
00047 |         if (hasAllowRules && !matchesAllow) {
00048 |             throw new BadRequestException("Login is not allowed from this IP address.");
00049 |         }
00050 |     }
```

050. src/main/java/com/citec/ems/iam/application/IpRuleService.java

```
00051 |
00052 | private boolean matches(UserLoginIpRule rule, String ipAddress) {
00053 |     if (ipAddress == null || ipAddress.isBlank()) {
00054 |         return false;
00055 |     }
00056 |     String clientIp = firstForwardedIp(ipAddress);
00057 |     if (rule.getIpAddress() != null && rule.getIpAddress().equals(clientIp)) {
00058 |         return true;
00059 |     }
00060 |     if (rule.getIpRange() != null) {
00061 |         return cidrMatches(rule.getIpRange(), clientIp);
00062 |     }
00063 |     return false;
00064 | }
00065 |
00066 | private String firstForwardedIp(String value) {
00067 |     int comma = value.indexOf(',');
00068 |     return comma >= 0 ? value.substring(0, comma).trim() : value.trim();
00069 | }
00070 |
00071 | private boolean cidrMatches(String cidr, String ipAddress) {
00072 |     if (!cidr.contains("/")) {
00073 |         return cidr.equals(ipAddress);
00074 |     }
00075 |     String[] parts = cidr.split("/");
00076 |     if (parts.length != 2) {
00077 |         return false;
00078 |     }
00079 |     try {
00080 |         byte[] networkBytes = InetAddress.getByName(parts[0]).getAddress();
00081 |         byte[] ipBytes = InetAddress.getByName(ipAddress).getAddress();
00082 |         if (networkBytes.length != 4 || ipBytes.length != 4) {
00083 |             return false;
00084 |         }
00085 |         int prefix = Integer.parseInt(parts[1]);
00086 |         if (prefix < 0 || prefix > 32) {
00087 |             return false;
00088 |         }
00089 |         long network = ipv4ToLong(networkBytes);
00090 |         long ip = ipv4ToLong(ipBytes);
00091 |         long mask = prefix == 0 ? 0 : (0xffffffffL << (32 - prefix)) & 0xffffffffL;
00092 |         return (network & mask) == (ip & mask);
00093 |     } catch (Exception ex) {
00094 |         return false;
00095 |     }
00096 | }
00097 |
00098 | private long ipv4ToLong(byte[] bytes) {
00099 |     return ((long) bytes[0] & 0xff) << 24
00100 |         | ((long) bytes[1] & 0xff) << 16
00101 |         | ((long) bytes[2] & 0xff) << 8
00102 |         | ((long) bytes[3] & 0xff);
00103 | }
00104 | }
00105 |
```

050. src/main/java/com/citec/ems/iam/application/lpRuleService.java

```
00106 |  
00107 |  
00108 |
```


051. src/main/java/com/citec/ems/iam/domain/ApplicationModule.java

src/main/java/com/citec/ems/iam/domain/ApplicationModule.java

39 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.GeneratedValue;
00006 | import jakarta.persistence.GenerationType;
00007 | import jakarta.persistence.Id;
00008 | import jakarta.persistence.Table;
00009 | import lombok.Getter;
00010 | import lombok.NoArgsConstructor;
00011 | import lombok.Setter;
00012 |
00013 | @Entity
00014 | @Table(name = "application_module", schema = "iam")
00015 | @Getter
00016 | @Setter
00017 | @NoArgsConstructor
00018 | public class ApplicationModule {
00019 |
00020 |     @Id
00021 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00022 |     @Column(name = "application_module_id")
00023 |     private Integer applicationModuleId;
00024 |
00025 |     @Column(name = "application_code", nullable = false, length = 50)
00026 |     private String applicationCode;
00027 |
00028 |     @Column(name = "application_name", nullable = false, length = 100)
00029 |     private String applicationName;
00030 |
00031 |     @Column(name = "description", length = 255)
00032 |     private String description;
00033 |
00034 |     @Column(name = "is_active", nullable = false)
00035 |     private Boolean active = true;
00036 | }
00037 |
00038 |
00039 |
```

052. src/main/java/com/citec/ems/iam/domain/ClientApplication.java

src/main/java/com/citec/ems/iam/domain/ClientApplication.java

33 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.Id;
00006 | import jakarta.persistence.Table;
00007 | import lombok.Getter;
00008 | import lombok.NoArgsConstructor;
00009 | import lombok.Setter;
00010 |
00011 | @Entity
00012 | @Table(name = "client_application", schema = "iam")
00013 | @Getter
00014 | @Setter
00015 | @NoArgsConstructor
00016 | public class ClientApplication {
00017 |
00018 |     @Id
00019 |     @Column(name = "client_application_id")
00020 |     private Short clientApplicationId;
00021 |
00022 |     @Column(name = "client_code", nullable = false, length = 50)
00023 |     private String clientCode;
00024 |
00025 |     @Column(name = "client_name", length = 100)
00026 |     private String clientName;
00027 |
00028 |     @Column(name = "is_active", nullable = false)
00029 |     private Boolean active = true;
00030 | }
00031 |
00032 |
00033 |
```

053. src/main/java/com/citec/ems/iam/domain/LoginAttempt.java

src/main/java/com/citec/ems/iam/domain/LoginAttempt.java

63 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDateTime;
00013 | import java.util.UUID;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 |
00019 | @Entity
00020 | @Table(name = "login_attempt", schema = "iam")
00021 | @Getter
00022 | @Setter
00023 | @NoArgsConstructor
00024 | public class LoginAttempt {
00025 |
00026 |     @Id
00027 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00028 |     @Column(name = "login_attempt_id")
00029 |     private Long loginAttemptId;
00030 |
00031 |     @Column(name = "username_or_email")
00032 |     private String usernameOrEmail;
00033 |
00034 |     @ManyToOne(fetch = FetchType.LAZY)
00035 |     @JoinColumn(name = "user_id")
00036 |     private User user;
00037 |
00038 |     @Column(name = "device_uuid")
00039 |     private UUID deviceUuid;
00040 |
00041 |     @Column(name = "success", nullable = false)
00042 |     private Boolean success;
00043 |
00044 |     @Column(name = "failure_reason", length = 100)
00045 |     private String failureReason;
00046 |
00047 |     @Column(name = "ip_address", length = 45)
00048 |     private String ipAddress;
00049 |
00050 |     @Column(name = "user_agent", columnDefinition = "text")
```

053. src/main/java/com/citec/ems/iam/domain/LoginAttempt.java

```
00051 |     private String userAgent;
00052 |
00053 |     @CreationTimestamp
00054 |     @Column(name = "created_at", nullable = false, updatable = false)
00055 |     private LocalDateTime createdAt;
00056 |
00057 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00058 |     @JoinColumn(name = "client_application_id", nullable = false)
00059 |     private ClientApplication clientApplication;
00060 | }
00061 |
00062 |
00063 |
```

054. src/main/java/com/citec/ems/iam/domain/Permission.java

src/main/java/com/citec/ems/iam/domain/Permission.java

52 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDateTime;
00013 | import lombok.Getter;
00014 | import lombok.NoArgsConstructor;
00015 | import lombok.Setter;
00016 | import org.hibernate.annotations.CreationTimestamp;
00017 |
00018 | @Entity
00019 | @Table(name = "permissions", schema = "iam")
00020 | @Getter
00021 | @Setter
00022 | @NoArgsConstructor
00023 | public class Permission {
00024 |
00025 |     @Id
00026 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00027 |     @Column(name = "permission_id")
00028 |     private Long permissionId;
00029 |
00030 |     @ManyToOne(fetch = FetchType.LAZY)
00031 |     @JoinColumn(name = "application_module_id")
00032 |     private ApplicationModule applicationModule;
00033 |
00034 |     @Column(name = "permission_code", nullable = false, length = 100)
00035 |     private String permissionCode;
00036 |
00037 |     @Column(name = "permission_name", nullable = false)
00038 |     private String permissionName;
00039 |
00040 |     @Column(name = "description")
00041 |     private String description;
00042 |
00043 |     @Column(name = "is_active", nullable = false)
00044 |     private Boolean active = true;
00045 |
00046 |     @CreationTimestamp
00047 |     @Column(name = "created_at", nullable = false, updatable = false)
00048 |     private LocalDateTime createdAt;
00049 | }
00050 |
```

054. src/main/java/com/citec/ems/iam/domain/Permission.java

```
00051 |  
00052 |
```

055. src/main/java/com/citec/ems/iam/domain/RefreshToken.java

src/main/java/com/citec/ems/iam/domain/RefreshToken.java

71 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDateTime;
00013 | import java.util.UUID;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 |
00019 | @Entity
00020 | @Table(name = "refresh_token", schema = "iam")
00021 | @Getter
00022 | @Setter
00023 | @NoArgsConstructor
00024 | public class RefreshToken {
00025 |
00026 |     @Id
00027 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00028 |     @Column(name = "refresh_token_id")
00029 |     private Long refreshTokenId;
00030 |
00031 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00032 |     @JoinColumn(name = "user_id", nullable = false)
00033 |     private User user;
00034 |
00035 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00036 |     @JoinColumn(name = "user_session_id", nullable = false)
00037 |     private UserSession userSession;
00038 |
00039 |     @Column(name = "token_hash", nullable = false)
00040 |     private String tokenHash;
00041 |
00042 |     @Column(name = "token_family_uuid", nullable = false)
00043 |     private UUID tokenFamilyUuid;
00044 |
00045 |     @CreationTimestamp
00046 |     @Column(name = "issued_at", nullable = false, updatable = false)
00047 |     private LocalDateTime issuedAt;
00048 |
00049 |     @Column(name = "expires_at", nullable = false)
00050 |     private LocalDateTime expiresAt;
```

055. src/main/java/com/citec/ems/iam/domain/RefreshToken.java

```
00051 |  
00052 |     @Column(name = "used_at")  
00053 |     private LocalDateTime usedAt;  
00054 |  
00055 |     @Column(name = "revoked_at")  
00056 |     private LocalDateTime revokedAt;  
00057 |  
00058 |     @Column(name = "revoke_reason")  
00059 |     private String revokeReason;  
00060 |  
00061 |     @ManyToOne(fetch = FetchType.LAZY)  
00062 |     @JoinColumn(name = "replaced_by_token_id")  
00063 |     private RefreshToken replacedByToken;  
00064 |  
00065 |     @CreationTimestamp  
00066 |     @Column(name = "created_at", nullable = false, updatable = false)  
00067 |     private LocalDateTime createdAt;  
00068 | }  
00069 |  
00070 |  
00071 |
```


056. src/main/java/com/citec/ems/iam/domain/Role.java

src/main/java/com/citec/ems/iam/domain/Role.java

45 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.GeneratedValue;
00006 | import jakarta.persistence.GenerationType;
00007 | import jakarta.persistence.Id;
00008 | import jakarta.persistence.Table;
00009 | import java.time.LocalDateTime;
00010 | import lombok.Getter;
00011 | import lombok.NoArgsConstructor;
00012 | import lombok.Setter;
00013 | import org.hibernate.annotations.CreationTimestamp;
00014 |
00015 | @Entity
00016 | @Table(name = "roles", schema = "iam")
00017 | @Getter
00018 | @Setter
00019 | @NoArgsConstructor
00020 | public class Role {
00021 |
00022 |     @Id
00023 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00024 |     @Column(name = "role_id")
00025 |     private Long roleId;
00026 |
00027 |     @Column(name = "role_code", nullable = false, length = 50)
00028 |     private String roleCode;
00029 |
00030 |     @Column(name = "role_name", nullable = false, length = 100)
00031 |     private String roleName;
00032 |
00033 |     @Column(name = "description")
00034 |     private String description;
00035 |
00036 |     @Column(name = "is_active", nullable = false)
00037 |     private Boolean active = true;
00038 |
00039 |     @CreationTimestamp
00040 |     @Column(name = "created_at", nullable = false, updatable = false)
00041 |     private LocalDateTime createdAt;
00042 | }
00043 |
00044 |
00045 |
```

057. src/main/java/com/citec/ems/iam/domain/RolePermission.java

src/main/java/com/citec/ems/iam/domain/RolePermission.java

62 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.EmbeddedId;
00005 | import jakarta.persistence.Entity;
00006 | import jakarta.persistence.FetchType;
00007 | import jakarta.persistence.JoinColumn;
00008 | import jakarta.persistence.ManyToOne;
00009 | import jakarta.persistence.MapId;
00010 | import jakarta.persistence.Table;
00011 | import java.time.LocalDate;
00012 | import java.time.LocalDateTime;
00013 | import lombok.Getter;
00014 | import lombok.NoArgsConstructor;
00015 | import lombok.Setter;
00016 | import org.hibernate.annotations.CreationTimestamp;
00017 | import org.hibernate.annotations.UpdateTimestamp;
00018 |
00019 | @Entity
00020 | @Table(name = "role_permissions", schema = "iam")
00021 | @Getter
00022 | @Setter
00023 | @NoArgsConstructor
00024 | public class RolePermission {
00025 |
00026 |     @EmbeddedId
00027 |     private RolePermissionId id;
00028 |
00029 |     @MapId("roleId")
00030 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00031 |     @JoinColumn(name = "role_id", nullable = false)
00032 |     private Role role;
00033 |
00034 |     @MapId("permissionId")
00035 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00036 |     @JoinColumn(name = "permission_id", nullable = false)
00037 |     private Permission permission;
00038 |
00039 |     @Column(name = "end_date")
00040 |     private LocalDate endDate;
00041 |
00042 |     @Column(name = "allowed", nullable = false)
00043 |     private Boolean allowed = true;
00044 |
00045 |     @Column(name = "is_active", nullable = false)
00046 |     private Boolean active = true;
00047 |
00048 |     @ManyToOne(fetch = FetchType.LAZY)
00049 |     @JoinColumn(name = "granted_by")
00050 |     private User grantedBy;
```

057. src/main/java/com/citec/ems/iam/domain/RolePermission.java

```
00051 |  
00052 |     @CreationTimestamp  
00053 |     @Column(name = "granted_at", nullable = false, updatable = false)  
00054 |     private LocalDateTime grantedAt;  
00055 |  
00056 |     @UpdateTimestamp  
00057 |     @Column(name = "updated_at")  
00058 |     private LocalDateTime updatedAt;  
00059 | }  
00060 |  
00061 |  
00062 |
```

058. src/main/java/com/citec/ems/iam/domain/RolePermissionId.java

src/main/java/com/citec/ems/iam/domain/RolePermissionId.java

32 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Embeddable;
00005 | import java.io.Serializable;
00006 | import java.time.LocalDate;
00007 | import lombok.AllArgsConstructor;
00008 | import lombok.EqualsAndHashCode;
00009 | import lombok.Getter;
00010 | import lombok.NoArgsConstructor;
00011 | import lombok.Setter;
00012 |
00013 | @Embeddable
00014 | @Getter
00015 | @Setter
00016 | @NoArgsConstructor
00017 | @AllArgsConstructor
00018 | @EqualsAndHashCode
00019 | public class RolePermissionId implements Serializable {
00020 |
00021 |     @Column(name = "role_id")
00022 |     private Long roleId;
00023 |
00024 |     @Column(name = "permission_id")
00025 |     private Long permissionId;
00026 |
00027 |     @Column(name = "start_date")
00028 |     private LocalDate startDate;
00029 | }
00030 |
00031 |
00032 |
```

059. src/main/java/com/citec/ems/iam/domain/User.java

src/main/java/com/citec/ems/iam/domain/User.java

54 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDateTime;
00013 | import lombok.Getter;
00014 | import lombok.NoArgsConstructor;
00015 | import lombok.Setter;
00016 | import org.hibernate.annotations.CreationTimestamp;
00017 | import org.hibernate.annotations.UpdateTimestamp;
00018 |
00019 | @Entity
00020 | @Table(name = "users", schema = "iam")
00021 | @Getter
00022 | @Setter
00023 | @NoArgsConstructor
00024 | public class User {
00025 |
00026 |     @Id
00027 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00028 |     @Column(name = "user_id")
00029 |     private Long userId;
00030 |
00031 |     @Column(name = "username", nullable = false, length = 50)
00032 |     private String username;
00033 |
00034 |     @Column(name = "password_hash", nullable = false)
00035 |     private String passwordHash;
00036 |
00037 |     @Column(name = "email", nullable = false)
00038 |     private String email;
00039 |
00040 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00041 |     @JoinColumn(name = "user_status_id", nullable = false)
00042 |     private UserStatus status;
00043 |
00044 |     @CreationTimestamp
00045 |     @Column(name = "created_at", nullable = false, updatable = false)
00046 |     private LocalDateTime createdAt;
00047 |
00048 |     @UpdateTimestamp
00049 |     @Column(name = "updated_at")
00050 |     private LocalDateTime updatedAt;
```

059. src/main/java/com/citec/ems/iam/domain/User.java

```
00051 | }  
00052 |  
00053 |  
00054 |
```

060. src/main/java/com/citec/ems/iam/domain/UserDevice.java

src/main/java/com/citec/ems/iam/domain/UserDevice.java

93 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDateTime;
00013 | import java.util.UUID;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 | import org.hibernate.annotations.UpdateTimestamp;
00019 |
00020 | @Entity
00021 | @Table(name = "user_device", schema = "iam")
00022 | @Getter
00023 | @Setter
00024 | @NoArgsConstructor
00025 | public class UserDevice {
00026 |
00027 |     public static final short AUTHORIZED = 1;
00028 |
00029 |     @Id
00030 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00031 |     @Column(name = "user_device_id")
00032 |     private Long userDeviceId;
00033 |
00034 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00035 |     @JoinColumn(name = "user_id", nullable = false)
00036 |     private User user;
00037 |
00038 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00039 |     @JoinColumn(name = "client_application_id", nullable = false)
00040 |     private ClientApplication clientApplication;
00041 |
00042 |     @Column(name = "device_uuid", nullable = false)
00043 |     private UUID deviceUuid;
00044 |
00045 |     @Column(name = "device_name")
00046 |     private String deviceName;
00047 |
00048 |     @Column(name = "platform", length = 50)
00049 |     private String platform;
00050 | }
```

060. src/main/java/com/citec/ems/iam/domain/UserDevice.java

```
00051 |     @Column(name = "os_version", length = 50)
00052 |     private String osVersion;
00053 |
00054 |     @Column(name = "app_version", length = 50)
00055 |     private String appVersion;
00056 |
00057 |     @Column(name = "push_token", columnDefinition = "text")
00058 |     private String pushToken;
00059 |
00060 |     @Column(name = "device_fingerprint_hash")
00061 |     private String deviceFingerprintHash;
00062 |
00063 |     @Column(name = "authorization_status", nullable = false)
00064 |     private Short authorizationStatus = 0;
00065 |
00066 |     @Column(name = "authorized_at")
00067 |     private LocalDateTime authorizedAt;
00068 |
00069 |     @ManyToOne(fetch = FetchType.LAZY)
00070 |     @JoinColumn(name = "authorized_by")
00071 |     private User authorizedBy;
00072 |
00073 |     @Column(name = "revoked_at")
00074 |     private LocalDateTime revokedAt;
00075 |
00076 |     @ManyToOne(fetch = FetchType.LAZY)
00077 |     @JoinColumn(name = "revoked_by")
00078 |     private User revokedBy;
00079 |
00080 |     @Column(name = "revoke_reason")
00081 |     private String revokeReason;
00082 |
00083 |     @CreationTimestamp
00084 |     @Column(name = "created_at", nullable = false, updatable = false)
00085 |     private LocalDateTime createdAt;
00086 |
00087 |     @UpdateTimestamp
00088 |     @Column(name = "updated_at")
00089 |     private LocalDateTime updatedAt;
00090 | }
00091 |
00092 |
00093 |
```


061. src/main/java/com/citec/ems/iam/domain/UserLoginIpRule.java

src/main/java/com/citec/ems/iam/domain/UserLoginIpRule.java

71 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDate;
00013 | import java.time.LocalDateTime;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 | import org.hibernate.annotations.UpdateTimestamp;
00019 |
00020 | @Entity
00021 | @Table(name = "user_login_ip_rule", schema = "iam")
00022 | @Getter
00023 | @Setter
00024 | @NoArgsConstructor
00025 | public class UserLoginIpRule {
00026 |
00027 |     public static final short ALLOW = 1;
00028 |     public static final short BLOCK = 2;
00029 |
00030 |     @Id
00031 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00032 |     @Column(name = "user_login_ip_rule_id")
00033 |     private Long userLoginIpRuleId;
00034 |
00035 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00036 |     @JoinColumn(name = "user_id", nullable = false)
00037 |     private User user;
00038 |
00039 |     @ManyToOne(fetch = FetchType.LAZY)
00040 |     @JoinColumn(name = "application_module_id")
00041 |     private ApplicationModule applicationModule;
00042 |
00043 |     @Column(name = "ip_address", length = 45)
00044 |     private String ipAddress;
00045 |
00046 |     @Column(name = "ip_range", length = 50)
00047 |     private String ipRange;
00048 |
00049 |     @Column(name = "rule_type", nullable = false)
00050 |     private Short ruleType = ALLOW;
```

061. src/main/java/com/citec/ems/iam/domain/UserLoginIpRule.java

```
00051 |  
00052 |     @Column(name = "is_active", nullable = false)  
00053 |     private Boolean active = true;  
00054 |  
00055 |     @Column(name = "start_date", nullable = false)  
00056 |     private LocalDate startDate = LocalDate.now();  
00057 |  
00058 |     @Column(name = "end_date")  
00059 |     private LocalDate endDate;  
00060 |  
00061 |     @CreationTimestamp  
00062 |     @Column(name = "created_at", nullable = false, updatable = false)  
00063 |     private LocalDateTime createdAt;  
00064 |  
00065 |     @UpdateTimestamp  
00066 |     @Column(name = "updated_at")  
00067 |     private LocalDateTime updatedAt;  
00068 | }  
00069 |  
00070 |  
00071 |
```

062. src/main/java/com/citec/ems/iam/domain/UserRole.java

src/main/java/com/citec/ems/iam/domain/UserRole.java

59 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.EmbeddedId;
00005 | import jakarta.persistence.Entity;
00006 | import jakarta.persistence.FetchType;
00007 | import jakarta.persistence.JoinColumn;
00008 | import jakarta.persistence.ManyToOne;
00009 | import jakarta.persistence.MapId;
00010 | import jakarta.persistence.Table;
00011 | import java.time.LocalDate;
00012 | import java.time.LocalDateTime;
00013 | import lombok.Getter;
00014 | import lombok.NoArgsConstructor;
00015 | import lombok.Setter;
00016 | import org.hibernate.annotations.CreationTimestamp;
00017 | import org.hibernate.annotations.UpdateTimestamp;
00018 |
00019 | @Entity
00020 | @Table(name = "user_roles", schema = "iam")
00021 | @Getter
00022 | @Setter
00023 | @NoArgsConstructor
00024 | public class UserRole {
00025 |
00026 |     @EmbeddedId
00027 |     private UserRoleId id;
00028 |
00029 |     @MapId("userId")
00030 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00031 |     @JoinColumn(name = "user_id", nullable = false)
00032 |     private User user;
00033 |
00034 |     @MapId("roleId")
00035 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00036 |     @JoinColumn(name = "role_id", nullable = false)
00037 |     private Role role;
00038 |
00039 |     @Column(name = "is_active", nullable = false)
00040 |     private Boolean active = true;
00041 |
00042 |     @Column(name = "end_date")
00043 |     private LocalDate endDate;
00044 |
00045 |     @ManyToOne(fetch = FetchType.LAZY)
00046 |     @JoinColumn(name = "assigned_by")
00047 |     private User assignedBy;
00048 |
00049 |     @CreationTimestamp
00050 |     @Column(name = "assigned_at", nullable = false, updatable = false)
```

062. src/main/java/com/citec/ems/iam/domain/UserRole.java

```
00051 |     private LocalDateTime assignedAt;
00052 |
00053 |     @UpdateTimestamp
00054 |     @Column(name = "updated_at")
00055 |     private LocalDateTime updatedAt;
00056 | }
00057 |
00058 |
00059 |
```

063. src/main/java/com/citec/ems/iam/domain/UserRoleId.java

src/main/java/com/citec/ems/iam/domain/UserRoleId.java

32 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Embeddable;
00005 | import java.io.Serializable;
00006 | import java.time.LocalDate;
00007 | import lombok.AllArgsConstructor;
00008 | import lombok.EqualsAndHashCode;
00009 | import lombok.Getter;
00010 | import lombok.NoArgsConstructor;
00011 | import lombok.Setter;
00012 |
00013 | @Embeddable
00014 | @Getter
00015 | @Setter
00016 | @NoArgsConstructor
00017 | @AllArgsConstructor
00018 | @EqualsAndHashCode
00019 | public class UserRoleId implements Serializable {
00020 |
00021 |     @Column(name = "user_id")
00022 |     private Long userId;
00023 |
00024 |     @Column(name = "role_id")
00025 |     private Long roleId;
00026 |
00027 |     @Column(name = "start_date")
00028 |     private LocalDate startDate;
00029 | }
00030 |
00031 |
00032 |
```

064. src/main/java/com/citec/ems/iam/domain/UserSession.java

src/main/java/com/citec/ems/iam/domain/UserSession.java

78 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.FetchType;
00006 | import jakarta.persistence.GeneratedValue;
00007 | import jakarta.persistence.GenerationType;
00008 | import jakarta.persistence.Id;
00009 | import jakarta.persistence.JoinColumn;
00010 | import jakarta.persistence.ManyToOne;
00011 | import jakarta.persistence.Table;
00012 | import java.time.LocalDateTime;
00013 | import java.util.UUID;
00014 | import lombok.Getter;
00015 | import lombok.NoArgsConstructor;
00016 | import lombok.Setter;
00017 | import org.hibernate.annotations.CreationTimestamp;
00018 |
00019 | @Entity
00020 | @Table(name = "user_session", schema = "iam")
00021 | @Getter
00022 | @Setter
00023 | @NoArgsConstructor
00024 | public class UserSession {
00025 |
00026 |     @Id
00027 |     @GeneratedValue(strategy = GenerationType.IDENTITY)
00028 |     @Column(name = "user_session_id")
00029 |     private Long userSessionId;
00030 |
00031 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00032 |     @JoinColumn(name = "user_id", nullable = false)
00033 |     private User user;
00034 |
00035 |     @ManyToOne(fetch = FetchType.LAZY, optional = false)
00036 |     @JoinColumn(name = "client_application_id", nullable = false)
00037 |     private ClientApplication clientApplication;
00038 |
00039 |     @ManyToOne(fetch = FetchType.LAZY)
00040 |     @JoinColumn(name = "user_device_id")
00041 |     private UserDevice userDevice;
00042 |
00043 |     @Column(name = "session_uuid", nullable = false)
00044 |     private UUID sessionUuid;
00045 |
00046 |     @Column(name = "ip_address", length = 45)
00047 |     private String ipAddress;
00048 |
00049 |     @Column(name = "user_agent", columnDefinition = "text")
00050 |     private String userAgent;
```

064. src/main/java/com/citec/ems/iam/domain/UserSession.java

```
00051 |  
00052 |     @CreationTimestamp  
00053 |     @Column(name = "started_at", nullable = false, updatable = false)  
00054 |     private LocalDateTime startedAt;  
00055 |  
00056 |     @Column(name = "last_seen_at")  
00057 |     private LocalDateTime lastSeenAt;  
00058 |  
00059 |     @Column(name = "expires_at", nullable = false)  
00060 |     private LocalDateTime expiresAt;  
00061 |  
00062 |     @Column(name = "revoked_at")  
00063 |     private LocalDateTime revokedAt;  
00064 |  
00065 |     @ManyToOne(fetch = FetchType.LAZY)  
00066 |     @JoinColumn(name = "revoked_by")  
00067 |     private User revokedBy;  
00068 |  
00069 |     @Column(name = "revoke_reason")  
00070 |     private String revokeReason;  
00071 |  
00072 |     @CreationTimestamp  
00073 |     @Column(name = "created_at", nullable = false, updatable = false)  
00074 |     private LocalDateTime createdAt;  
00075 | }  
00076 |  
00077 |  
00078 |
```

065. src/main/java/com/citec/ems/iam/domain/UserStatus.java

src/main/java/com/citec/ems/iam/domain/UserStatus.java

32 lines

```
00001 | package com.citec.ems.iam.domain;
00002 |
00003 | import jakarta.persistence.Column;
00004 | import jakarta.persistence.Entity;
00005 | import jakarta.persistence.Id;
00006 | import jakarta.persistence.Table;
00007 | import lombok.Getter;
00008 | import lombok.NoArgsConstructor;
00009 | import lombok.Setter;
00010 |
00011 | @Entity
00012 | @Table(name = "user_status", schema = "iam")
00013 | @Getter
00014 | @Setter
00015 | @NoArgsConstructor
00016 | public class UserStatus {
00017 |
00018 |     public static final String ACTIVE = "ACTIVE";
00019 |
00020 |     @Id
00021 |     @Column(name = "user_status_id")
00022 |     private Short userStatusId;
00023 |
00024 |     @Column(name = "status_code", nullable = false, length = 50)
00025 |     private String statusCode;
00026 |
00027 |     @Column(name = "description")
00028 |     private String description;
00029 | }
00030 |
00031 |
00032 |
```


066. src/main/java/com/citec/ems/iam/infrastructure/ApplicationModuleRepository.java

src/main/java/com/citec/ems/iam/infrastructure/ApplicationModuleRepository.java

14 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface ApplicationModuleRepository extends JpaRepository<ApplicationModule, Integer> {
00009 |     Optional<ApplicationModule> findByApplicationCodeIgnoreCase(String applicationCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

067. src/main/java/com/citec/ems/iam/infrastructure/ClientApplicationRepository.java

src/main/java/com/citec/ems/iam/infrastructure/ClientApplicationRepository.java

14 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface ClientApplicationRepository extends JpaRepository<ClientApplication, Short> {
00009 |     Optional<ClientApplication> findByClientCodeIgnoreCase(String clientCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

068. src/main/java/com/citec/ems/iam/infrastructure/LoginAttemptRepository.java

src/main/java/com/citec/ems/iam/infrastructure/LoginAttemptRepository.java

12 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import org.springframework.data.jpa.repository.JpaRepository;
00006 |
00007 | public interface LoginAttemptRepository extends JpaRepository<LoginAttempt, Long> {
00008 | }
00009 |
00010 |
00011 |
00012 |
```

069. src/main/java/com/citec/ems/iam/infrastructure/PermissionRepository.java

src/main/java/com/citec/ems/iam/infrastructure/PermissionRepository.java

14 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface PermissionRepository extends JpaRepository<Permission, Long> {
00009 |     Optional<Permission> findByPermissionCodeIgnoreCase(String permissionCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

070. src/main/java/com/citec/ems/iam/infrastructure/RefreshTokenRepository.java

src/main/java/com/citec/ems/iam/infrastructure/RefreshTokenRepository.java

14 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface RefreshTokenRepository extends JpaRepository<RefreshToken, Long> {
00009 |     Optional<RefreshToken> findByTokenHash(String tokenHash);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

071. src/main/java/com/citec/ems/iam/infrastructure/RolePermissionRepository.java

src/main/java/com/citec/ems/iam/infrastructure/RolePermissionRepository.java

34 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.time.LocalDate;
00006 | import java.util.List;
00007 | import org.springframework.data.jpa.repository.JpaRepository;
00008 | import org.springframework.data.jpa.repository.Query;
00009 | import org.springframework.data.repository.query.Param;
00010 |
00011 | public interface RolePermissionRepository extends JpaRepository<RolePermission, RolePermissionId> {
00012 |
00013 |     @Query("""
00014 |         select distinct p.permissionCode
00015 |         from RolePermission rp
00016 |         join rp.permission p
00017 |         join rp.role r
00018 |         join UserRole ur on ur.role = r
00019 |         where ur.user.userId = :userId
00020 |         and ur.active = true
00021 |         and rp.active = true
00022 |         and rp.allowed = true
00023 |         and p.active = true
00024 |         and ur.id.startDate <= :today
00025 |         and rp.id.startDate <= :today
00026 |         and (ur.endDate is null or ur.endDate >= :today)
00027 |         and (rp.endDate is null or rp.endDate >= :today)
00028 |         """)
00029 |     List<String> findAllowedPermissionCodesByUserId(@Param("userId") Long userId, @Param("today") LocalDate today);
00030 | }
00031 |
00032 |
00033 |
00034 |
```

072. src/main/java/com/citec/ems/iam/infrastructure/RoleRepository.java

src/main/java/com/citec/ems/iam/infrastructure/RoleRepository.java

14 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface RoleRepository extends JpaRepository<Role, Long> {
00009 |     Optional<Role> findByRoleCodeIgnoreCase(String roleCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

073. src/main/java/com/citec/ems/iam/infrastructure/UserDeviceRepository.java

src/main/java/com/citec/ems/iam/infrastructure/UserDeviceRepository.java

15 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.util.Optional;
00006 | import java.util.UUID;
00007 | import org.springframework.data.jpa.repository.JpaRepository;
00008 |
00009 | public interface UserDeviceRepository extends JpaRepository<UserDevice, Long> {
00010 |     Optional<UserDevice> findByDeviceUuid(UUID deviceUuid);
00011 | }
00012 |
00013 |
00014 |
00015 |
```


074. src/main/java/com/citec/ems/iam/infrastructure/UserLoginIpRuleRepository.java

src/main/java/com/citec/ems/iam/infrastructure/UserLoginIpRuleRepository.java

31 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.time.LocalDate;
00006 | import java.util.List;
00007 | import org.springframework.data.jpa.repository.JpaRepository;
00008 | import org.springframework.data.jpa.repository.Query;
00009 | import org.springframework.data.repository.query.Param;
00010 |
00011 | public interface UserLoginIpRuleRepository extends JpaRepository<UserLoginIpRule, Long> {
00012 |
00013 |     @Query("""
00014 |         select rule
00015 |         from UserLoginIpRule rule
00016 |         left join rule.applicationModule module
00017 |         where rule.user.userId = :userId
00018 |             and rule.active = true
00019 |             and rule.startDate <= :today
00020 |             and (rule.endDate is null or rule.endDate >= :today)
00021 |             and (:applicationModuleId is null or module.applicationModuleId is null or module.applicationModuleId = :applicationModuleId)
00022 |         """)
00023 |     List<UserLoginIpRule> findActiveRules(
00024 |         @Param("userId") Long userId,
00025 |         @Param("applicationModuleId") Integer applicationModuleId,
00026 |         @Param("today") LocalDate today);
00027 | }
00028 |
00029 |
00030 |
00031 |
```

075. src/main/java/com/citec/ems/iam/infrastructure/UserRepository.java

src/main/java/com/citec/ems/iam/infrastructure/UserRepository.java

27 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 | import org.springframework.data.jpa.repository.Query;
00008 | import org.springframework.data.repository.query.Param;
00009 |
00010 | public interface UserRepository extends JpaRepository<User, Long> {
00011 |     Optional<User> findByUsernameIgnoreCase(String username);
00012 |
00013 |     Optional<User> findByEmailIgnoreCase(String email);
00014 |
00015 |     @Query("""
00016 |         select user
00017 |         from User user
00018 |         join fetch user.status
00019 |         where lower(user.username) = lower(:identifier)
00020 |             or lower(user.email) = lower(:identifier)
00021 |         """)
00022 |     Optional<User> findByUsernameOrEmail(@Param("identifier") String identifier);
00023 | }
00024 |
00025 |
00026 |
00027 |
```

076. src/main/java/com/citec/ems/iam/infrastructure/UserRoleRepository.java

src/main/java/com/citec/ems/iam/infrastructure/UserRoleRepository.java

28 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.time.LocalDate;
00006 | import java.util.List;
00007 | import org.springframework.data.jpa.repository.JpaRepository;
00008 | import org.springframework.data.jpa.repository.Query;
00009 | import org.springframework.data.repository.query.Param;
00010 |
00011 | public interface UserRoleRepository extends JpaRepository<UserRole, UserRoleId> {
00012 |
00013 |     @Query("""
00014 |         select distinct r.roleCode
00015 |         from UserRole ur
00016 |         join ur.role r
00017 |         where ur.user.userId = :userId
00018 |             and ur.active = true
00019 |             and r.active = true
00020 |             and ur.id.startDate <= :today
00021 |             and (ur.endDate is null or ur.endDate >= :today)
00022 |         """)
00023 |     List<String> findActiveRoleCodesByUserId(@Param("userId") Long userId, @Param("today") LocalDate today);
00024 | }
00025 |
00026 |
00027 |
00028 |
```

077. src/main/java/com/citec/ems/iam/infrastructure/UserSessionRepository.java

src/main/java/com/citec/ems/iam/infrastructure/UserSessionRepository.java

15 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.util.Optional;
00006 | import java.util.UUID;
00007 | import org.springframework.data.jpa.repository.JpaRepository;
00008 |
00009 | public interface UserSessionRepository extends JpaRepository<UserSession, Long> {
00010 |     Optional<UserSession> findBySessionUuid(UUID sessionUuid);
00011 | }
00012 |
00013 |
00014 |
00015 |
```

078. src/main/java/com/citec/ems/iam/infrastructure/UserStatusRepository.java

src/main/java/com/citec/ems/iam/infrastructure/UserStatusRepository.java

14 lines

```
00001 | package com.citec.ems.iam.infrastructure;
00002 |
00003 |
00004 | import com.citec.ems.iam.domain.*;
00005 | import java.util.Optional;
00006 | import org.springframework.data.jpa.repository.JpaRepository;
00007 |
00008 | public interface UserStatusRepository extends JpaRepository<UserStatus, Short> {
00009 |     Optional<UserStatus> findByStatusCodeIgnoreCase(String statusCode);
00010 | }
00011 |
00012 |
00013 |
00014 |
```

079. src/main/java/com/citec/ems/iam/package-info.java

src/main/java/com/citec/ems/iam/package-info.java

4 lines

```
00001 | @org.springframework.modulith.ApplicationModule(  
00002 |     displayName = "Identity and Access Management",  
00003 |     allowedDependencies = {})  
00004 | package com.citec.ems.iam;
```

080. src/main/java/com/citec/ems/iam/security/AccessTokenAuthenticationFilter.java

src/main/java/com/citec/ems/iam/security/AccessTokenAuthenticationFilter.java

80 lines

```
00001 | package com.citec.ems.iam.security;
00002 |
00003 | import com.citec.ems.iam.domain.UserSession;
00004 | import com.citec.ems.iam.infrastructure.UserSessionRepository;
00005 | import jakarta.servlet.FilterChain;
00006 | import jakarta.servlet.ServletException;
00007 | import jakarta.servlet.http.HttpServletRequest;
00008 | import jakarta.servlet.http.HttpServletResponse;
00009 | import java.io.IOException;
00010 | import java.time.LocalDateTime;
00011 | import java.util.ArrayList;
00012 | import java.util.List;
00013 | import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
00014 | import org.springframework.security.core.authority.SimpleGrantedAuthority;
00015 | import org.springframework.security.core.context.SecurityContextHolder;
00016 | import org.springframework.security.web.authentication.WebAuthenticationDetailsSource;
00017 | import org.springframework.stereotype.Component;
00018 | import org.springframework.web.filter.OncePerRequestFilter;
00019 |
00020 | @Component
00021 | public class AccessTokenAuthenticationFilter extends OncePerRequestFilter {
00022 |
00023 |     private final AccessTokenService accessTokenService;
00024 |     private final UserSessionRepository userSessionRepository;
00025 |
00026 |     public AccessTokenAuthenticationFilter(
00027 |         AccessTokenService accessTokenService,
00028 |         UserSessionRepository userSessionRepository) {
00029 |         this.accessTokenService = accessTokenService;
00030 |         this.userSessionRepository = userSessionRepository;
00031 |     }
00032 |
00033 |     @Override
00034 |     protected void doFilterInternal(
00035 |         HttpServletRequest request,
00036 |         HttpServletResponse response,
00037 |         FilterChain filterChain) throws ServletException, IOException {
00038 |         String header = request.getHeader("Authorization");
00039 |         if (header == null || !header.startsWith("Bearer ")) {
00040 |             filterChain.doFilter(request, response);
00041 |             return;
00042 |         }
00043 |
00044 |         try {
00045 |             AccessTokenClaims claims = accessTokenService.parse(header.substring(7));
00046 |             UserSession session = userSessionRepository.findBySessionUuid(claims.sessionUuid()).orElse(null);
00047 |             if (session != null
00048 |                 && session.getRevokedAt() == null
00049 |                 && session.getExpiresAt().isAfter(LocalDateTime.now())
00050 |                 && SecurityContextHolder.getContext().getAuthentication() == null) {
```

080. src/main/java/com/citec/ems/iam/security/AccessTokenAuthenticationFilter.java

```
00051 |         AuthenticatedUser principal = new AuthenticatedUser(  
00052 |             claims.userId(),  
00053 |             claims.username(),  
00054 |             claims.sessionUuid(),  
00055 |             claims.roles(),  
00056 |             claims.permissions());  
00057 |         UsernamePasswordAuthenticationToken authentication = new UsernamePasswordAuthenticationToken(  
00058 |             principal,  
00059 |             null,  
00060 |             authorities(claims.roles(), claims.permissions()));  
00061 |         authentication.setDetails(new WebAuthenticationDetailsSource().buildDetails(request));  
00062 |         SecurityContextHolder.getContext().setAuthentication(authentication);  
00063 |     }  
00064 | } catch (RuntimeException ex) {  
00065 |     SecurityContextHolder.clearContext();  
00066 | }  
00067 |  
00068 |     filterChain.doFilter(request, response);  
00069 | }  
00070 |  
00071 | private List<SimpleGrantedAuthority> authorities(List<String> roles, List<String> permissions) {  
00072 |     List<SimpleGrantedAuthority> authorities = new ArrayList<>();  
00073 |     roles.forEach(role -> authorities.add(new SimpleGrantedAuthority("ROLE_" + role)));  
00074 |     permissions.forEach(permission -> authorities.add(new SimpleGrantedAuthority("PERMISSION_" + permission)));  
00075 |     return authorities;  
00076 | }  
00077 | }  
00078 |  
00079 |  
00080 |
```


081. src/main/java/com/citec/ems/iam/security/AccessTokenClaims.java

src/main/java/com/citec/ems/iam/security/AccessTokenClaims.java

17 lines

```
00001 | package com.citec.ems.iam.security;
00002 |
00003 | import java.time.Instant;
00004 | import java.util.List;
00005 | import java.util.UUID;
00006 |
00007 | public record AccessTokenClaims(
00008 |     Long userId,
00009 |     String username,
00010 |     UUID sessionUuid,
00011 |     Instant expiresAt,
00012 |     List<String> roles,
00013 |     List<String> permissions) {
00014 | }
00015 |
00016 |
00017 |
```

082. src/main/java/com/citec/ems/iam/security/AccessTokenService.java

src/main/java/com/citec/ems/iam/security/AccessTokenService.java

175 lines

```
00001 | package com.citec.ems.iam.security;
00002 |
00003 | import com.citec.ems.iam.domain.User;
00004 | import java.nio.charset.StandardCharsets;
00005 | import java.time.Instant;
00006 | import java.time.LocalDateTime;
00007 | import java.time.ZoneId;
00008 | import java.util.Base64;
00009 | import java.util.List;
00010 | import java.util.UUID;
00011 | import java.util.regex.Matcher;
00012 | import java.util.regex.Pattern;
00013 | import javax.crypto.Mac;
00014 | import javax.crypto.spec.SecretKeySpec;
00015 | import org.springframework.security.authentication.BadCredentialsException;
00016 | import org.springframework.stereotype.Service;
00017 |
00018 | @Service
00019 | public class AccessTokenService {
00020 |
00021 |     private static final Base64.Encoder ENCODER = Base64.getUrlEncoder().withoutPadding();
00022 |     private static final Base64.Decoder DECODER = Base64.getUrlDecoder();
00023 |     private static final String HEADER = "{\"alg\":\"HS256\",\"typ\":\"JWT\"}";
00024 |
00025 |     private final EmsSecurityProperties properties;
00026 |
00027 |     public AccessTokenService(EmsSecurityProperties properties) {
00028 |         this.properties = properties;
00029 |     }
00030 |
00031 |     public GeneratedAccessToken generate(
00032 |         User user,
00033 |         UUID sessionUuid,
00034 |         List<String> roles,
00035 |         List<String> permissions,
00036 |         LocalDateTime now) {
00037 |         Instant issuedAt = now.atZone(ZoneId.systemDefault()).toInstant();
00038 |         Instant expiresAt = issuedAt.plus(properties.accessTokenTtl());
00039 |
00040 |         String payload = "{"
00041 |             + "\"sub\":\"" + user.getUserId()
00042 |             + "\",\"username\":\"" + escape(user.getUsername()) + "\""
00043 |             + "\",\"sid\":\"" + sessionUuid + "\""
00044 |             + "\",\"iat\":\"" + issuedAt.getEpochSecond()
00045 |             + "\",\"exp\":\"" + expiresAt.getEpochSecond()
00046 |             + "\",\"roles\":\"" + jsonArray(roles)
00047 |             + "\",\"permissions\":\"" + jsonArray(permissions)
00048 |             + "\"}";
00049 |
00050 |         String unsignedToken = encode(HEADER) + "." + encode(payload);
```

082. src/main/java/com/citec/ems/iam/security/AccessTokenService.java

```
00051 |     String signature = sign(unsignedToken);
00052 |     return new GeneratedAccessToken(unsignedToken + "." + signature, LocalDateTime.ofInstant(expiresAt, ZoneId.systemDefault()));
00053 | }
00054 |
00055 | public AccessTokenClaims parse(String token) {
00056 |     String[] parts = token.split("\\.");
00057 |     if (parts.length != 3) {
00058 |         throw new BadCredentialsException("Invalid access token.");
00059 |     }
00060 |
00061 |     String unsignedToken = parts[0] + "." + parts[1];
00062 |     if (!constantTimeEquals(sign(unsignedToken), parts[2])) {
00063 |         throw new BadCredentialsException("Invalid access token signature.");
00064 |     }
00065 |
00066 |     try {
00067 |         String payload = new String(DECODER.decode(parts[1]), StandardCharsets.UTF_8);
00068 |         Instant expiresAt = Instant.ofEpochSecond(number(payload, "exp"));
00069 |         if (expiresAt.isBefore(Instant.now())) {
00070 |             throw new BadCredentialsException("Access token has expired.");
00071 |         }
00072 |
00073 |         return new AccessTokenClaims(
00074 |             number(payload, "sub"),
00075 |             string(payload, "username"),
00076 |             UUID.fromString(string(payload, "sid")),
00077 |             expiresAt,
00078 |             stringArray(payload, "roles"),
00079 |             stringArray(payload, "permissions"));
00080 |     } catch (BadCredentialsException ex) {
00081 |         throw ex;
00082 |     } catch (Exception ex) {
00083 |         throw new BadCredentialsException("Invalid access token.", ex);
00084 |     }
00085 | }
00086 |
00087 | private String encode(String value) {
00088 |     return ENCODER.encodeToString(value.getBytes(StandardCharsets.UTF_8));
00089 | }
00090 |
00091 | private String sign(String value) {
00092 |     try {
00093 |         Mac mac = Mac.getInstance("HmacSHA256");
00094 |         mac.init(new SecretKeySpec(properties.tokenSecret().getBytes(StandardCharsets.UTF_8), "HmacSHA256"));
00095 |         return ENCODER.encodeToString(mac.doFinal(value.getBytes(StandardCharsets.UTF_8)));
00096 |     } catch (Exception ex) {
00097 |         throw new IllegalStateException("Unable to sign access token.", ex);
00098 |     }
00099 | }
00100 |
00101 | private boolean constantTimeEquals(String left, String right) {
00102 |     byte[] leftBytes = left.getBytes(StandardCharsets.UTF_8);
00103 |     byte[] rightBytes = right.getBytes(StandardCharsets.UTF_8);
00104 |     if (leftBytes.length != rightBytes.length) {
00105 |         return false;
```

082. src/main/java/com/citec/ems/iam/security/AccessTokenService.java

```
00106 |     }
00107 |     int result = 0;
00108 |     for (int i = 0; i < leftBytes.length; i++) {
00109 |         result |= leftBytes[i] ^ rightBytes[i];
00110 |     }
00111 |     return result == 0;
00112 | }
00113 |
00114 | private String jsonArray(List<String> values) {
00115 |     return values.stream()
00116 |         .map(value -> "\"" + escape(value) + "\"")
00117 |         .collect(java.util.stream.Collectors.joining(", ", "[", "]"));
00118 | }
00119 |
00120 | private long number(String json, String field) {
00121 |     Matcher matcher = Pattern.compile "\"" + Pattern.quote(field) + "\"\\s*:\\s*(\\d+)\".matcher(json);
00122 |     if (!matcher.find()) {
00123 |         throw new BadCredentialsException("Invalid access token payload.");
00124 |     }
00125 |     return Long.parseLong(matcher.group(1));
00126 | }
00127 |
00128 | private String string(String json, String field) {
00129 |     Matcher matcher = Pattern.compile "\"" + Pattern.quote(field) + "\"\\s*:\\s*\\\"(?:\\\\\\\\.|[^\"])*\\\".matcher(json);
00130 |     if (!matcher.find()) {
00131 |         throw new BadCredentialsException("Invalid access token payload.");
00132 |     }
00133 |     return unescape(matcher.group(1));
00134 | }
00135 |
00136 | private List<String> stringArray(String json, String field) {
00137 |     Matcher matcher = Pattern.compile "\"" + Pattern.quote(field) + "\"\\s*:\\s*\\[(.*?)\\].matcher(json);
00138 |     if (!matcher.find()) {
00139 |         return List.of();
00140 |     }
00141 |     Matcher itemMatcher = Pattern.compile "\"((?:\\\\\\\\.|[^\"])*)\".matcher(matcher.group(1));
00142 |     java.util.ArrayList<String> values = new java.util.ArrayList<>();
00143 |     while (itemMatcher.find()) {
00144 |         values.add(unescape(itemMatcher.group(1)));
00145 |     }
00146 |     return values;
00147 | }
00148 |
00149 | private String escape(String value) {
00150 |     return value == null ? "" : value.replace("\\\\", "\\\\").replace("\"", "\\\"");
00151 | }
00152 |
00153 | private String unescape(String value) {
00154 |     StringBuilder result = new StringBuilder(value.length());
00155 |     boolean escaped = false;
00156 |     for (int i = 0; i < value.length(); i++) {
00157 |         char current = value.charAt(i);
00158 |         if (escaped) {
00159 |             result.append(current);
00160 |             escaped = false;
```

082. src/main/java/com/citec/ems/iam/security/AccessTokenService.java

```
00161 |         } else if (current == '\\') {
00162 |             escaped = true;
00163 |         } else {
00164 |             result.append(current);
00165 |         }
00166 |     }
00167 |     return result.toString();
00168 | }
00169 |
00170 | public record GeneratedAccessToken(String token, LocalDateTime expiresAt) {
00171 | }
00172 | }
00173 |
00174 |
00175 |
```

083. src/main/java/com/citec/ems/iam/security/AuthenticatedUser.java

src/main/java/com/citec/ems/iam/security/AuthenticatedUser.java

15 lines

```
00001 | package com.citec.ems.iam.security;
00002 |
00003 | import java.util.List;
00004 | import java.util.UUID;
00005 |
00006 | public record AuthenticatedUser(
00007 |     Long userId,
00008 |     String username,
00009 |     UUID sessionUuid,
00010 |     List<String> roles,
00011 |     List<String> permissions) {
00012 | }
00013 |
00014 |
00015 |
```

084. src/main/java/com/citec/ems/iam/security/EmsSecurityProperties.java

src/main/java/com/citec/ems/iam/security/EmsSecurityProperties.java

26 lines

```
00001 | package com.citec.ems.iam.security;
00002 |
00003 | import java.time.Duration;
00004 | import org.springframework.boot.context.properties.ConfigurationProperties;
00005 |
00006 | @ConfigurationProperties(prefix = "ems.security")
00007 | public record EmsSecurityProperties(
00008 |     String tokenSecret,
00009 |     Duration accessTokenTtl,
00010 |     Duration refreshTokenTtl) {
00011 |
00012 |     public EmsSecurityProperties {
00013 |         if (tokenSecret == null || tokenSecret.isBlank()) {
00014 |             tokenSecret = "change-this-local-development-secret-before-production";
00015 |         }
00016 |         if (accessTokenTtl == null) {
00017 |             accessTokenTtl = Duration.ofMinutes(15);
00018 |         }
00019 |         if (refreshTokenTtl == null) {
00020 |             refreshTokenTtl = Duration.ofDays(7);
00021 |         }
00022 |     }
00023 | }
00024 |
00025 |
00026 |
```

085. src/main/java/com/citec/ems/iam/security/RefreshTokenService.java

src/main/java/com/citec/ems/iam/security/RefreshTokenService.java

33 lines

```
00001 | package com.citec.ems.iam.security;
00002 |
00003 | import java.nio.charset.StandardCharsets;
00004 | import java.security.MessageDigest;
00005 | import java.security.SecureRandom;
00006 | import java.util.Base64;
00007 | import org.springframework.stereotype.Service;
00008 |
00009 | @Service
00010 | public class RefreshTokenService {
00011 |
00012 |     private static final int TOKEN_BYTES = 64;
00013 |     private final SecureRandom secureRandom = new SecureRandom();
00014 |
00015 |     public String generateRawToken() {
00016 |         byte[] bytes = new byte[TOKEN_BYTES];
00017 |         secureRandom.nextBytes(bytes);
00018 |         return Base64.getUrlEncoder().withoutPadding().encodeToString(bytes);
00019 |     }
00020 |
00021 |     public String hash(String token) {
00022 |         try {
00023 |             MessageDigest digest = MessageDigest.getInstance("SHA-256");
00024 |             byte[] hashed = digest.digest(token.getBytes(StandardCharsets.UTF_8));
00025 |             return Base64.getUrlEncoder().withoutPadding().encodeToString(hashed);
00026 |         } catch (Exception ex) {
00027 |             throw new IllegalStateException("Unable to hash refresh token.", ex);
00028 |         }
00029 |     }
00030 | }
00031 |
00032 |
00033 |
```


086. src/main/java/com/citec/ems/iam/security/SecurityConfig.java

src/main/java/com/citec/ems/iam/security/SecurityConfig.java

70 lines

```
00001 | package com.citec.ems.iam.security;
00002 |
00003 | import org.springframework.context.annotation.Bean;
00004 | import org.springframework.context.annotation.Configuration;
00005 | import org.springframework.http.HttpMethod;
00006 | import org.springframework.security.config.annotation.method.configuration.EnableMethodSecurity;
00007 | import org.springframework.security.config.annotation.web.builders.HttpSecurity;
00008 | import org.springframework.security.config.annotation.web.configurers.AbstractHttpConfigurer;
00009 | import org.springframework.security.config.http.SessionCreationPolicy;
00010 | import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;
00011 | import org.springframework.security.crypto.password.PasswordEncoder;
00012 | import org.springframework.security.core.userdetails.UserDetailsService;
00013 | import org.springframework.security.core.userdetails.UsernameNotFoundException;
00014 | import org.springframework.security.web.SecurityFilterChain;
00015 | import org.springframework.security.web.authentication.UsernamePasswordAuthenticationFilter;
00016 |
00017 | @Configuration
00018 | @EnableMethodSecurity
00019 | public class SecurityConfig {
00020 |
00021 |     @Bean
00022 |     SecurityFilterChain securityFilterChain(
00023 |         HttpSecurity http,
00024 |         AccessTokenAuthenticationFilter accessTokenAuthenticationFilter) throws Exception {
00025 |         return http
00026 |             .csrf(AbstractHttpConfigurer::disable)
00027 |             .httpBasic(AbstractHttpConfigurer::disable)
00028 |             .formLogin(AbstractHttpConfigurer::disable)
00029 |             .sessionManagement(session -> session.sessionCreationPolicy(SessionCreationPolicy.STATELESS))
00030 |             .authorizeHttpRequests(auth -> auth
00031 |                 .requestMatchers("/api/auth/**").permitAll()
00032 |                 .requestMatchers(HttpMethod.GET, "/api/hr/**")
00033 |                 .hasAnyAuthority(
00034 |                     "ROLE_SUPER_ADMIN",
00035 |                     "ROLE_HR_MANAGER",
00036 |                     "PERMISSION_EMPLOYEE_VIEW",
00037 |                     "PERMISSION_ORG_STRUCTURE_MANAGE",
00038 |                     "PERMISSION_POSITION_MANAGE")
00039 |                 .requestMatchers("/api/hr/**")
00040 |                 .hasAnyAuthority(
00041 |                     "ROLE_SUPER_ADMIN",
00042 |                     "ROLE_HR_MANAGER",
00043 |                     "PERMISSION_EMPLOYEE_EDIT",
00044 |                     "PERMISSION_ORG_STRUCTURE_MANAGE",
00045 |                     "PERMISSION_POSITION_MANAGE")
00046 |                 .requestMatchers("/api/iam/**")
00047 |                 .hasAnyAuthority(
00048 |                     "ROLE_SUPER_ADMIN",
00049 |                     "PERMISSION_USER_MANAGE",
00050 |                     "PERMISSION_ROLE_MANAGE")
```

086. src/main/java/com/citec/ems/iam/security/SecurityConfig.java

```
00051 |         .anyRequest().authenticated()
00052 |         .addFilterBefore(accessTokenAuthenticationFilter, UsernamePasswordAuthenticationFilter.class)
00053 |         .build();
00054 |     }
00055 |
00056 |     @Bean
00057 |     PasswordEncoder passwordEncoder() {
00058 |         return new BCryptPasswordEncoder();
00059 |     }
00060 |
00061 |     @Bean
00062 |     UserDetailsService userDetailsService() {
00063 |         return username -> {
00064 |             throw new UsernameNotFoundException(username);
00065 |         };
00066 |     }
00067 | }
00068 |
00069 |
00070 |
```

087. src/main/java/com/citec/ems/iam/web/AuthController.java

src/main/java/com/citec/ems/iam/web/AuthController.java

51 lines

```
00001 | package com.citec.ems.iam.web;
00002 |
00003 |
00004 | import com.citec.ems.iam.application.*;
00005 | import com.citec.ems.iam.web.IamDtos.AuthResponse;
00006 | import com.citec.ems.iam.web.IamDtos.LoginRequest;
00007 | import com.citec.ems.iam.web.IamDtos.LogoutRequest;
00008 | import com.citec.ems.iam.web.IamDtos.RefreshRequest;
00009 | import com.citec.ems.iam.security.AuthenticatedUser;
00010 | import jakarta.servlet.http.HttpServletRequest;
00011 | import jakarta.validation.Valid;
00012 | import org.springframework.http.HttpStatus;
00013 | import org.springframework.security.core.annotation.AuthenticationPrincipal;
00014 | import org.springframework.web.bind.annotation.PostMapping;
00015 | import org.springframework.web.bind.annotation.RequestBody;
00016 | import org.springframework.web.bind.annotation.RequestMapping;
00017 | import org.springframework.web.bind.annotation.ResponseStatus;
00018 | import org.springframework.web.bind.annotation.RestController;
00019 |
00020 | @RestController
00021 | @RequestMapping("/api/auth")
00022 | public class AuthController {
00023 |
00024 |     private final AuthService authService;
00025 |
00026 |     public AuthController(AuthService authService) {
00027 |         this.authService = authService;
00028 |     }
00029 |
00030 |     @PostMapping("/login")
00031 |     public AuthResponse login(@Valid @RequestBody LoginRequest request, HttpServletRequest servletRequest) {
00032 |         return authService.login(request, servletRequest);
00033 |     }
00034 |
00035 |     @PostMapping("/refresh")
00036 |     public AuthResponse refresh(@Valid @RequestBody RefreshRequest request) {
00037 |         return authService.refresh(request.refreshToken());
00038 |     }
00039 |
00040 |     @PostMapping("/logout")
00041 |     @ResponseStatus(HttpStatus.NO_CONTENT)
00042 |     public void logout(
00043 |         @AuthenticationPrincipal AuthenticatedUser authenticatedUser,
00044 |         @RequestBody(required = false) LogoutRequest request) {
00045 |         authService.logout(authenticatedUser, request == null ? null : request.refreshToken());
00046 |     }
00047 | }
00048 |
00049 |
00050 |
```

00051 |

088. src/main/java/com/citec/ems/iam/web/IamAdminController.java

src/main/java/com/citec/ems/iam/web/IamAdminController.java

97 lines

```
00001 | package com.citec.ems.iam.web;
00002 |
00003 |
00004 | import com.citec.ems.iam.application.*;
00005 | import com.citec.ems.iam.web.IamDtos.PermissionCreateRequest;
00006 | import com.citec.ems.iam.web.IamDtos.PermissionResponse;
00007 | import com.citec.ems.iam.web.IamDtos.RoleCreateRequest;
00008 | import com.citec.ems.iam.web.IamDtos.RolePermissionGrantRequest;
00009 | import com.citec.ems.iam.web.IamDtos.RoleResponse;
00010 | import com.citec.ems.iam.web.IamDtos.UserCreateRequest;
00011 | import com.citec.ems.iam.web.IamDtos.UserDetail;
00012 | import com.citec.ems.iam.web.IamDtos.UserRoleAssignRequest;
00013 | import com.citec.ems.iam.web.IamDtos.UserSummary;
00014 | import jakarta.validation.Valid;
00015 | import java.util.List;
00016 | import org.springframework.data.domain.Page;
00017 | import org.springframework.data.domain.Pageable;
00018 | import org.springframework.http.HttpStatus;
00019 | import org.springframework.web.bind.annotation.GetMapping;
00020 | import org.springframework.web.bind.annotation.PathVariable;
00021 | import org.springframework.web.bind.annotation.PostMapping;
00022 | import org.springframework.web.bind.annotation.RequestBody;
00023 | import org.springframework.web.bind.annotation.RequestMapping;
00024 | import org.springframework.web.bind.annotation.ResponseStatus;
00025 | import org.springframework.web.bind.annotation.RestController;
00026 |
00027 | @RestController
00028 | @RequestMapping("/api/iam")
00029 | public class IamAdminController {
00030 |
00031 |     private final IamAdminService iamAdminService;
00032 |
00033 |     public IamAdminController(IamAdminService iamAdminService) {
00034 |         this.iamAdminService = iamAdminService;
00035 |     }
00036 |
00037 |     @GetMapping("/users")
00038 |     public Page<UserSummary> listUsers(Pageable pageable) {
00039 |         return iamAdminService.listUsers(pageable);
00040 |     }
00041 |
00042 |     @PostMapping("/users")
00043 |     @ResponseStatus(HttpStatus.CREATED)
00044 |     public UserSummary createUser(@Valid @RequestBody UserCreateRequest request) {
00045 |         return iamAdminService.createUser(request);
00046 |     }
00047 |
00048 |     @GetMapping("/users/{userId}")
00049 |     public UserDetail getUser(@PathVariable Long userId) {
00050 |         return iamAdminService.getUser(userId);
```

088. src/main/java/com/citec/ems/iam/web/IamAdminController.java

```
00051 |     }
00052 |
00053 |     @PostMapping("/users/{userId}/roles")
00054 |     @ResponseStatus(HttpStatus.NO_CONTENT)
00055 |     public void assignRole(@PathVariable Long userId, @Valid @RequestBody UserRoleAssignRequest request) {
00056 |         iamAdminService.assignRole(userId, request);
00057 |     }
00058 |
00059 |     @GetMapping("/roles")
00060 |     public Page<RoleResponse> listRoles(Pageable pageable) {
00061 |         return iamAdminService.listRoles(pageable);
00062 |     }
00063 |
00064 |     @PostMapping("/roles")
00065 |     @ResponseStatus(HttpStatus.CREATED)
00066 |     public RoleResponse createRole(@Valid @RequestBody RoleCreateRequest request) {
00067 |         return iamAdminService.createRole(request);
00068 |     }
00069 |
00070 |     @PostMapping("/roles/{roleId}/permissions")
00071 |     @ResponseStatus(HttpStatus.NO_CONTENT)
00072 |     public void grantPermission(
00073 |         @PathVariable Long roleId,
00074 |         @Valid @RequestBody RolePermissionGrantRequest request) {
00075 |         iamAdminService.grantPermission(roleId, request);
00076 |     }
00077 |
00078 |     @GetMapping("/permissions")
00079 |     public Page<PermissionResponse> listPermissions(Pageable pageable) {
00080 |         return iamAdminService.listPermissions(pageable);
00081 |     }
00082 |
00083 |     @PostMapping("/permissions")
00084 |     @ResponseStatus(HttpStatus.CREATED)
00085 |     public PermissionResponse createPermission(@Valid @RequestBody PermissionCreateRequest request) {
00086 |         return iamAdminService.createPermission(request);
00087 |     }
00088 |
00089 |     @GetMapping("/application-modules/{applicationModuleId}/permissions")
00090 |     public List<PermissionResponse> applicationPermissions(@PathVariable Integer applicationModuleId) {
00091 |         return iamAdminService.applicationPermissions(applicationModuleId);
00092 |     }
00093 | }
00094 |
00095 |
00096 |
00097 |
```

089. src/main/java/com/citec/ems/iam/web/IamDtos.java

src/main/java/com/citec/ems/iam/web/IamDtos.java

144 lines

```
00001 | package com.citec.ems.iam.web;
00002 |
00003 |
00004 | import com.citec.ems.iam.application.*;
00005 | import jakarta.validation.Valid;
00006 | import jakarta.validation.constraints.Email;
00007 | import jakarta.validation.constraints.NotBlank;
00008 | import jakarta.validation.constraints.NotNull;
00009 | import jakarta.validation.constraints.Size;
00010 | import java.time.LocalDate;
00011 | import java.time.LocalDateTime;
00012 | import java.util.List;
00013 | import java.util.UUID;
00014 |
00015 | public final class IamDtos {
00016 |
00017 |     private IamDtos() {
00018 |     }
00019 |
00020 |     public record LoginRequest(
00021 |         @NotBlank String usernameOrEmail,
00022 |         @NotBlank String password,
00023 |         String clientCode,
00024 |         String applicationCode,
00025 |         UUID deviceUuid) {
00026 |     }
00027 |
00028 |     public record RefreshRequest(@NotBlank String refreshToken) {
00029 |     }
00030 |
00031 |     public record LogoutRequest(String refreshToken) {
00032 |     }
00033 |
00034 |     public record AuthResponse(
00035 |         String tokenType,
00036 |         String accessToken,
00037 |         String refreshToken,
00038 |         LocalDateTime accessTokenExpiresAt,
00039 |         LocalDateTime refreshTokenExpiresAt,
00040 |         AuthUserResponse user,
00041 |         List<String> roles,
00042 |         List<String> permissions) {
00043 |     }
00044 |
00045 |     public record AuthUserResponse(Long userId, String username, String email, String statusCode) {
00046 |     }
00047 |
00048 |     public record UserSummary(
00049 |         Long userId,
00050 |         String username,
```

089. src/main/java/com/citec/ems/iam/web/IamDtos.java

```
00051 |         String email,
00052 |         String statusCode,
00053 |         LocalDateTime createdAt,
00054 |         LocalDateTime updatedAt) {
00055 |     }
00056 |
00057 |     public record UserCreateRequest(
00058 |         @NotBlank @Size(max = 50) String username,
00059 |         @NotBlank @Email @Size(max = 255) String email,
00060 |         @NotBlank @Size(min = 8, max = 128) String password,
00061 |         Short userStatusId) {
00062 |     }
00063 |
00064 |     public record UserRoleAssignRequest(
00065 |         @NotNull Long roleId,
00066 |         LocalDate startDate,
00067 |         LocalDate endDate,
00068 |         Long assignedByUserId) {
00069 |     }
00070 |
00071 |     public record RoleResponse(Long roleId, String roleCode, String roleName, String description, Boolean active) {
00072 |     }
00073 |
00074 |     public record RoleCreateRequest(
00075 |         @NotBlank @Size(max = 50) String roleCode,
00076 |         @NotBlank @Size(max = 100) String roleName,
00077 |         @Size(max = 255) String description,
00078 |         Boolean active) {
00079 |     }
00080 |
00081 |     public record PermissionResponse(
00082 |         Long permissionId,
00083 |         Integer applicationModuleId,
00084 |         String permissionCode,
00085 |         String permissionName,
00086 |         String description,
00087 |         Boolean active) {
00088 |     }
00089 |
00090 |     public record PermissionCreateRequest(
00091 |         Integer applicationModuleId,
00092 |         @NotBlank @Size(max = 100) String permissionCode,
00093 |         @NotBlank @Size(max = 255) String permissionName,
00094 |         @Size(max = 255) String description,
00095 |         Boolean active) {
00096 |     }
00097 |
00098 |     public record RolePermissionGrantRequest(
00099 |         @NotNull Long permissionId,
00100 |         LocalDate startDate,
00101 |         LocalDate endDate,
00102 |         Boolean allowed,
00103 |         Boolean active,
00104 |         Long grantedByUserId) {
00105 |     }
```


089. src/main/java/com/citec/ems/iam/web/IamDtos.java

```
00106 |
00107 |     public record UserDetails(
00108 |         Long userId,
00109 |         String username,
00110 |         String email,
00111 |         String statusCode,
00112 |         List<String> roles,
00113 |         List<String> permissions,
00114 |         LocalDateTime createdAt,
00115 |         LocalDateTime updatedAt) {
00116 |     }
00117 |
00118 |     public record LoginIpRuleCreateRequest(
00119 |         @NotNull Long userId,
00120 |         Integer applicationModuleId,
00121 |         String ipAddress,
00122 |         String ipRange,
00123 |         @NotNull Short ruleType,
00124 |         Boolean active,
00125 |         LocalDate startDate,
00126 |         LocalDate endDate) {
00127 |     }
00128 |
00129 |     public record DeviceRegisterRequest(
00130 |         @NotNull Long userId,
00131 |         @NotNull Short clientApplicationId,
00132 |         @NotNull UUID deviceUuid,
00133 |         String deviceName,
00134 |         String platform,
00135 |         String osVersion,
00136 |         String appVersion,
00137 |         String pushToken,
00138 |         String deviceFingerprintHash) {
00139 |     }
00140 | }
00141 |
00142 |
00143 |
00144 |
```

090. src/main/java/com/citec/ems/shared/ApiExceptionHandler.java

src/main/java/com/citec/ems/shared/ApiExceptionHandler.java

50 lines

```
00001 | package com.citec.ems.shared;
00002 |
00003 | import java.time.Instant;
00004 | import java.util.Map;
00005 | import org.springframework.http.HttpStatus;
00006 | import org.springframework.http.ResponseEntity;
00007 | import org.springframework.security.access.AccessDeniedException;
00008 | import org.springframework.security.authentication.BadCredentialsException;
00009 | import org.springframework.web.bind.MethodArgumentNotValidException;
00010 | import org.springframework.web.bind.annotation.ExceptionHandler;
00011 | import org.springframework.web.bind.annotation.RestControllerAdvice;
00012 |
00013 | @RestControllerAdvice
00014 | public class ApiExceptionHandler {
00015 |
00016 |     @ExceptionHandler(NotFoundException.class)
00017 |     ResponseEntity<Map<String, Object>> handleNotFound(NotFoundException ex) {
00018 |         return error(HttpStatus.NOT_FOUND, ex.getMessage());
00019 |     }
00020 |
00021 |     @ExceptionHandler({BadRequestException.class, BadCredentialsException.class})
00022 |     ResponseEntity<Map<String, Object>> handleBadRequest(RuntimeException ex) {
00023 |         return error(HttpStatus.BAD_REQUEST, ex.getMessage());
00024 |     }
00025 |
00026 |     @ExceptionHandler(AccessDeniedException.class)
00027 |     ResponseEntity<Map<String, Object>> handleForbidden(AccessDeniedException ex) {
00028 |         return error(HttpStatus.FORBIDDEN, "You do not have permission to perform this action.");
00029 |     }
00030 |
00031 |     @ExceptionHandler(MethodArgumentNotValidException.class)
00032 |     ResponseEntity<Map<String, Object>> handleValidation(MethodArgumentNotValidException ex) {
00033 |         String message = ex.getBindingResult().getFieldErrors().stream()
00034 |             .findFirst()
00035 |             .map(error -> error.getField() + " " + error.getDefaultMessage())
00036 |             .orElse("Validation failed.");
00037 |         return error(HttpStatus.BAD_REQUEST, message);
00038 |     }
00039 |
00040 |     private ResponseEntity<Map<String, Object>> error(HttpStatus status, String message) {
00041 |         return ResponseEntity.status(status).body(Map.of(
00042 |             "timestamp", Instant.now().toString(),
00043 |             "status", status.value(),
00044 |             "error", status.getReasonPhrase(),
00045 |             "message", message));
00046 |     }
00047 | }
00048 |
00049 |
00050 |
```

091. src/main/java/com/citec/ems/shared/BadRequestException.java

src/main/java/com/citec/ems/shared/BadRequestException.java

11 lines

```
00001 | package com.citec.ems.shared;
00002 |
00003 | public class BadRequestException extends RuntimeException {
00004 |
00005 |     public BadRequestException(String message) {
00006 |         super(message);
00007 |     }
00008 | }
00009 |
00010 |
00011 |
```

092. src/main/java/com/citec/ems/shared/NotFoundException.java

src/main/java/com/citec/ems/shared/NotFoundException.java

11 lines

```
00001 | package com.citec.ems.shared;
00002 |
00003 | public class NotFoundException extends RuntimeException {
00004 |
00005 |     public NotFoundException(String message) {
00006 |         super(message);
00007 |     }
00008 | }
00009 |
00010 |
00011 |
```

093. src/main/java/com/citec/ems/shared/package-info.java

src/main/java/com/citec/ems/shared/package-info.java

4 lines

```
00001 | @org.springframework.modulith.ApplicationModule(  
00002 |     displayName = "Shared",  
00003 |     allowedDependencies = {})  
00004 | package com.citec.ems.shared;
```

094. src/test/java/com/citec/ems/EmsApplicationTests.java

src/test/java/com/citec/ems/EmsApplicationTests.java

19 lines

```
00001 | package com.citec.ems;
00002 |
00003 | import org.junit.jupiter.api.Test;
00004 | import org.springframework.boot.test.context.SpringBootTest;
00005 | import org.springframework.modulith.core.ApplicationModules;
00006 |
00007 | @SpringBootTest
00008 | class EmsApplicationTests {
00009 |
00010 |     @Test
00011 |     void contextLoads() {
00012 |     }
00013 |
00014 |     @Test
00015 |     void verifiesModularStructure() throws ClassNotFoundException {
00016 |         ApplicationModules.of(Class.forName("com.citec.ems.EmsApplication")).verify();
00017 |     }
00018 |
00019 | }
```